MODE-S Wind on a phone

The mobile view `/app`: Skew-T, time–height section, ILS approach and approach conditions in your pocket

Otso Laakso (OH2GAX)
Tuusula, Finland · near Helsinki-Vantaa airport (EFHK)

In short. The full MODE-S Wind pages are made for a computer screen. The mobile view is a light page for phones: open the address of MODE-S Wind with `/app` at the end. It shows *one product at a time*, filling the screen:

- **Skew-T · Area now** — the temperature and wind profile measured by the aircraft over the last 30 minutes to 3 hours, with the radiosonde and the MEPS model if you like;
- **Skew-T · Area hour** — any stored hourly profile;
- **Time–height** — the hourly profiles side by side: fronts, inversions and jet streams as they pass;
- **Calendar** — weeks at a glance: one row per day, one column per hour, coloured by the inversion, the low-level jet, the wind, the windshear reports or the turbulence of that hour;
- **ILS approach** — the aircraft on final on the 3° glideslope and on a map, live, with the wind measured along the approach, any windshear alert and, if you like, the ACI + windshear chart;
- **EDR · WS · ACI · ST** — the approach conditions index and windshear of every landing and departure (also with the turbulence, EDR), the space–time diagram of the wind changes, or the boundary layer (low-level jet, inversion; also flight by flight), with the flights of the period as a list;
- **History** — the landings or the departures of 1 hour to a day, or of any date, as cards with their winds per height band and their reports, with a search; and the METARs and pilot reports;
- **Statistics** — what the receiver hears: message rate, downlink formats, meteo yield, range rings and more, the sections of your choice;
- **Weather** — the MEPS weather maps of the newest model run, hour by hour from 6 hours back to 24 or 48 hours ahead, with an animation slider, the FMI radar at the present hour and, if you like, the live air traffic on the map;
- **GNSS** — for users with the *Full view* right: aircraft with degraded GPS now and hour by hour, and the spoofing indicators (Section 14).

Everything is chosen from the ≡ menu at the top left. Tap a diagram for its values. *Add to Home Screen* gives MODE-S Wind its own icon, and it opens full screen like an app. The diagrams are the same as on the full site, so the other user guides explain how to read them; this guide explains how to use them on a phone.

Contents

1 Getting started **2**

2 The menu **3**

3 Touch, rotation and updates **4**

4 Skew-T: Area now and Area hour **6**

5 Time–height **8**

6 Calendar **12**

7 ILS approach **13**

8 EDR · WS · ACI · ST **22**

9 History **29**

10 Statistics **35**

11 Weather **38**

12 Mast **47**

13 Meteogram **49**

14 GNSS (Full view) **54**

15 Pilot (Pilot right) **57**

16 When something looks wrong **62**

17 Quick reference **63**

1 Getting started

1.1 Opening the mobile view

Open the phone's browser and go to the address you use for MODE-S Wind on a computer, with `/app` added at the end — for example `http://your-address/app`. The phone has to reach MODE-S Wind the same way a computer does (your own network, or the connection set up for it). The browser asks for the same user name and password as the other pages and normally remembers them.

The mobile view opens the product you looked at last. All choices — the view, its options, the theme — are remembered on the phone. The Skew-T options and a chosen Light or Dark theme are the same settings as on the Live Map and the Sounding page *of the same browser*: if you also open the full site on the phone, it starts with the same choices (the full site has no Auto).

1.2 Add to Home Screen

Android (Chrome): open `/app`, tap the browser menu ⋮ and choose Add to Home screen (on some phones Install app), then Add.

iPhone and iPad (Safari): open `/app`, tap the Share button (the square with an arrow) and choose Add to Home Screen, then Add.

the icon

The MODE-S Wind icon (right) appears among your apps. It opens the mobile view full screen, without the browser's address bar, and can be switched to like any other app. The first time it is opened the phone may ask for the user name and password once more. From 28 October 2026 the login is then **remembered for 30 days** (renewed while you use MODE-S Wind) — also after the phone's screen has been off or the browser has been closed.

Getting a new version (from 11 November 2026). An installed app has no browser *Refresh*, so the app looks after it itself. When you come back to it and the server has a newer version, it reloads at once. When a new version arrives *while the app is open* (it checks every five minutes, and when you open What's new), a bar `New version available` appears at the bottom: tap Reload, or × to do it later. You can also reload by hand at any time: ≡ → General → Reload. The view you were in and all your options are kept. Without a connection nothing is reloaded (the button says `No connection`).

1.3 The screen

A slim bar at the top holds five things (Fig. 1):

- ≡ at the left opens the menu (Section 2);
- the **name of the view** and under it a short line: what is shown and when it was updated — e.g. `last 60 min · 15:55 UTC`, `24.10.2026 15–16 UTC`, or on the ILS view the number of aircraft on the approach, the QNH and any active windshear alert;
- the **current time in UTC** (from 28 October 2026), hours and minutes — all times of MODE-S Wind are UTC, two hours behind Finnish winter time and three behind summer time (the one exception is the Meteogram's Time option, Section 13);
- the **theme button** (from 28 October 2026): *moon* — dark theme, *sun* — light theme, *half-filled circle* — **Auto**: the theme follows the phone's own dark / light setting, and changes with it while the page is open (e.g. when the phone turns dark in the evening). Each tap steps *Dark* → *Light* → *Auto* → *Dark*; a short note under the button names the new setting. Auto is the setting until you choose one; your choice is remembered on the phone;

- a **dot** at the right: the system health of MODE-S Wind, the same indicators as on the Statistics page. *Green*: all parts work. *Amber*: something is late, retrying or using a fallback (e.g. no new MEPS model data). *Red*: a part has stopped. The menu's General section says which part; the Statistics page on the full site has the details.

Below the bar is the product. All times are UTC.

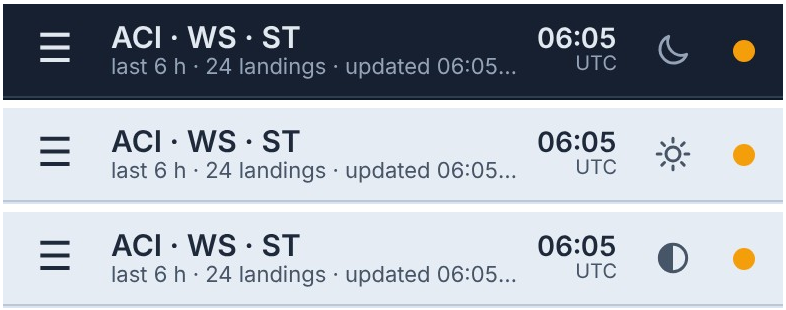


Figure 1. The top bar with the theme set to Dark (moon), Light (sun) and Auto (half circle; here the phone is set to light): the view and its status line, the current UTC time, the theme button and the system-health dot.

2 The menu

Tap ≡. The menu slides in from the left (Fig. 2); × or a tap beside it closes it.

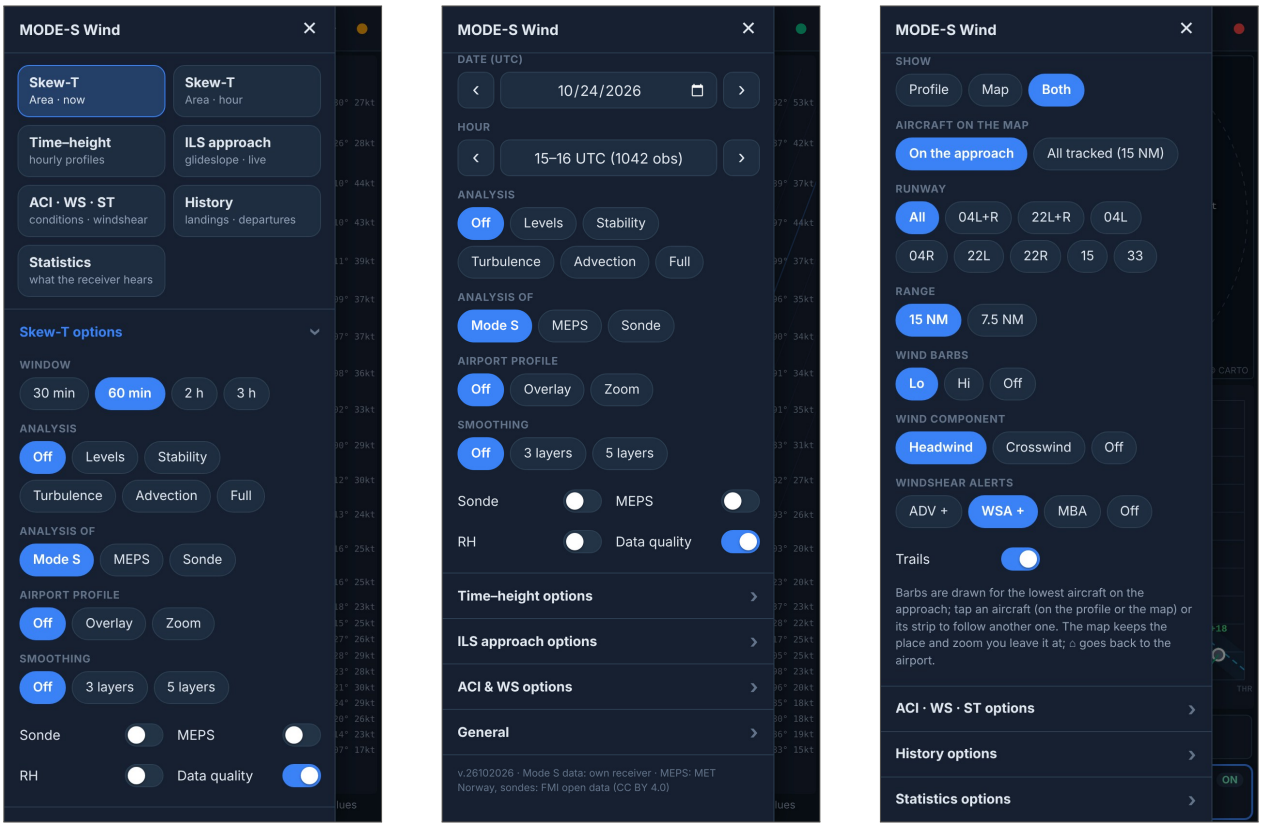


Figure 2. The menu. Left: the views as tiles (the shown one highlighted) and the options of the shown view, here Skew-T, opened below them. Middle: Area · hour with its date and hour steppers and the switches. Right: the ILS approach options (from 28 October 2026 with Show and Aircraft on the map); the other products' options are folded but one tap away.

Views. The tiles at the top choose the product (Weather from 30 October 2026, Mast from 1 November 2026, Section 12); users with the *Full view* right also have GNSS (from 3 November 2026, Section 14), and users with the *Pilot* right Pilot (from 5 November 2026, Section 15); Meteogram is for everyone (from 5 November 2026, Section 13), and so is Calendar (from 10 November 2026, Section 6). A tap switches to it and closes the menu.

Options. Below the tiles each product has a folding section: Skew-T options, Time–height options, Calendar options, ILS approach options, EDR · WS · ACI · ST options, Mast options, Meteogram options, Weather options, History options and Statistics options. The shown view's section opens by itself (its title in blue); the others stay folded and open with a tap, so you

can, for example, set the Skew-T options while the History is shown. The menu scrolls on its own.

Controls. Choices are buttons: the highlighted one is in use, a tap on another changes it at once. On / off settings are switches. Dates have a calendar field and, for stored hours, ⟨ ⟩ buttons that step one day or one hour back and forward (over midnight to the next or previous day). Changes take effect immediately; there is no *Apply* button.

General. The system health in words, What's new, About and User guides (from 28 October 2026, Fig. 3), a link to the full site and to this guide. What's new shows the same short notes of each new version as the full site's *What's new* window — the newest one open, the older ones folded (tap to open). What's new is also at the top of the menu, next to the *MODE-S Wind* title (from 30 October 2026), so it is easy to find when there is something new. At the top of What's new there are User guides and About buttons, at the top of User guides What's new and About, and at the top of About the other two (About from 31 October 2026), so you can move between the three directly. A small **red dot** on ≡ and on both buttons tells that there is a version you have not read yet; it goes away when you open What's new (also when you have read it on the full site in the same browser). About shows the same text as the About window of the full site: who made MODE-S Wind and how to reach him, the receiver site, how the observations are made, the products, the windshear detection and the credits. User guides lists all the guides of the full site — e.g. the Windshear guide or the windshear study log — with their version, date, size and contents (Contents); Open PDF opens one in the phone's PDF viewer, Download saves it. Both open over the view; × or the phone's *Back* closes them.

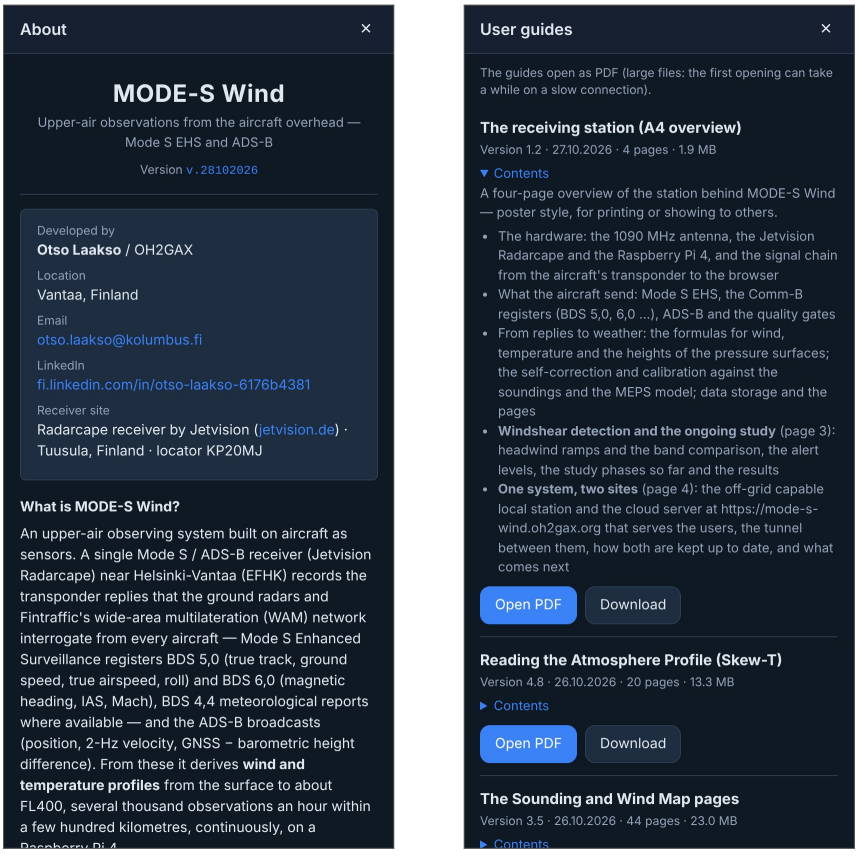


Figure 3. About (left) and User guides (right, with the contents of the first guide opened), from the menu's General section.

3 Touch, rotation and updates

Tap for values. On the full site the values appear when the mouse is over a diagram. On the phone, *tap* the diagram: a box shows the values at that point — on the Skew-T the level's temperature and wind (and the sonde's and model's), on the time–height section the hour and layer (or the strip under it), on the hodograph the layer's wind, on the Calendar the hour's values (a second tap on the same hour opens that day in Time–height), on the ACI & WS chart the nearest landing, departure or go-around, on the space–time diagram the flights of that cell. *Drag* sideways or up and down to move along; tap elsewhere to move the box. On the ILS view a tap on an aircraft chooses whose wind barbs are drawn (Section 7).

Scrolling. When a view is taller than the screen — EDR · WS · ACI · ST with its list of flights, the History, the ILS view with many aircraft, or a Skew-T turned to landscape — swipe up and down to scroll, also when the swipe starts on the diagram. A short tap still gives the values.

Portrait and landscape. In portrait each diagram takes the full width and as much of the height as it needs. Turn the

phone for more room across: the time–height section gets a longer time axis, the ILS profile spreads over the screen (Fig. 4); the Skew-T keeps its shape and the page scrolls.

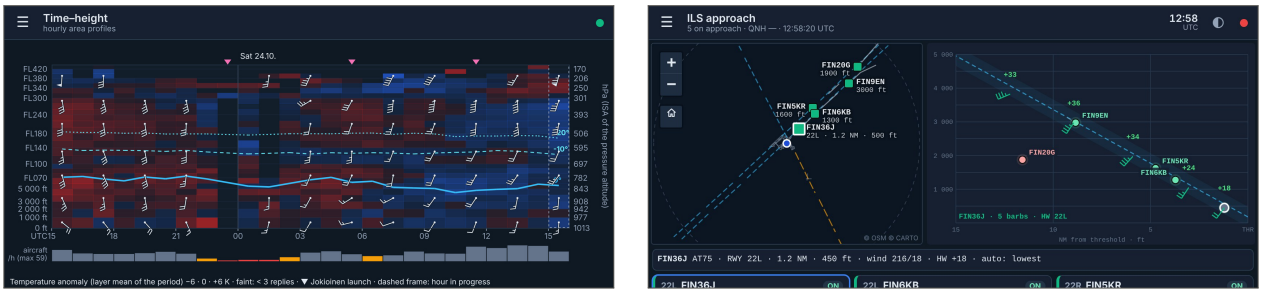


Figure 4. Landscape: the time–height section (24 hours) and the ILS approach with Show: Both — the map beside the profile. The strips of the ILS view are below; scroll down to see them. (The ILS picture: approach traffic replayed from the stored observations; the map tiles are not shown, see Fig. 17.)

Updates. Only the view on the screen is updated, and nothing while the phone's screen is off or another app is in front; when you come back the view is updated at once (and the app itself is reloaded when the server has a newer version, Section 1.2). Table 1 gives the intervals. The ILS view is the only one that updates every few seconds: it loads only the aircraft on the approach and the active alerts, compressed — about 1–5 kB each time, depending on the traffic — so watching it for an hour on mobile data uses roughly 1–4 megabytes; with All tracked (30 NM) the map's list is longer, a few kB more each time in busy hours. Its map adds the aircraft positions (well under 1 kB each time) and, once, the map program and the map tiles (about half a megabyte; the phone keeps them). The Statistics view uses about 7 megabytes an hour (Section 10); the Weather view about 1–3 megabytes when opened, and 1–3 megabytes an hour more with the air traffic on (Section 11); the other views use little.

View	Updated	Data
Skew-T · Area now	every 2 min	the area profile; METAR every 10 min
Skew-T · Area hour	when you choose an hour	stored hourly profile
Time–height	every 10 min	hourly profiles of 12 h – 14 days; with strips, BL turbulence, MEPS fields or the forecast also their data (about 10–50 kB)
Calendar	every 15 min	one line of figures per hour of 4 weeks – 120 days (about 10–25 kB)
ILS approach	every 5 s	live approach traffic (with the map: their positions), alerts; METAR and QNH every 10 min
EDR · WS · ACI · ST	every minute	landings, departures, go-arounds of 3 h – 1 day; space–time diagram
History	every minute (not for a date)	landings or departures of 1 h – 1 day or of one date
Statistics	every 30 s	the Statistics page's data of the window (about 55 kB)
Weather	maps when opened; run every 10 min	the model hours (about 20–70 kB each); with the FMI radar its newest image every 4 min; with air traffic its positions every 6 s
Mast	every 5 min	the mast's 10-minute steps, its profile, the boundary layer and the EFHK now box (the comparison every 15 min)
Meteogram	every 10 min	one place's model hours (a few kB)
GNSS (Full view)	every 30 s; chart every 2 min	degraded aircraft now and the 24 h figures (about 15 kB, 45 kB with the heatmap); the hours or days and spoofing indicators of the window (5–55 kB)
Health dot	every minute	the system health indicators

Table 1. When the views are updated (only while shown and the screen is on).

4 Skew-T: Area now and Area hour

The Skew-T is the same diagram as the Live Map's *Atmosphere Profile* and the Sounding page (Fig. 5): temperature against pressure, with the isotherms tilted to the right, and the wind as barbs at the right. The grey curve and the grey barbs with their values are the *aircraft* — the average of all aircraft within 150 km. With Sonde the Jokioinen radiosonde is drawn in pink (barbs inside the right edge of the plot), with MEPS the MEPS weather model in teal (barbs in the margin, just left of the aircraft's). The labels at the top left give the data quality, the sounding's and the model's time and how far the aircraft are from them.

Figure 5. Left: Skew-T · Area now, the last 60 minutes, with Sonde (pink) and MEPS (teal) on; the labels give how far the aircraft are from them: on average 0.0 K from the 12 UTC sounding and −0.3 K from the model, wind 10 and 7 kt RMS. Right: Area · hour 24 October 2026 15–16 UTC with Analysis: Levels (tropopause, 0 °C level, maximum wind); a tap at 447 hPa shows the aircraft (−29.7 °C, 191°/35 kt), the sonde and the model at that level.

Area now is the live profile, updated every 2 minutes, with the latest METAR at the ground (ground line, temperature and dewpoint). **Area hour** shows one stored hour: choose the date and the hour in the menu, or step with ⟨ ⟩. The METAR of that hour is stored with it.

Option	Choices	What it does
Window	30 min, 60 min, 2 h, 3 h	period of Area now; longer = more aircraft, less up to date
Date, Hour	calendar, ⟨ ⟩	the stored hour of Area hour
Analysis	Off, Levels, Stability, Turbulence, Advection, Full	automatic marks: tropopause, isotherms, jets, inversions, stability, turbulence risk, warm / cold advection, contrails, surface parcel
Analysis of	Mode S, MEPS, Sonde	which profile the analysis is computed from; MEPS and Sonde add likely cloud layers
Airport profile	Off, Overlay, Zoom	the fine low-level profile of the landings and departures at EFHK; Zoom shows the lowest 5 000 ft; from 29 October 2026 with the hour's low-level jet (◊ `LLJ`) and surface inversion (red `SFC INV` bracket)
Smoothing	Off, 3 layers, 5 layers	vertical smoothing of the aircraft profile
Sonde	switch	the Jokioinen radiosonde
MEPS	switch	the MEPS weather model over the same area
RH	switch	the trial moisture strip with the sonde's and the model's humidity
Data quality	switch	the data badge and the rings at layers with only 1–2 aircraft
Hodograph	switch	the hodograph under the diagram (from 10 November 2026, below)
Hodograph: previous hours	switch	the two hours before, as faint dashed curves
Hodograph range	All levels, Below FL200, Below 10 000 ft, Below 5 000 ft	how high the hodograph reaches

Table 2. Skew-T options. The *Reading the Atmosphere Profile (Skew-T)* guide explains each of them.

4.1 The hodograph

From 10 November 2026 the Skew-T view can draw a *hodograph* under the diagram: switch Hodograph on in the Skew-T options and scroll down (Fig. 6). It works in both Area now and Area hour. A hodograph shows the wind of every layer as *one point* — the tip of the wind vector, north up and east to the right, so a north-westerly wind lies to the lower right of the centre. The points are joined from the ground upwards and coloured by height, from red at the ground to violet at the top; the rings are the wind speed in knots, and the labels around the edge tell where the wind blows *from*.

- **The length of the curve** between two heights is the vertical wind shear between them; the total from the lowest to the highest point is written at the bottom, with the turning.
- **The turning.** A curve that turns clockwise with height is a wind veering with height: warm air is arriving. Anticlockwise is backing: cold air.
- **A low-level jet** is a loop that goes out and comes back within the lowest few thousand feet.
- **The disc** around a point is how well the layer's mean wind is known (the scatter of its winds divided by the root of the number of aircraft); a hollow point has fewer than three aircraft. The thick line marked `low` is the airport's low-level profile.

Hodograph: previous hours adds the two hours before as faint dashed curves, so you see which way the profile is moving. Hodograph range cuts the top: Below 10 000 ft or Below 5 000 ft opens up the lowest layers, which otherwise crowd near the centre under a strong jet stream. *Tap* a point for the layer, its wind, the number of aircraft and how good the mean is. The area profile is the mean of the aircraft within 150 km over an hour, not one ascent as from a balloon; that is what makes the curve smooth enough to read. The hodograph is off by default on the phone; the *Sounding and Wind Map pages* guide has more on reading it.

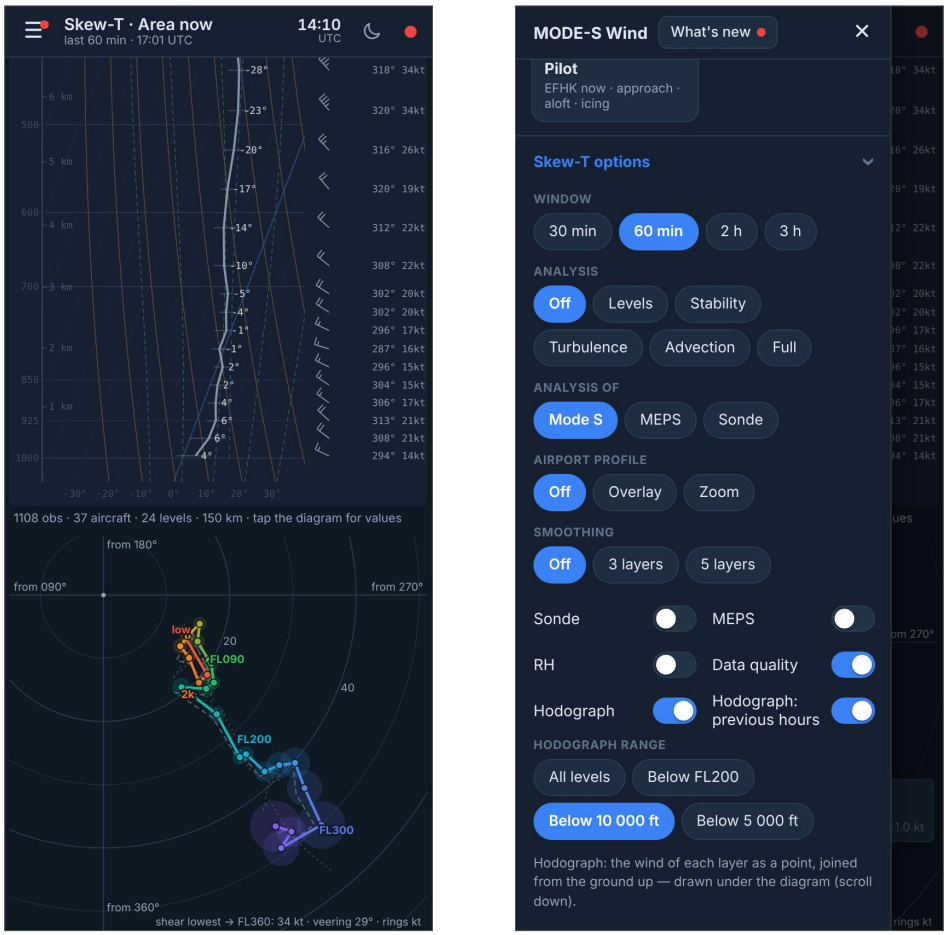


Figure 6. Left: Area now on 8 November 2026 at 14:10 UTC, scrolled down to the hodograph: a north-westerly wind that grows from 13 kt in the lowest layer to 48 kt at FL300–FL360 (35 kt of vector shear from the lowest layer to FL360, veering 29°), with the two previous hours dashed. Right: the Skew-T options with the three hodograph controls at the bottom.

5 Time–height

The time–height section puts the hourly profiles side by side: time runs to the right, height up (Fig. 7). The colour is the chosen field, the barbs the hourly wind; lines mark the 0, −10 and −20 °C levels, the tropopause and the maximum wind when switched on. The strip at the bottom gives the aircraft per hour; a pink triangle at the top marks a Jokioinen launch, and the last column (dashed frame) is the hour in progress. Tap a cell for its values.

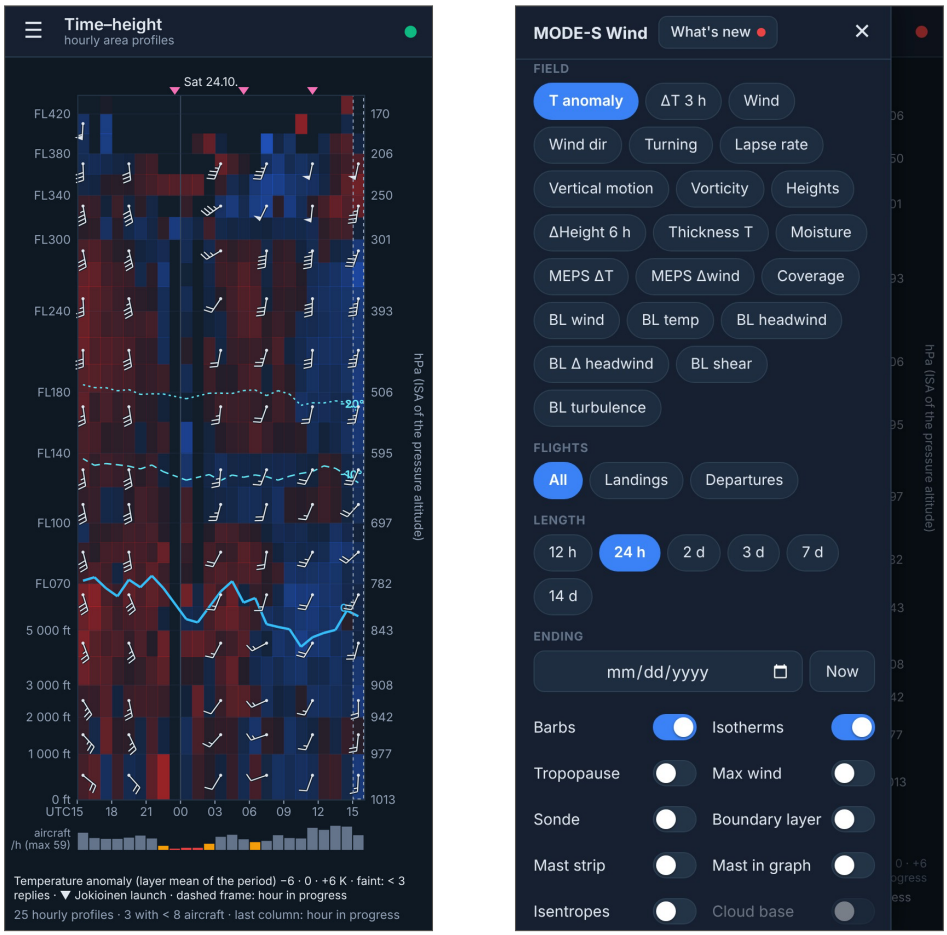


Figure 7. Time–height, 24 hours to 24 October 2026 15 UTC, temperature anomaly (blue colder, red warmer than the period's mean in that layer), with barbs and isotherms; right: its options as of 10 November 2026 — the fields, the flights, the length and the first switches (the rest are in Fig. 10).

Options. Field: temperature anomaly, temperature change over 3 h, wind speed, lapse rate, vertical motion, vorticity, heights of the pressure surfaces, height change over 6 h, thickness temperature check, the moisture trial and the *boundary layer* — BL wind and BL temp, the lowest 5 000 ft at the airport on a linear scale (Fig. 8) — and, from 31 October 2026, the boundary layer *flight by flight*: BL headwind, BL Δ headwind and BL shear, every landing and departure as its own thin column with its headwind, its change against the flights of the previous 45 minutes or its change with height (Fig. 9); Flights chooses all, the landings or the departures, and the coloured line along the top gives the landing runway. Length: 12 h to 14 days — 24 h by default on a phone; landscape gives the longer periods room. Ending: a past date, or Now. Switches for barbs, isotherms, tropopause, maximum wind, the sonde as its own column and the Boundary layer — the low-level jet (◊ with its speed in kt, joined hour to hour) and the surface inversion (red, from the ground to its top) on any field. Switches that do not apply to the BL fields are greyed out. Save picture (PNG) stores the section as an image on the phone. The *Sounding and Wind Map pages* guide explains the fields.

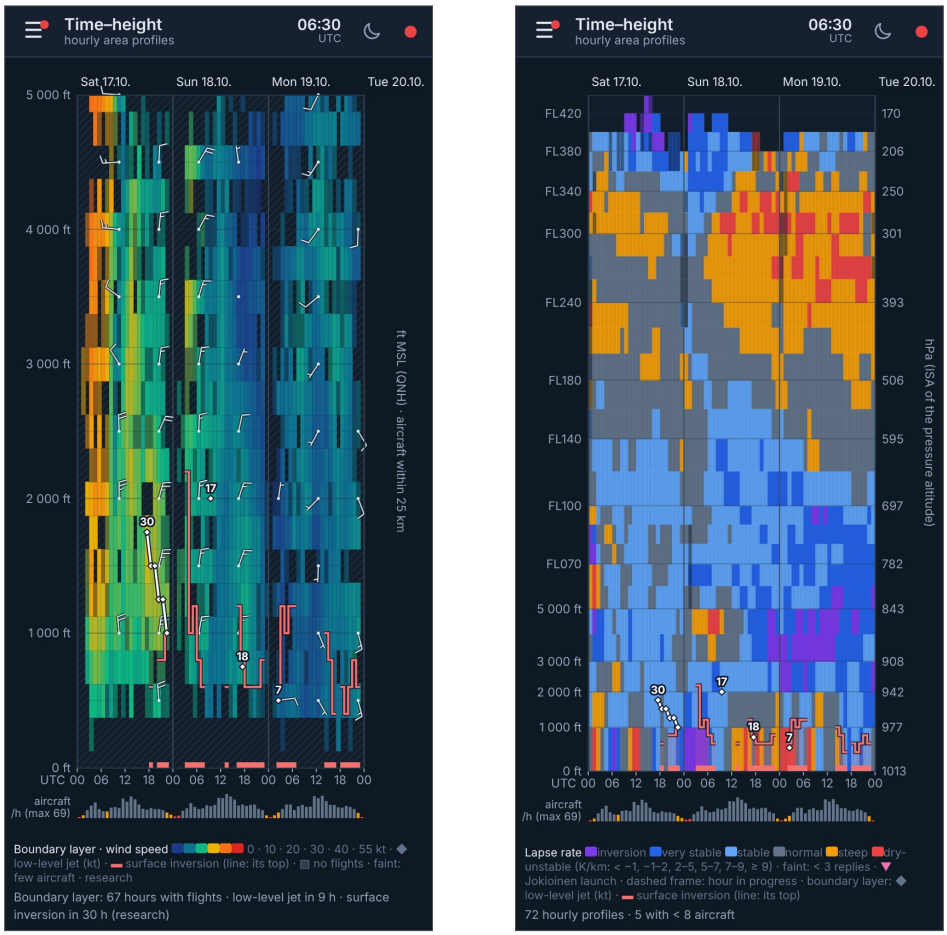


Figure 8. Left: BL wind, 72 hours to the end of 19 October 2026 — the wind of the aircraft within 25 km in 250-ft steps (blue calm, green, yellow, red 55 kt), the low-level jet of the evening of 17 October (30 kt at 1 750 ft, sinking) and the surface inversions under the high (red bar; the red line is the top, mostly at 600–1 200 ft — drawn again on 10 November 2026 with the corrected top). Hatched: no flights. Right: Lapse rate with the Boundary layer switch on — the jet and the inversions in the lowest 3 000 ft.

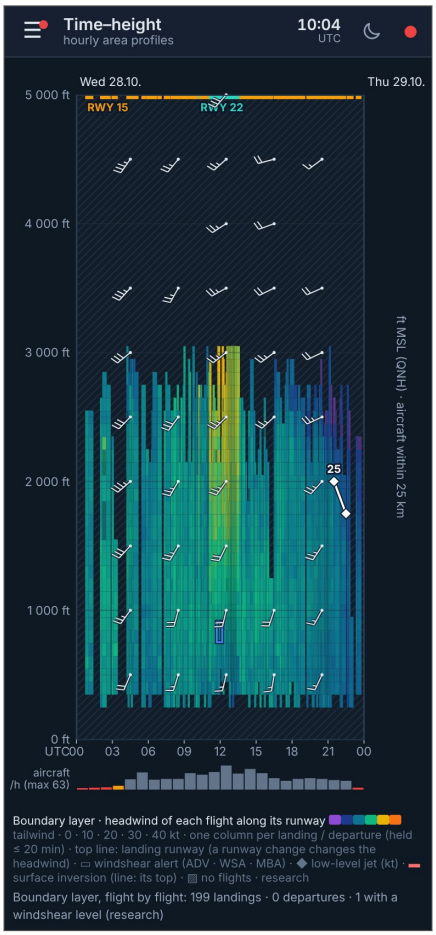


Figure 9. BL headwind, landings only, 28 October 2026 (the station's data): one column per landing, runway 15 most of the day (10–20 kt of headwind), the landings on runway 22 around midday with 25–35 kt — the runway changed, not the wind (line along the top) — and a low-level jet of 25 kt at 2 000 ft in the evening.

5.1 More fields, overlays and strips

From 10 November 2026 the phone has the same additions as the full site's time–height section (Fig. 10). Six more **fields**:

- Wind dir — each cell coloured by where the wind blows from, on a colour wheel (north red, east yellow-green, south cyan, west violet); a wind shift is a sharp change of colour.
- Turning — how the wind turns with height: veering red (warm air arriving), backing blue (cold air), as the warming or cooling it would bring in K per hour. Faint below 3 000 ft, where friction turns the wind too.
- MEPS ΔT and MEPS Δwind — the hourly profile minus the MEPS model's mean over the same area: where and when the model was off. A comparison only; nothing is corrected with it.
- Coverage — the number of aircraft behind each cell.
- BL turbulence (research) — the turbulence the landing and departing aircraft met in the lowest 5 000 ft, in 500-ft bands, as EDR coloured by the ICAO grades (smooth, slight, light, moderate, severe). Users with the *Pilot* right also see the pilot reports and SIGMETs as triangles at the top.

And more **switches**:

- Isentropes — lines of equal potential temperature every 5 K. Air that neither gains nor loses heat moves along them; where they are packed the air is stable, and where they slope through the hours a front is passing.
- Cloud base — on the boundary-layer fields, the airport's METAR ceiling (solid) and lowest cloud (dotted).
- Forecast (MEPS) — the model continues the section to the right of the present, hatched and behind a dashed line, 12 hours ahead for a 24-hour section and up to 48 hours for the longer ones. It is a *forecast, not a measurement*, and is there only when the period ends now.
- Average day — folds the period into the 24 UTC hours of the day: what repeats daily stays, single weather systems average out. Best with 7 or 14 days (Fig. 10, right).
- Strip: runway and Strip: weather — strips under the aircraft strip: the most used landing and departure runway of each hour, and the METAR visibility / ceiling class with the present weather. Users with *Full view* also have Strip: GNSS, the share of aircraft with a degraded GNSS position.

The legend under the graph grows with what is switched on, and the graph makes room for it; with several strips and overlays on, landscape or a 24-hour period keeps it readable. Tap a cell or a strip for its values. The switches are remembered and shared with the full site in the same browser. The *Sounding and Wind Map pages* guide explains each of them in full.

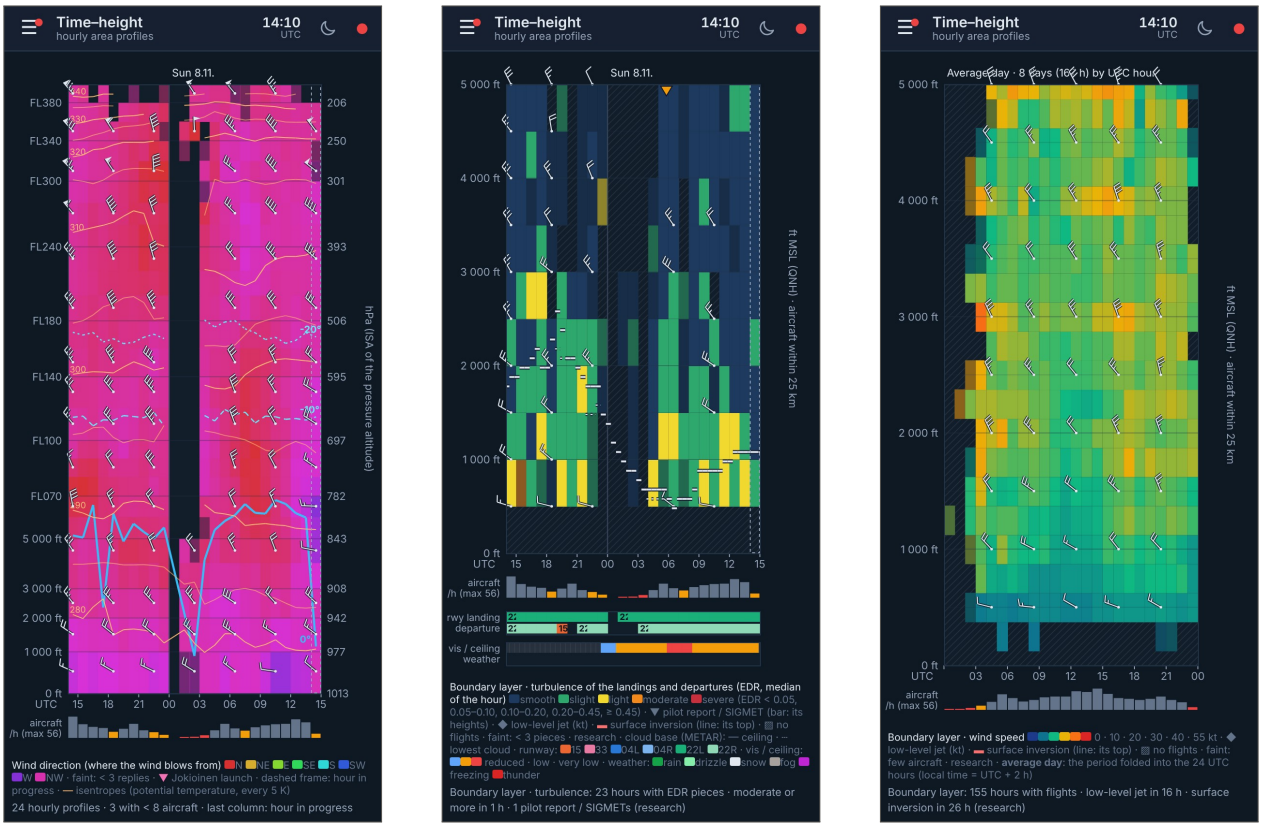


Figure 10. Left: Wind dir with Isentropes, 24 hours to 8 November 2026 15 UTC — a north-westerly to northerly flow (pink to red) through the whole depth; the empty columns are night hours without aircraft. Middle: BL turbulence with Cloud base and the runway and weather strips, the same period: mostly smooth to slight, light (yellow) in places, mostly below 1 500 ft, on a day of low cloud (amber and red in the weather strip). Right: Average day of BL wind, the 168 hours to 8 November 2026 15 UTC.

6 Calendar

The time–height section shows days; the Calendar (from 10 November 2026) shows weeks (Fig. 11). It has **one row per UTC day**, the newest at the top, and **one column per UTC hour**; each cell is coloured by one quantity of that hour, chosen in the Calendar options:

- Inversion — how much warmer the air at 600–1 000 ft is than at the surface, from the boundary layer at the airport;
- Low-level jet — the speed of a jet found below 5 000 ft;
- Wind 2 000 ft — the wind at 2 000–3 000 ft from the hourly area profile;
- Windshear — the landings and departures of the hour that got a windshear level;
- Turbulence — the median EDR of the hour's flights (research);
- Aircraft — how many aircraft the hour's area profile had;
- GNSS (only with *Full view*) — the share of aircraft with a degraded GNSS position.

A dark cell means there were flights but no inversion, jet or report; an empty cell has no data — mostly the night hours without traffic. A line marks the start of each week, and the dashed frame is the present hour. Period: 4 weeks, 8 weeks or 120 days.

Tap, and tap again. A tap on a cell shows all the values of that hour. A *second tap on the same hour* opens that day in the Time–height view, 24 hours long, in the field that goes with the quantity — BL temp for the inversion, BL wind for the jet, BL Δ headwind for the windshear, BL turbulence, Wind or Coverage. To return to the present there, tap Now in the Time–height options. The view is updated every 15 minutes while it is shown.

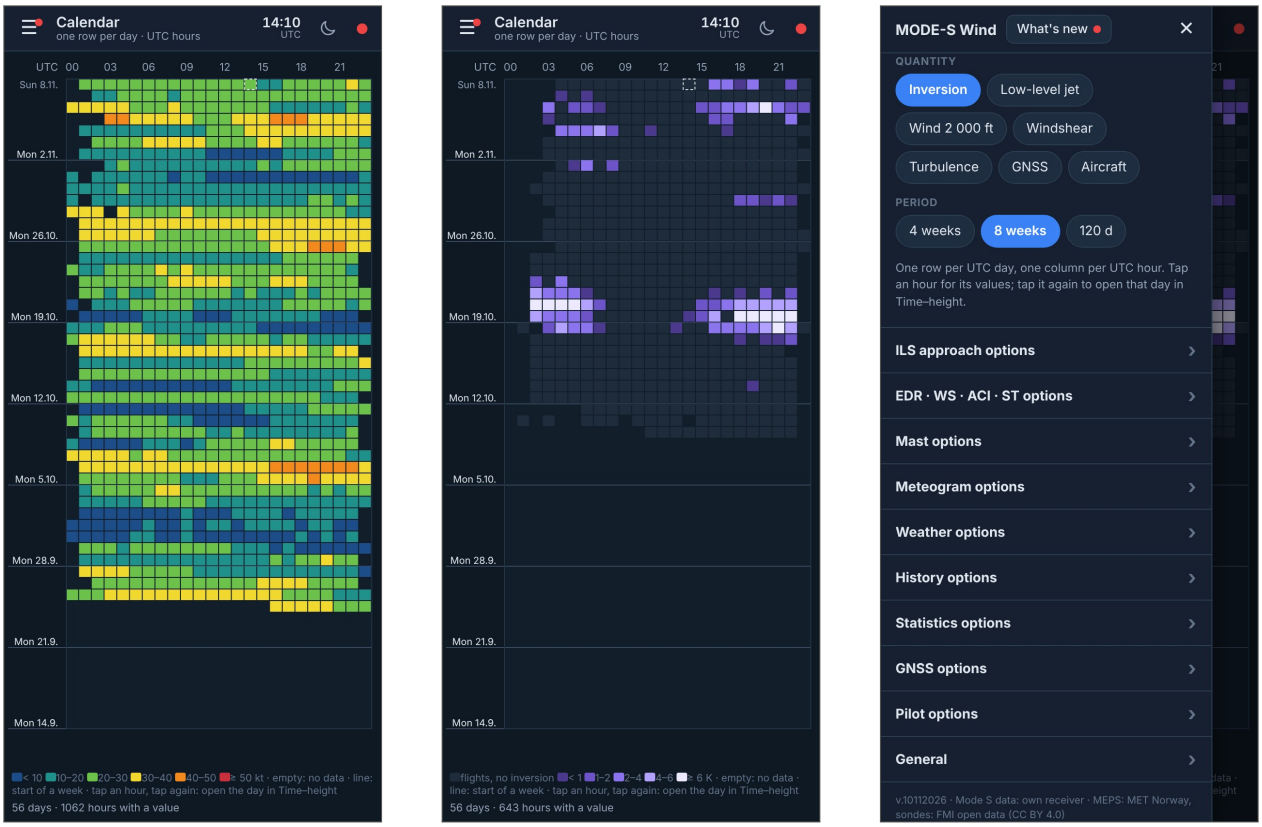


Figure 11. The Calendar, the 8 weeks to 8 November 2026 (the rows at the bottom are older than the stored data). Left: Wind 2 000 ft — windy spells of a few days (yellow and orange, 30–50 kt) between calmer ones (blue). Middle: Inversion — surface inversions belong to the evening, night and morning hours and come in spells, here around 18–22 October and again on 4–8 November. Right: the Calendar options.

7 ILS approach

The ILS view is the Windshear page's vertical profile of the final approach, live (Fig. 12): distance from the runway threshold to the right (15 NM to the threshold), altitude up (to 5 000 ft). It is redrawn every 5 seconds.

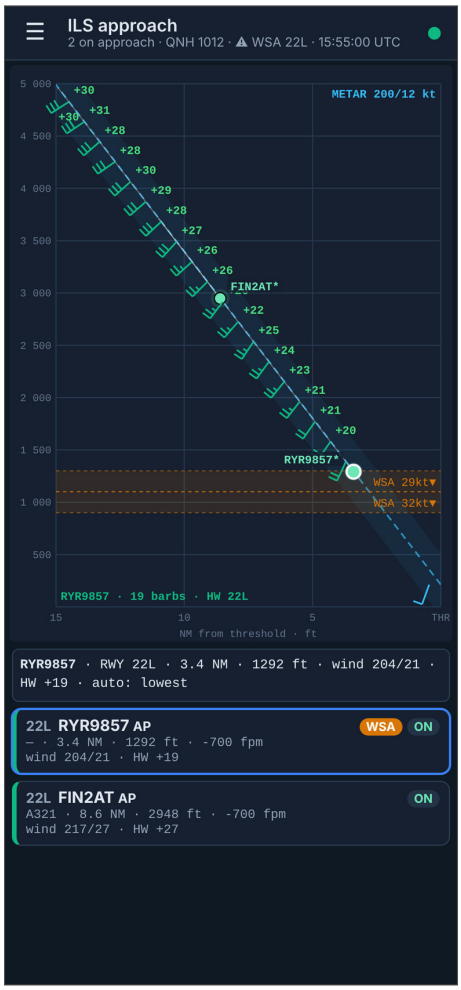


Figure 12. ILS approach, runway 22L. Two aircraft on the glideslope (green `ON`); the barbs are those of the lower one, RYR9857, measured along its approach from 15 NM in, with the headwind component of each in kt above it; a windshear alert (`WSA`) is active at 900–1 300 ft. The METAR surface wind is the blue barb at the threshold. Under the profile the chosen aircraft and the strips. *Illustration with simulated approach traffic.*

What is drawn.

- **The glideslope**: the dashed line is the 3° path (318.5 ft per NM above the threshold, corrected for the QNH), the shaded band ±300 ft around it.
- **Aircraft** in the approach corridors as dots with their callsigns (⋆: autopilot engaged): green *on* the glideslope, amber *high*, red *low*; the faint line behind each is its path over the last minute.
- **Wind barbs** of one aircraft, measured by it on the way down, coloured by where the wind comes from (green: computed from the aircraft's speed and heading replies; blue, orange, violet: other report types). The number above each barb is the headwind component for that runway (Headwind; positive = headwind) or the crosswind (Crosswind). Small hollow circles mark points where no wind could be computed (amber: the aircraft was turning; grey: a gap in its position data). The line at the bottom left names the aircraft and the number of barbs.
- **Windshear alerts**: an active alert is a band across the profile at its altitudes, blue `ADV` (advisory), amber `WSA` (windshear alert) or red `MBA` (microburst alert), with the change in kt (▼ loss, ▲ gain of headwind).
- **METAR**: the surface wind as a blue barb at the threshold, its text at the top right.

The map (from 28 October 2026). With Show: Both — the default — a map of the approaches is above the profile (beside it in landscape), with Map only the map, with Profile only the profile as before (Fig. 17). The map shows the aircraft as on a radar screen: an open square framed in the colour of the wind source (as the barbs) with the callsign and altitude; the aircraft whose barbs are drawn has its square framed thinly in white (in the light theme in black, which stands out on the light map), from 30 October 2026 the same size as the other approach traffic and with the same label (in the pictures below still larger).

Departures (from 29 October 2026). The climb-outs are not shown by default, as on the Windshear page. With Departures on the map: Shown they are drawn like the other traffic, with their runway and an arrow in the label, e.g. `A016 · 22R↑` — useful to see a departure climbing through the layers the arrivals descend in. A go-around is not a departure:

it stays with the approaches.

Boundary layer line (from 29 October 2026; research). Under the selected aircraft a line always tells what the lowest air is doing (Fig. 13): *Boundary layer, last hour* with a chip `LLJ 750 ft · 18 kt 023°` when there is a low-level jet and a red chip `surface INV +2 K` when the air at 600–1 000 ft is warmer than at the ground; otherwise grey *no jet · no inversion*, and *no recent flights* at night when nothing flew. A jet means the headwind changes on final below its height; an inversion means calm, smooth air near the ground (Section 8).

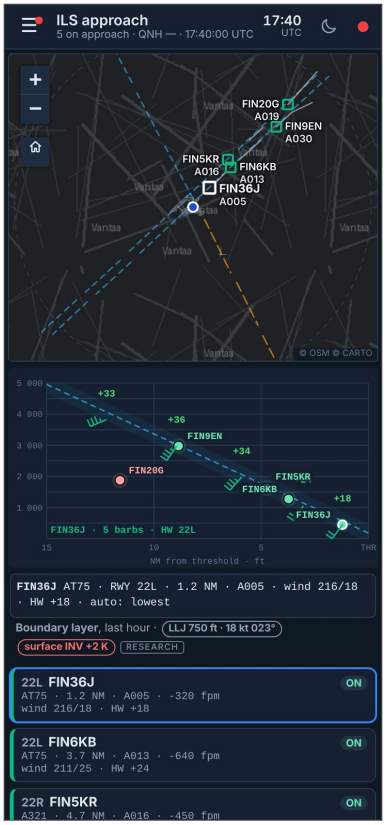


Figure 13. The boundary-layer line under the selected aircraft: a low-level jet at 750 ft and a surface inversion of +2 K (18 October 2026 17 UTC).

Turbulence on the map (from 29 October 2026; research). The option Turbulence on the map (EDR, research) (Off by default, 30 min, 1 h) colours the straight pieces (about 2 km each) of the paths of the landings of that period — and of the departures when Departures on the map is Shown (from 30 October 2026) — with thin lines by their turbulence, the EDR (Section 8): green smooth, yellow light (from 0.10), orange moderate (from 0.20), red severe (from 0.45). Where several flights flew the same path the strongest is drawn on top, so the colour shows the worst piece. A line under the selected aircraft sums it up, e.g. *Turbulence (EDR), last 60 min · 9 landings, 3 departures · max 0.26 moderate at 1400 ft (FIN8MC 22L) · median below 1000 ft 0.15, 1000–3000 ft 0.12* — the departures are counted only when they are shown (Fig. 14).

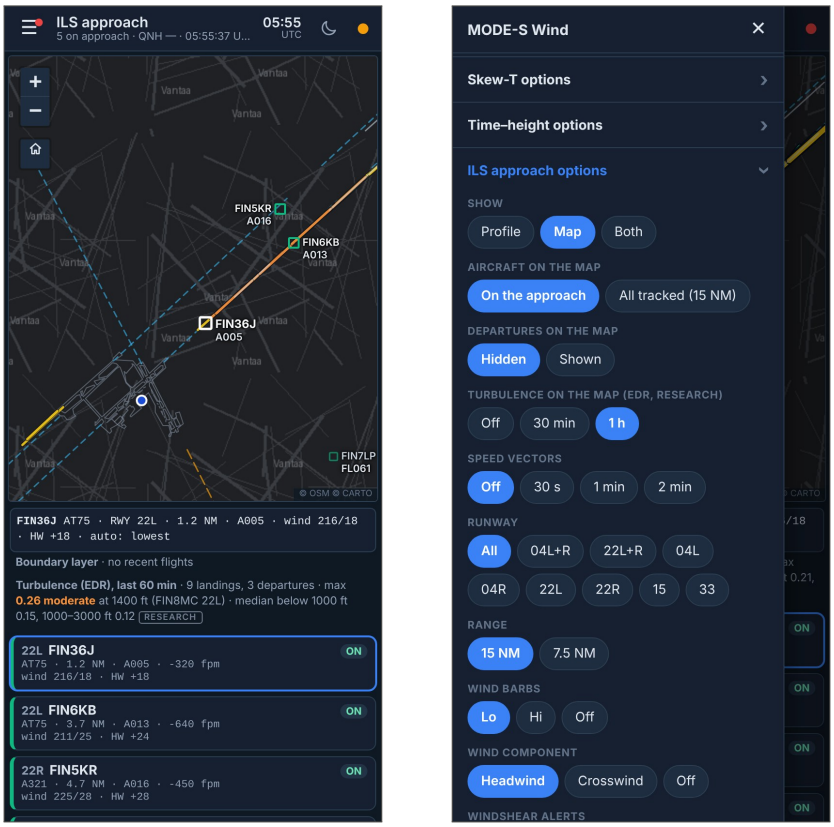


Figure 14. Left: Turbulence on the map 1 h with Departures on the map: Shown — the final approach to 22L in orange (moderate) and yellow (light), a departure at the lower left; the flights of 28 October 2026 replayed in a test. Right: the option in the menu.

Speed vectors (from 29 October 2026). The option Speed vectors (Off by default, 30 s, 1 min, 2 min) gives each aircraft a line to where it will be after that time at its ground speed along its track, as on a radar screen — in the colour of its square, white for the selected aircraft. A faster aircraft has a longer line: below 5 000 ft, at 140–250 kt, 30 s is 1.2–2.1 NM and 1 minute 2.3–4.2 NM, about the spacing between the aircraft on final. Start with 30 s; 1 min shows the spacing to the aircraft ahead. A landed aircraft gets none.

Altitudes as a controller sees them (from 29 October 2026). On the map, the strips and the line under the profile the altitude is written as on an air traffic controller's radar label: `A023` is 2 300 ft above sea level on the airport's QNH (the METAR pressure), and above the transition altitude of 5 000 ft the flight level, e.g. `FL061`. Aircraft transponders report their altitude on the standard pressure of 1013.25 hPa, so on a low-pressure day the raw value is several hundred feet too high: at QNH 1003 hPa an aircraft reporting 1 275 ft is really at about 1 000 ft, `A010`. The profile's height axis and the glideslope already use the QNH.

ATC labels (from 30 October 2026). The option Labels on the map: ATC (Standard by default) gives every aircraft on the map the three-line label of the ATC style of the Live Map and the Windshear page, in a small monospaced font: the callsign; the level in hundreds of feet — the altitude on the METAR QNH up to the transition altitude of 5 000 ft, the flight level above, without the letters — with ↑ or ↓ when climbing or descending at 300 ft/min or more and the selected level when the aircraft has sent it (`007↓020`: at 700 ft, descending, 2 000 ft set); the ground speed in tens of knots and the type (`14 A320`), for a departure followed by its runway (`04R↑`). The labels are placed as the standard ones (below); the squares are drawn a little smaller. The options Aircraft type on the map and Selected altitude change only the standard labels (the selected altitude still appears in the strips and the selected aircraft's line). **Tap for the extended label** (from 31 October 2026; Fig. 15). With the ATC labels, *tap an aircraft* — any one on the map — and its label grows into the six-line *extended label* of the big pages' Hover, on a dark backing (light in the light theme) with a thin frame, the other labels moving out of its way: the callsign, type and registration; the level with the trend, the selected level, the vertical rate and `AP` / `APP` when the aircraft reports the autopilot; the ground speed, true and indicated airspeed and Mach number; the magnetic heading and track, the drift and the bank angle; the wind with its head- or tailwind and crosswind and the temperature; the distance and bearing from the airport, the NACp and the squawk. The values are what the receiver has just heard from the aircraft (a value not heard within 20 s shows as `---`), read again every 2 seconds while the label is open — about 0.5 kB each time. It stays open until you tap an empty place on the map (a tap on the label itself keeps it), switch the labels to Standard, or the aircraft leaves the map. A tap on an aircraft on the approach still also follows it for the barbs, as before. The Windshear guide explains the lines in detail.

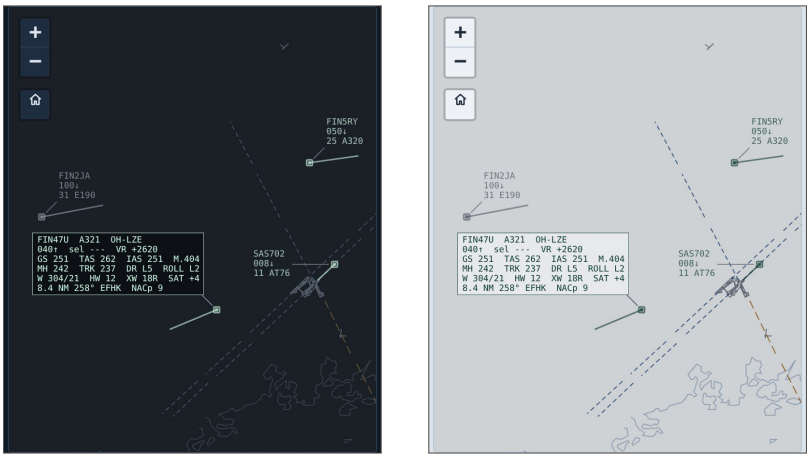


Figure 15. The extended label of FIN47U after a tap, ATC map style, dark and light theme. A test with the stored observations of 30 October 2026, 08:29:44–51 UTC (four aircraft: positions, levels, speeds and FIN47U's BDS 5,0 / 6,0 values as stored) and the QNH 1018 of that time: FIN47U climbing to the south-west through 4 000 ft at 2 600 ft/min, IAS 251 kt, Mach 0.404, magnetic heading 242°, the wind 304/21 with 12 kt headwind and 18 kt crosswind from the right, 8.4 NM from the airport. No selected altitude was stored, hence `sel ---`.

ATC map style (from the evening of 30 October 2026; Fig. 16). The option Map style: ATC (research) (Street by default) gives the map the radar-display look of the Live Map and the Windshear page: no street map but a plain background — dark grey in the dark theme, light grey in the light one — with the coastline, the airport layout and the ILS (short dashes) and RNP (long dashes) finals. The squares, trails, speed vectors and labels are monochrome: light below 6 000 ft, grey above; the aircraft whose barbs the profile shows is not marked on this map (the line under the profile tells which one it is). It goes well with Labels on the map: ATC, but works with either. The first time it is used the phone loads the drawing code and the coastline, about 0.8 MB; after that the browser keeps them and only checks whether they have changed. Like the ATC style of the big pages it is a research display, not for air traffic control: the positions are those received by this station.

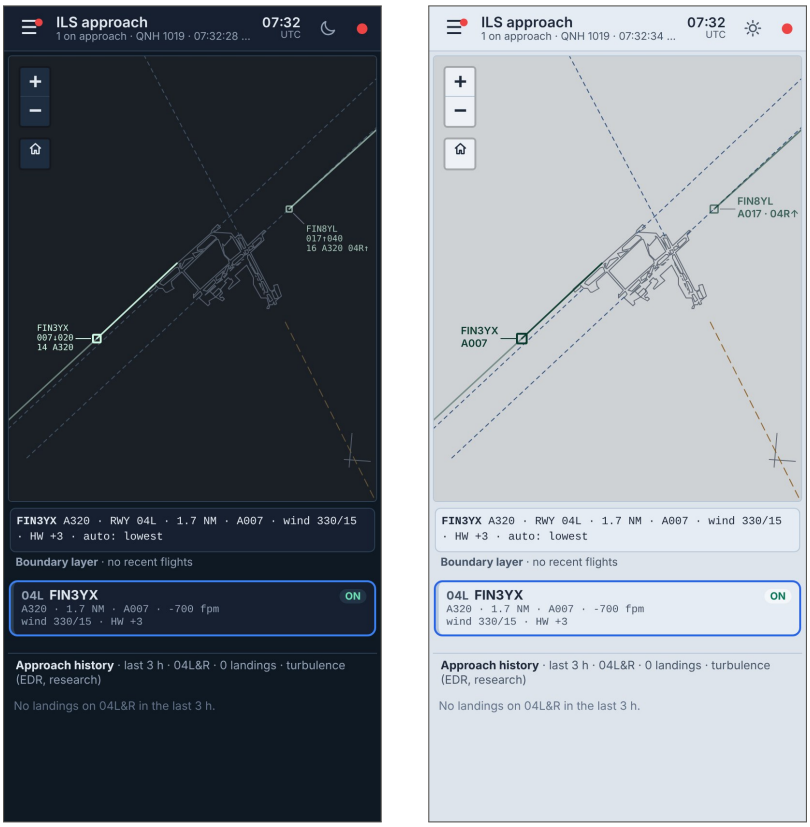


Figure 16. Map style: ATC, Show: Map, Runway: 04L+R, All tracked, departures shown, Speed vectors: 1 min. The traffic is the live approach state of the production site of 30 October 2026, 14:22 UTC (its list of the other aircraft was still the earlier one, within 15 NM below 5 000 ft), with the METAR QNH 1019 of that time, drawn in a test with the label placement of 31 October 2026: FIN3YX at 700 ft on final to 04L — its label to the left, off the approach line —, FIN8YL climbing out of 04R. Left: dark theme with Labels on the map: ATC; right: light theme with the standard labels.

Selected altitude (from 30 October 2026). With the option Selected altitude: On (Off by default) the altitude is followed by the altitude the pilots have set on the autopilot panel (the MCP or FCU), when the aircraft has sent it in the last 2 minutes — from its answers to the radars' Enhanced Surveillance questions (register BDS 4,0) or its own ADS-B. It is written as on the Live Map (from the evening of 30 October 2026): + climbing, – descending (300 ft/min or more), = level, then the selected altitude as `A` and hundreds of feet up to the transition altitude of 5 000 ft and as a flight level above. `A023+A050` is an aircraft at 2 300 ft, climbing, with 5 000 ft set; `FL120-A030` one at FL120 descending, with 3 000 ft set — usually the altitude it has been cleared to; `FL390=FL390` one level at its selected FL390. The selected altitude is as the pilots set it (on the QNH below the transition altitude). Not every aircraft sends it, and the radars ask for it only now and then, so it is missing for some aircraft; it is never guessed. The option Aircraft type on the map (Off by default) adds the ICAO type after the callsign, e.g. `FIN456 AT75`. With either option on, the map's labels and squares are drawn a little smaller so that the longer labels fit; with both off they are as before.

Label placement (from 31 October 2026). The labels — standard or ATC — are placed as on the Live Map and the Windshear page: each is joined to its square by a thin *leader line* and moved to where it covers least — not over another label, another square, the trails, the speed vectors or the approach lines — and it stays there while that place stays free, so the labels do not jump. The approach traffic is placed first. Before, a label stood at a fixed place beside its square (to the right, to the left for 22R and 04R), so on 04L and 04R — where the aircraft fly up and to the right — the first aircraft's label lay over the approach line ahead of it. On the street map the labels have a dark outline (a light one in the light theme) to stay readable over the map. Behind them the ILS centrelines (blue, dashed; the RNP approach to 33 amber), the airport, the tracking range (dashed circle) and, with Trails, each aircraft's path over the last 2 minutes — in the ATC map style (from 31 October 2026) instead the *history plots* of the big pages' ATC style: small dots of the last five positions, about 6 seconds apart and fainter with age, behind *every* aircraft on the map (an aircraft on the approach has them at once, the others after half a minute). Aircraft on the map: On the approach (default; those of the chosen runway) or All tracked (30 NM) — also every other aircraft the receiver hears within 30 NM of the airport, at any level (from 31 October 2026; before within 15 NM and below 5 000 ft), dimmed. Departures on the map: Shown alone adds the climb-outs within 15 NM below 5 000 ft. *Tap* an aircraft on the approach to follow it, as on the profile. Zoom with two fingers or + −, move with one finger; the map keeps the place and zoom you leave it at, and the house button under them takes you back to the airport. To scroll the page down to the strips, swipe on the profile or the strips, not on

the map. The map is light or dark with the theme.

A landed aircraft (from 29 October 2026). After touchdown the receiver still hears an aircraft for a while, so before it stayed in the view for about one and a half minutes at its last position, about 0.6 NM before the threshold. Now an aircraft within 2 NM of the threshold and below 1 000 ft whose position has not changed for 15 seconds is shown *faded*: a grey square with `landed` on the map, a grey dot on the profile and a dimmed strip with `LANDED?` and `no new position for 22 s`. After 30 seconds it is no longer shown, and Auto has already moved the barbs to the next aircraft. Only the display changes; the landing is recorded as before.

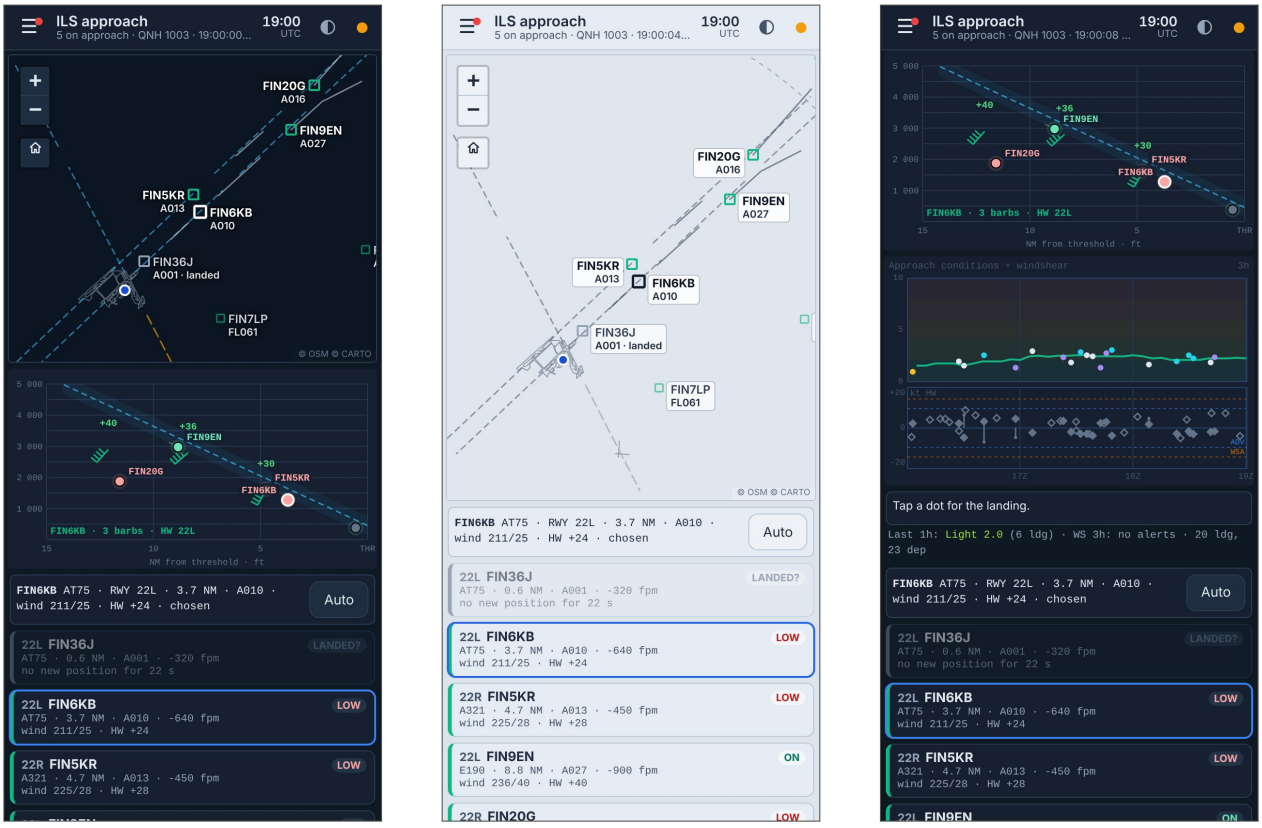


Figure 17. The ILS approach with the map (the labels still at their fixed place of before 31 October 2026): five aircraft on 22L and 22R (approach traffic replayed from the stored observations of 27 October 2026, with QNH 1003 hPa; FIN36J is shown as just landed; in these pictures the map tiles — streets, coast, lakes — could not be loaded, on the phone they are under the aircraft). Left: Show: Both — the map, the lower profile and the strips; FIN6KB has been tapped, so its square is white and the profile shows its barbs; Auto goes back to the lowest. Middle: Show: Map in the light theme, with room for four strips. Right: the chart WS + ACI (3 h) under the profile, scrolled down; the chart with the stored landings and departures of 28 October 2026 16–19 UTC.

Sizes. In portrait with Show: Both the map takes about 40 % of the screen height and the profile about a quarter, so that the first strips are visible under them; with Map the map takes 58 % and leaves room for about four strips.

Chart under the view (from 29 October 2026; from 30 October on by default). The option Chart under the view — WS + EDR (the default), WS + ACI or Off — adds a chart of the EDR · WS · ACI · ST view to the ILS view (Fig. 17, right: WS + ACI): on top the turbulence (EDR, Section 8) of every landing (dot) and departure (ring), or how rough each landing was (the Approach Conditions Index), and below the windshear report of every landing and departure as a change of headwind in kt, for the last 3, 6 or 12 hours (Chart range, 3 h by default). It is under the profile, or under the map with Show: Map, and the strips follow it. Tap a dot for that landing or departure; the line under it gives the last hour's turbulence or conditions and the windshear alerts of the period. Section 8 explains the chart. Its data are fetched at most once a minute.

Boundary layer below (from 31 October 2026; research). The option Boundary layer below — Off (the default), BL wind, BL temp, BL headwind, BL Δ headwind or BL shear — adds one of the boundary-layer charts of the EDR · WS · ACI · ST view (Section 8) after that chart, on the same Chart range and the same time axis (Fig. 18), so the turbulence and windshear of the flights and the wind layers they flew through are on one screen; it works also with Chart under the view: Off. Tap it for an hour's cell, a jet or a flight at a height.

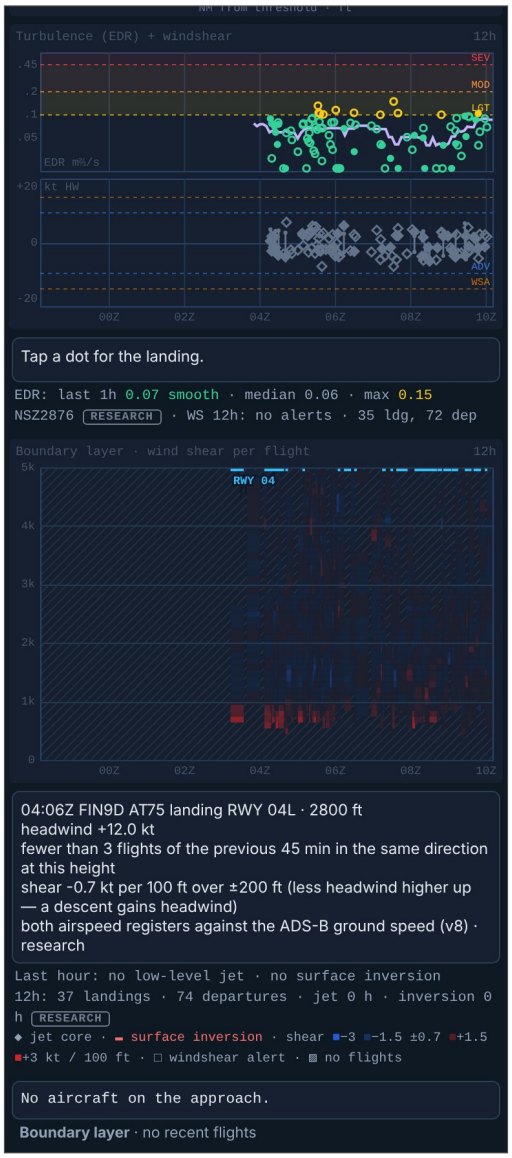


Figure 18. The ILS view scrolled down (stored data to 31 October 2026 10:10 UTC, drawn in a test with no aircraft on the approach): WS + EDR for 12 hours and, under it, Boundary layer below: BL shear with a tapped landing (FIN9D, 2 800 ft).

Approach history under the strips (from the evening of 30 October 2026; Fig. 20). Under the strips the ILS view lists the landings of the last 3 hours (Approach history under the strips: Off, 3 h — the default — or 6 h) of the runways chosen, newest first, with the same cards as the History view (Section 9): time, callsign, type, runway, the ACI, a windshear level, the EDR and a go-around as chips, and a second line chosen with Approach history shows — by default Turbulence: the EDR of the landing with its class, below and above 1 000 ft and its roughest piece; or the headwind, crosswind or wind per height band, the ACI or the windshear report. Tap a card for all its bands, its reports and the boundary layer at the time; tap again to close it. So the traffic now on final and how rough the last landings found the air are on one screen. The list is fetched at most once a minute. **With the departures** (from 3 November 2026): when Departures on the map is Shown, the list also has the departures of the same hours, in one list with the landings, newest first, as List: Both of the History view — a departure's runway is in a light-blue box, its card has its own chips and line, and tapped open it shows its headwind profile and its bands (Fig. 19). The title then reads e.g. `Landings and departures · last 3 h · 22 landings, 46 departures (blue runway)`. With Hidden (the default) the list has the landings only, as before.

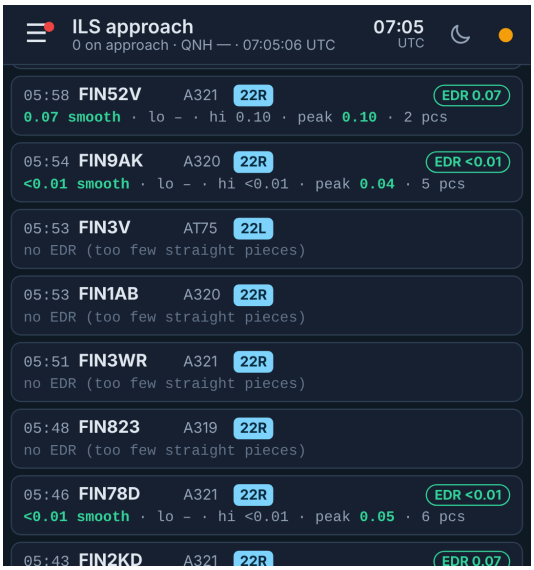


Figure 19. The approach history under the strips with Departures on the map: Shown (3 h, Turbulence; dark theme). A test on the stored data of 3 November 2026 to 07:05 UTC: the departures from 22R and 22L (light-blue runway boxes) between the landings (TAY7LL, BCS3612 on 22L).

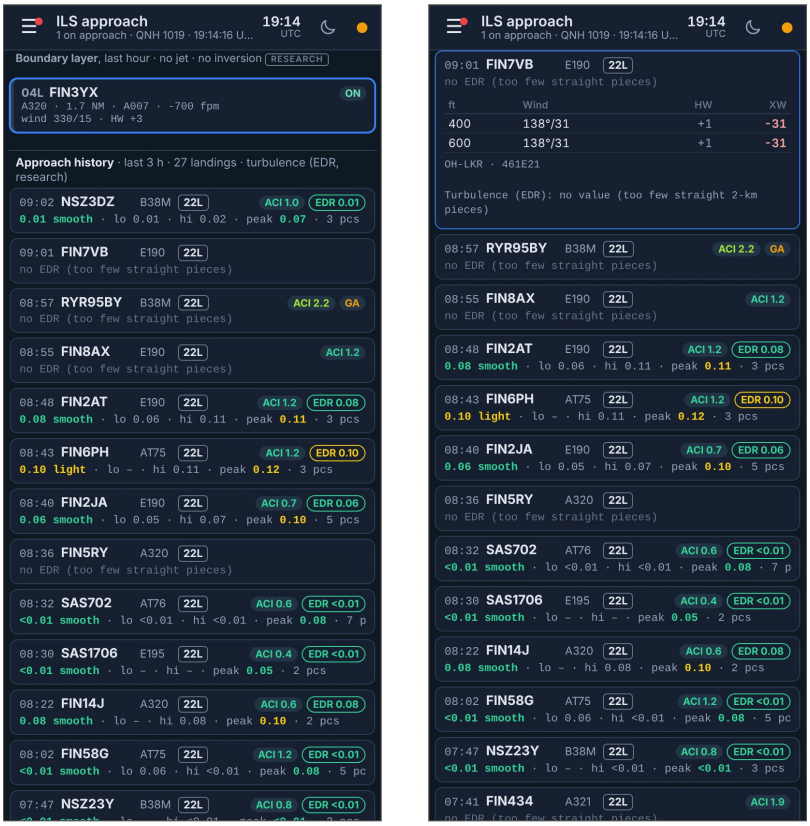


Figure 20. The approach history under the strips, scrolled down (3 h, Turbulence). A test: the stored landings of 30 October 2026, 06:05–09:05 UTC (all on 22L), moved to the present, under the approach traffic of 14:22 UTC. Left: the list — most landings smooth, FIN6PH light (0.10); landings without enough straight pieces have no EDR. Right: FIN7VB tapped open: its band winds, registration and turbulence.

Whose barbs. By default the barbs follow the *lowest* aircraft on the approach, the one closest to the threshold — the freshest picture of the wind it is flying through — and change to the next aircraft when it lands. Tap another aircraft on the profile or the map, or its strip, to follow that one; the line under the profile then says `chosen` and an Auto button takes you back to the lowest.

The strips under the profile list the aircraft on the approach, nearest first: runway, callsign (`AP`: autopilot), any windshear alert of that aircraft, the glideslope status; aircraft type, distance, altitude (`A010`, see above) and vertical speed; the latest

wind and its headwind (`HW`) or tailwind (`TW`) component. The coloured left edge is the wind source as for the barbs. The top bar gives the count, the QNH used for the glideslope and the active alerts.

Option	Choices	What it does
Show	Profile, Map, Both	the profile, the map, or both (default)
Aircraft on the map	On the approach, All tracked (30 NM)	only the aircraft on the approach (default), or also all others within 30 NM at any level
Departures on the map	Hidden, Shown	the climb-outs, marked with their runway and ↑ (default Hidden)
Turbulence on the map	Off, 30 min, 1 h	the paths of the latest landings and departures coloured by their EDR, and a summary line (default Off; research)
Map style	Street, ATC	the street map, or the radar-display look of the Live Map and the Windshear page (default Street; research)
Labels on the map	Standard, ATC	callsign / altitude, or the ATC style's three lines `FIN3YX / 007↓020 / 14 A320` (default Standard)
Aircraft type on the map	Off, On	the ICAO type after the callsign, e.g. `FIN456 AT75` (default Off)
Selected altitude	Off, On	the altitude set on the autopilot panel after the altitude, e.g. `FL120–A030` (default Off)
Speed vectors	Off, 30 s, 1 min, 2 min	a line to where each aircraft will be after that time on the map (default Off)
Trails	switch	each aircraft's path over the last minute (profile) and 2 minutes (map)
Runway	All, 04L+R, 22L+R, 04L … 33	the approaches shown (profile, strips and map)
Range	15 NM, 7.5 NM	7.5 NM spreads the last part of the approach
Wind barbs	Lo, Hi, Off	Lo: one barb per 400 ft / 0.5 NM; Hi: per 150 ft / 0.2 NM (denser, more scatter)
Wind component	Headwind, Crosswind, Off	the number above each barb
Windshear alerts	ADV +, WSA +, MBA, Off	the lowest alert level drawn (the same setting as on the Windshear page of the same browser)
Chart under the view	WS + EDR, WS + ACI, Off	turbulence (EDR) or approach conditions (ACI) above the windshear of each landing and departure, under the profile or the map (default WS + EDR)
Chart range	3 h, 6 h, 12 h	the period of that chart (default 3 h)
Boundary layer below	Off, BL wind, BL temp, BL headwind, BL Δ headwind, BL shear	a boundary-layer chart after that chart, on its range (default Off)
Approach history under the strips	Off, 3 h, 6 h	the landings of that period as History cards under the strips (default 3 h); with Departures on the map: Shown also the departures (blue runway)
Approach history shows	Turbulence, Headwind, Crosswind, Wind, ACI, Windshear	the second line of those cards (default Turbulence)

Table 3. ILS approach options. The *Windshear detection and the Windshear page* guide explains the alerts and the barbs.

8 EDR · WS · ACI · ST

The approach conditions and windshear chart of the Windshear page (Fig. 21). The upper part is the *Approach Conditions Index* (ACI, 0–10) of every landing: how rough the last part of the approach was, from the aircraft's own replies — below 2 smooth, 2–4 light, 4–6 choppy, 6–8 rough, 8 and above very rough. The dots are coloured by the aircraft class (`N` narrowbody, `R` regional jet, `T` turboprop, `W` widebody), the line is the running median. With ACI + windshear the lower part shows the windshear report of each landing and departure on the same time axis, as a change of headwind in kt: the strongest headwind ramp of a landing as a stem from zero, the change against the previous 45 minutes as a filled diamond (landings) or a hollow one (departures), go-arounds as triangles, and the advisory and alert levels as dashed lines.

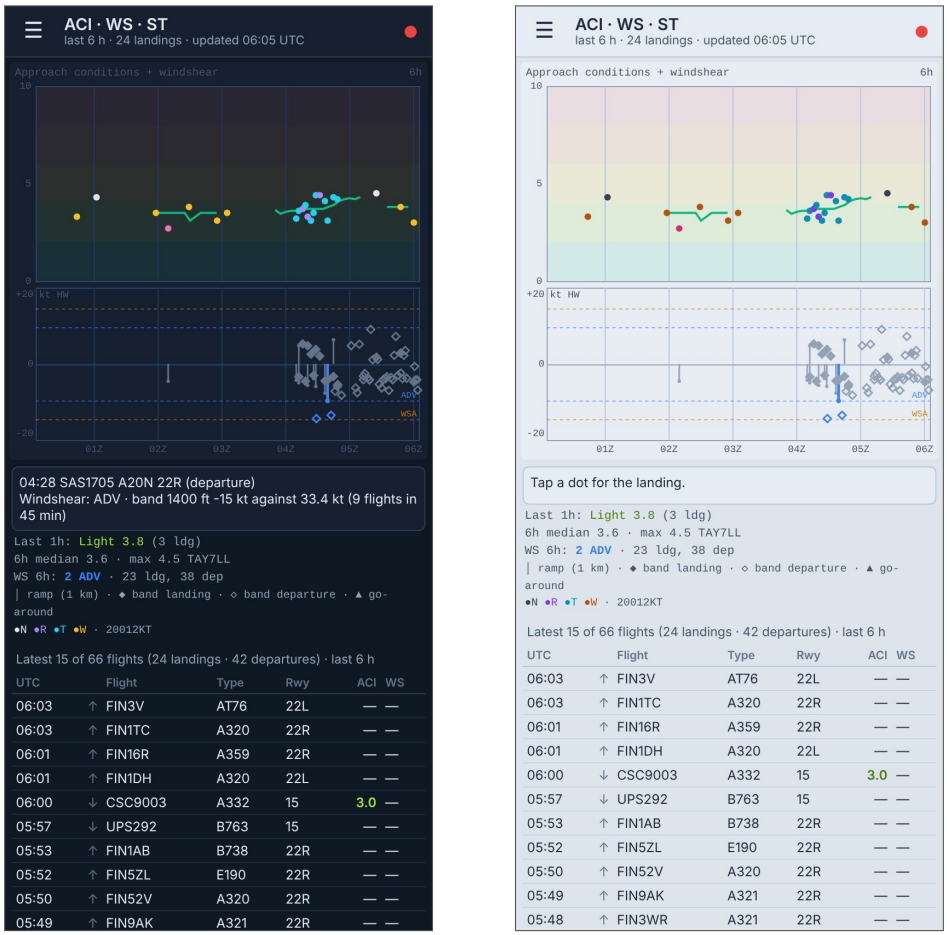


Figure 21. ACI + windshear, the last 6 hours to 28 October 2026 06:05 UTC: landings on runway 15, departures from 22L/22R in a south-westerly low-level jet, ACI mostly 3–4.5. Left: a tap on a diamond shows that flight (here the departure SAS1705, an advisory: 15 kt less headwind at 1 400 ft than the flights before); right: the light theme. The list under the chart is in Fig. 26.

ACI + EDR + WS (from 29 October 2026; research). The fourth Show choice puts the **turbulence** between the two (Fig. 22). For every landing (dot) and departure (ring) MODE-S Wind estimates from the aircraft's own replies the **EDR** — the turbulence measure of ICAO, which calls a peak EDR of 0.10 light, 0.20 moderate and 0.45 severe for a medium-size aircraft. The middle strip has those three lines on a logarithmic scale, the dots coloured green (smooth), yellow, orange or red, and the running median in purple. **Tap a flight** to see its EDR over the whole approach or climb, below and above 1 000 ft and its strongest 2-km piece; a line marks it through all three strips. The summary adds *EDR: last 1h 0.13 light · median 0.12 · max 0.19*. The EDR is an average over the approach from replies about 2 s apart, not the ICAO one-minute peak: a research value, to be compared with the pilot reports. The *Windshear* guide explains how it is computed.

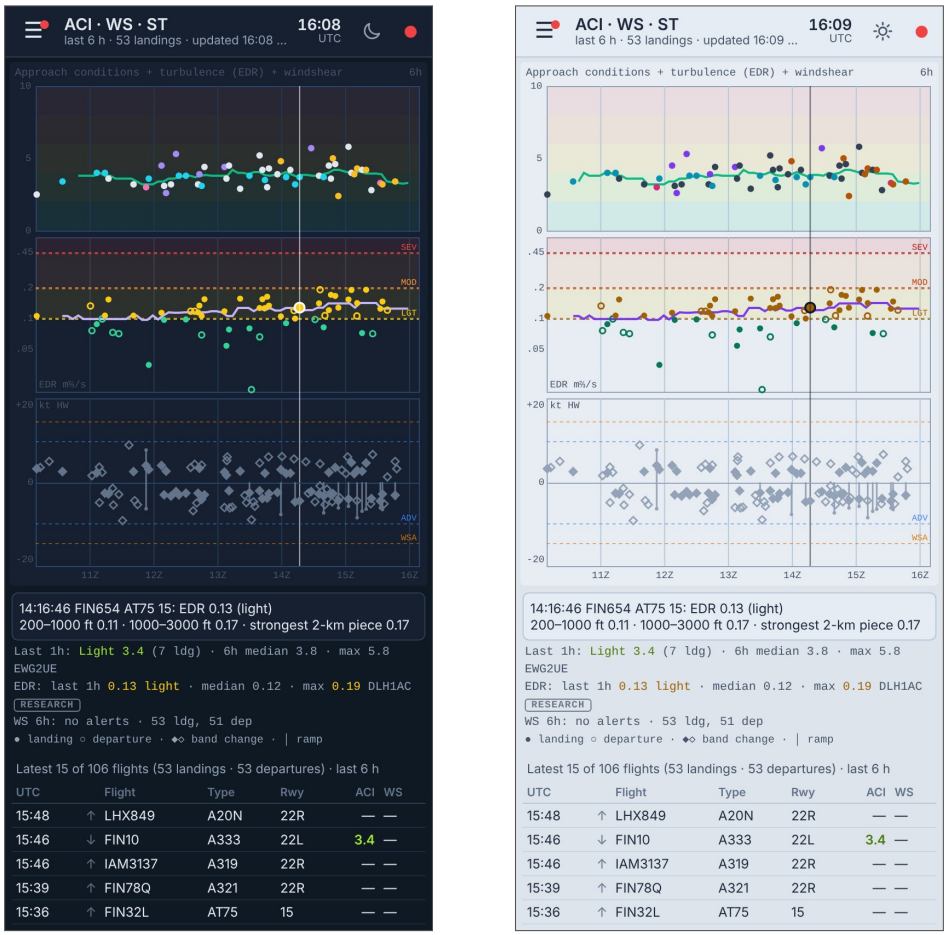


Figure 22. ACI + EDR + WS, 6 hours: the flights of 28 October 2026 07–12 UTC replayed in a test (the time axis shows the replay time). A tapped landing (FIN654, EDR 0.13, light) is marked through the three strips; most landings were light, the departures (rings) a little lower. Right: the light theme.

EDR + WS (from 31 October 2026) is the same without the ACI part: the turbulence and the windshear of every landing and departure, a little taller each.

Boundary layer below (from 31 October 2026; research). With ACI only, ACI + windshear, ACI + EDR + WS or EDR + WS the option Boundary layer below adds one of the boundary-layer charts (described next: BL wind, BL temp, BL headwind, BL Δ headwind or BL shear) *under* the chart, on the same hours and the same time axis (Fig. 23) — so you can see, for example, whether the turbulence and the windshear of the flights came with a low-level jet, an inversion or a change of the headwind. It has its own tap read-out and summary lines; the list of flights follows under it. Off (the default) leaves the chart alone; with Space–time and the boundary-layer views the option is hidden.

Figure 23. Boundary layer below, 12 hours of stored data to 31 October 2026 10:10 UTC (drawn in a test; the stored flights start at about 04 UTC). Left: EDR + WS with BL headwind under it and a tapped departure (FIN935, 3 200 ft, 16 ktheadwind). Right: ACI + EDR + WS with BL wind (Light theme). The hours line up through all the parts.

Boundary layer (from 29 October 2026; research). The choices BL wind and BL temp (until 31 October 2026 Boundary layer · wind and · temperature) show the lowest 5 000 ft at the airport hour by hour (Fig. 24): the wind speed in colour, or the temperature against the ground (red: warmer than at the ground — an *inversion*, calm and smooth air; blue: colder). A **low-level jet** — a band of strong wind a few hundred metres up, usually at night — is marked by a diamond at its height with its speed, joined hour to hour; a **surface inversion** by a red bar along the bottom. Hatched: no aircraft flew. Tap a cell for its wind or temperature, a diamond for the jet, a red bar for the inversion. The summary gives the last hour, e.g. *low-level jet 750 ft 18 kt from 023° · surface inversion +2 K*. The *Windshear* guide explains how they are found.

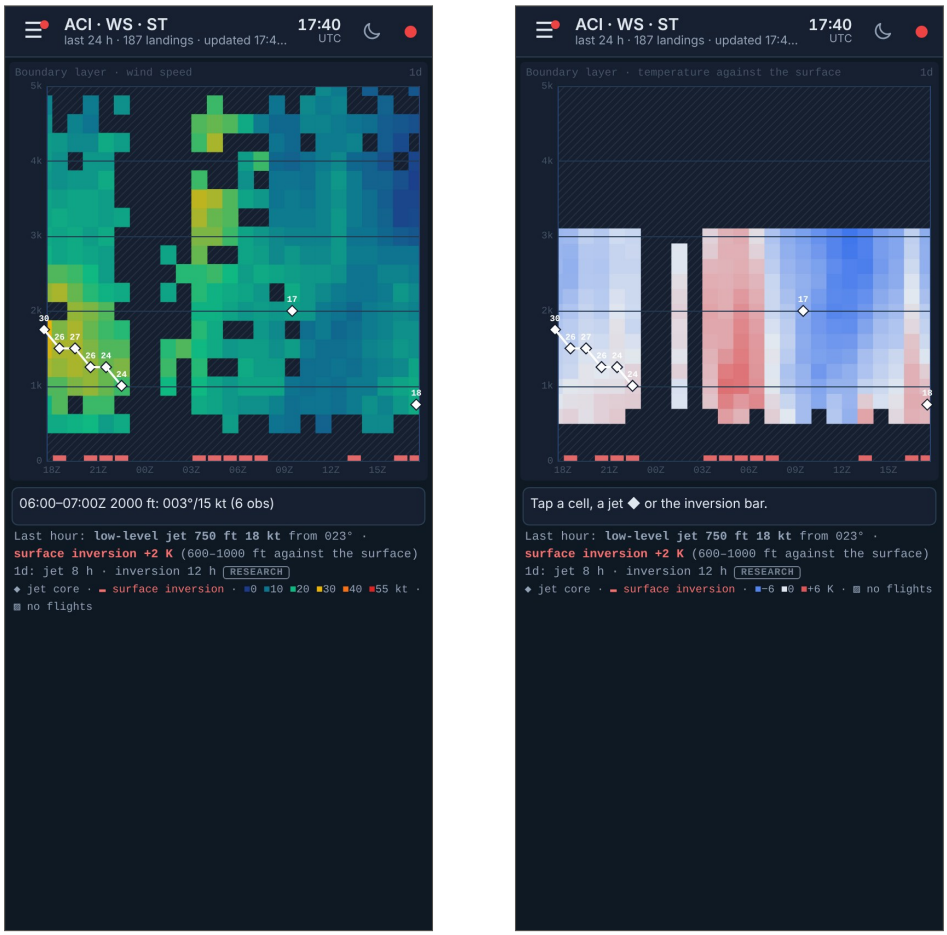


Figure 24. Boundary layer, 17 October 18 UTC – 18 October 17 UTC. Left, wind: the evening jet sinking from 1 800 to 1 000 ft (30 → 24 kt), a tapped cell above. Right, temperature: warmer aloft in the night and morning (red), colder by day (blue); the red bars are the hours with a surface inversion.

Flight by flight (from 31 October 2026; research). BL headwind, BL Δ headwind and BL shear draw every landing and departure as its own column, held until the next flight, in 100-ft steps — the same fields as in the Time–height view (Section 5) and on the Windshear page: the headwind along the flight's runway (violet a tailwind); the change against the flights of the previous 45 minutes in the same direction (blue less, red more headwind); and the vertical shear of the flight's own headwind, red where it grows with height (a descent loses headwind), blue where it falls (Fig. 25). A coloured line along the top gives the landing runway, windshear alerts are outlined, and the jet and inversion marks stay on top. Tap a column for the flight at that height: its headwind, the change, the shear and its alert.

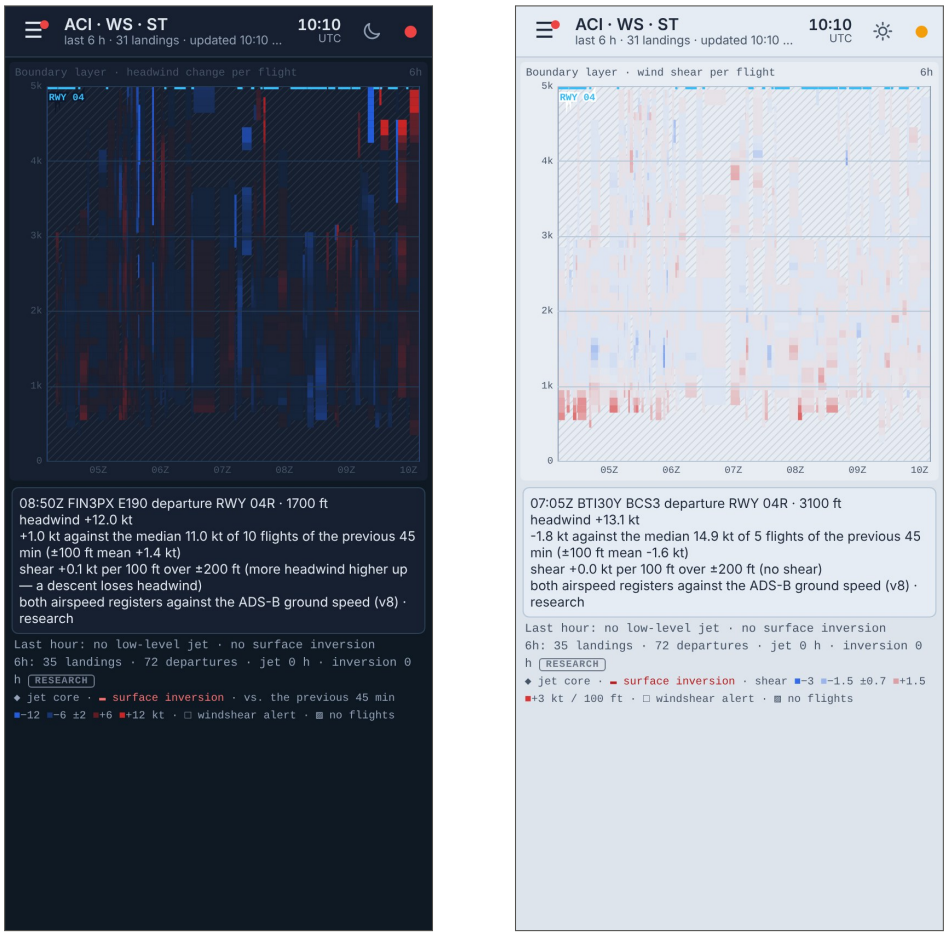


Figure 25. Boundary layer flight by flight, 31 October 2026 04–10 UTC, a calm morning. Left, BL Δ headwind with a tapped departure: FIN3PX at 1 700 ft, 1 kt more headwind than the median of the 10 flights before it. Right, BL shear (Light theme), a tapped climb-out at 3 100 ft with hardly any shear.

Official reports on the chart (from 29 October 2026). When a METAR reports windshear (`WS R22L`) the windshear part has an amber bar along its top from that METAR to the next one, with the runway; a pilot report near the airport (a WXREP or ARS within 30 NM, at or below 10 000 ft) is a teal **P** at the time it was observed (Fig. 33, left). Tap them for the whole message. The summary adds e.g. `METAR WS R22L · 1 pilot report near`. The same marks are on the WS + ACI chart of the ILS view. The messages are listed in the History (Section 9.1).

Under the chart: the box with the tapped landing, departure or go-around; the summary lines of the Windshear page (last hour, median and maximum of the period, the windshear alerts and the counts); and the **list of the flights of the period** (from 28 October 2026; Fig. 26), newest first. Its heading gives the count and the period (*Latest 15 of 59 flights (17 landings · 42 departures) · last 3 h*); 15 flights are shown, Show all opens the rest. From 1 November 2026 each flight is a card as in the History view: the time, ↓ landing or ↑ departure, the callsign, the aircraft type, the runway and the chips **ACI** (landings), the **windshear level**, **EDR** (turbulence) and **GA** (before a table). The list follows the chart: with ACI only the landings, with the windshear and EDR charts the landings and the departures, and with Space–time the flights of the diagram with their largest band change and its height (*−7 @1200*). Landings in History and Departures in History open the History view (Section 9) with the same period, for the bands and reports of each flight. **Options**: the period (3 h, 6 h, 12 h, 1 day) and ACI + windshear, ACI only, ACI + EDR + WS, Space–time, BL wind, BL temp, BL headwind, BL Δ headwind or BL shear. The *Windshear* guide and the *Approach Conditions Index* article explain them.

Figure 26. The list under the chart (scrolled down; Light theme), ACI + windshear, the day to 31 October 2026 10:10 UTC from the stored history (drawn in a test): the latest 15 of 39 flights — landings (↓) on 04L / 04R and departures (↑) — as History cards with their ACI, EDR and GA chips, Show all 39 and the History buttons.

Space–time (from 28 October 2026; Fig. 27) shows how the wind changes move through the successive flights: time to the right, height (400–1 800 ft) upwards, and in each cell the median change of the headwind of the landings and departures in that 5–30 minutes against the flights of the previous 45 minutes — **blue** less headwind, **red** more, from 3 to 15 kt. A front or a shear layer is a coloured streak through several flights and bands, turbulence scattered single cells. A dot marks a flight at or beyond its advisory limit, a triangle at the top a flight with a windshear level. Tap a cell: the box below shows the cell, and the list its flights with their change and wind (All flights × goes back to the whole period). Turn the phone for shorter time cells.

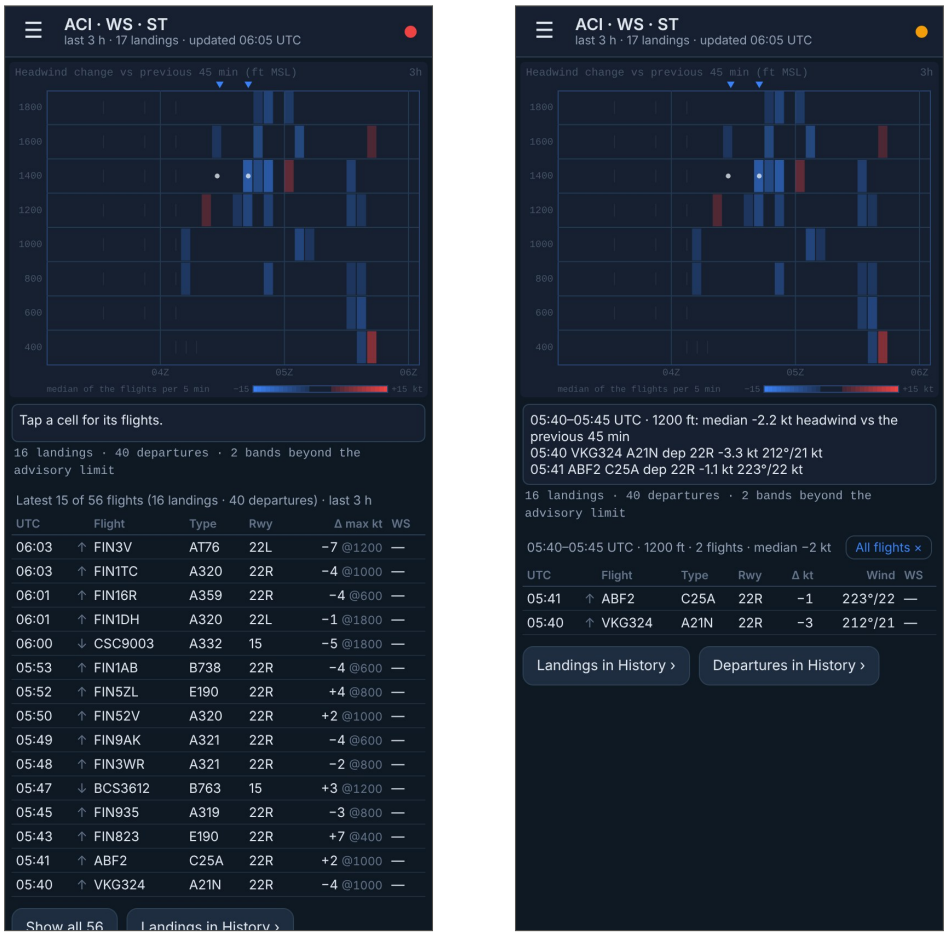


Figure 27. Space–time, the last 3 hours to 28 October 2026 06:05 UTC: the morning departures from 22L/22R climbing into a weakening south-westerly low-level jet — blue cells (less headwind) at 1 200–1 600 ft. Left: the diagram and its flights, each with its largest band change; right: a tapped cell (05:40–05:45 UTC, 1 200 ft) and its two departures in the list.

9 History

The Approach History of the Windshear page made for a phone (Fig. 28): the **landings** or the **departures**, newest first, as cards.

- **Line 1**: time (UTC; earlier days with the date), callsign, aircraft type, runway (in a thin frame; the columns line up from card to card), and chips: the **ACI** value, the **windshear level** (`ADV`, `WSA`, `MBA`), ***trial*** (a departure whose headwind ramps along the climb would have given that level — a trial, no alert), **EDR** (from 29 October 2026: the turbulence of the flight, outlined in its colour — green smooth, yellow light from 0.10, orange moderate from 0.20, red severe from 0.45) and **GA** (go-around).
- **Line 2**: depends on Show. Headwind and Crosswind: one cell per height band (the heights are in the row under the search field) with the component along the runway in kt — headwind green, tailwind red; crosswind + from the right, − from the left. Wind: an arrow (where the wind blows to) and the speed. ACI: the index and its two segments with the METAR. Windshear: the strongest headwind ramp and the band change. Turbulence (from 29 October 2026): the EDR with its class, below (lo) and above 1 000 ft (hi), the roughest 2-km piece (peak) and the number of pieces (Fig. 29).
- **Tap a card** to open it: all its height bands with wind, headwind and crosswind, the approach conditions in detail, the windshear report and the turbulence in words — the texts the full site shows when the mouse is over a row — and, from 29 October 2026, the **boundary layer** of that hour: *Boundary layer then: low-level jet 1750 ft 27 kt from 134°* (or a surface inversion, or neither). Tap again to close.

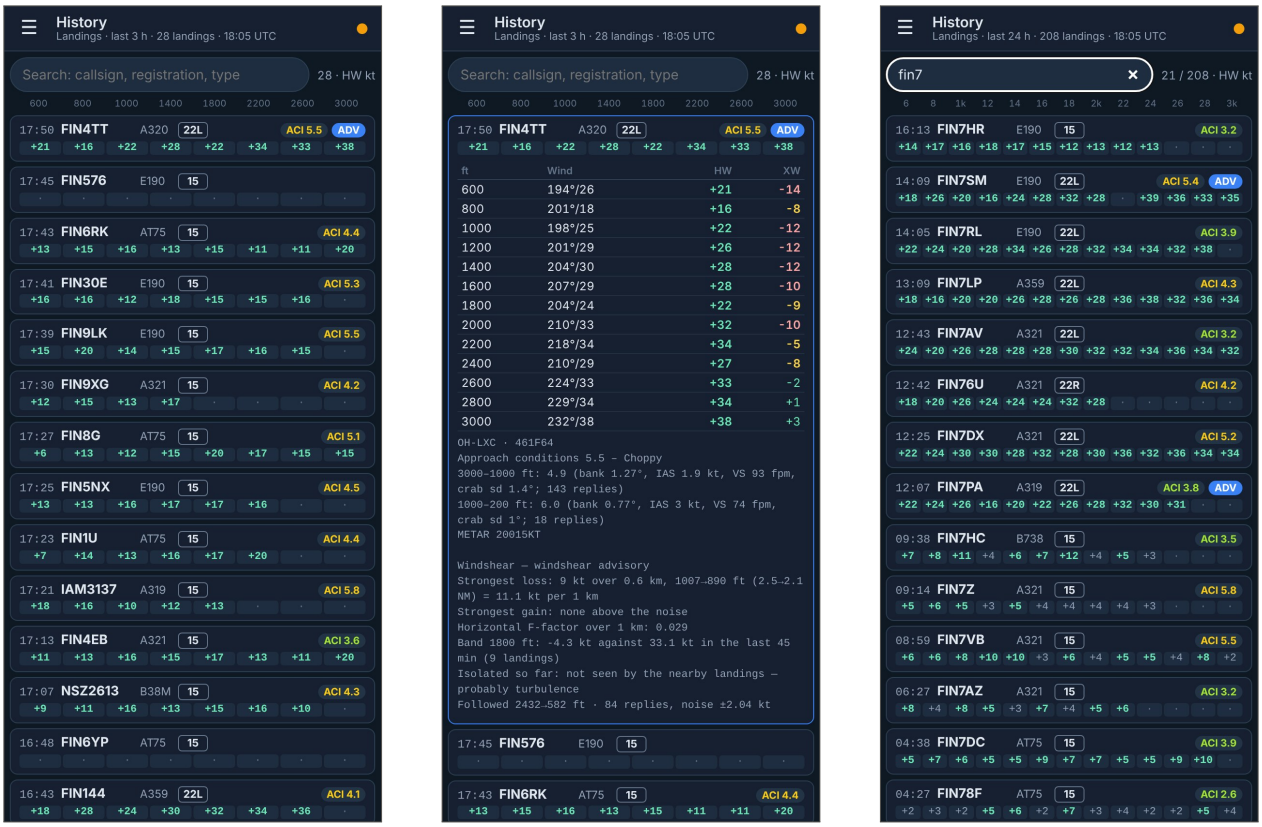


Figure 28. History, 27 October 2026. Left: the landings of the last 3 hours, headwind per band (a strong south-westerly: +13 to +38 kt). Middle: a card opened — the bands, the approach conditions and the windshear report. Right: the search `fin7` over a day, with the 13 height bands of Hi.

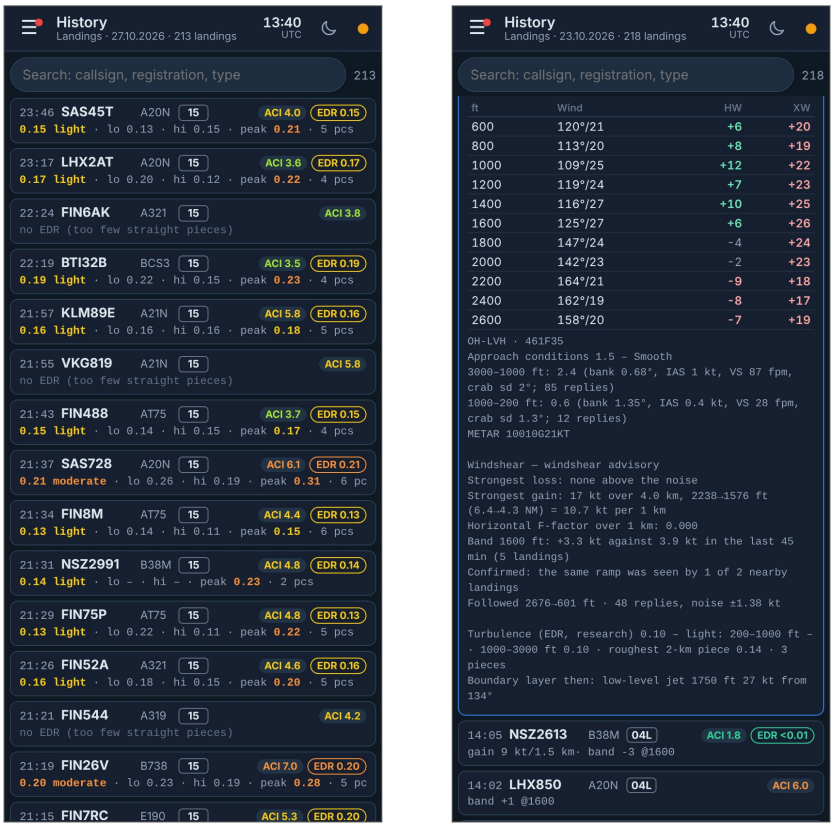


Figure 29. Left: Show: Turbulence, the landings of 27 October 2026 — most light, SAS728 and FIN26V moderate; “no EDR” when the approach had too few straight pieces. Right: an opened card of 23 October with the turbulence and the boundary layer then — FIN440 had a headwind-gain advisory while a low-level jet blew at 1
,750
,ft. The EDR is a research value: an average over the approach, not the ICAO one-minute peak.

The headwind profile (from 31 October 2026). An opened card begins with the flight’s headwind every 100 ft against the flights of the previous 45 minutes on the same runway direction — their median (dashed) and middle half (grey band) — with the ramps of its windshear report drawn thick (blue loss, red gain), the band change as a diamond and its turbulence pieces in a strip at the right (Fig. 30). A flight that leaves the grey band and comes back met a gust of its own; when the band moves with it, the air mass changed. The *Windshear* guide explains the chart.

Figure 30. An opened card of 27 October 2026, FIN9DZ on runway 15 with an advisory (the station's data): a loss of 10 kt over 0.4 km at about 600 ft where the six flights before it kept their headwind, and moderate turbulence there (orange piece at the right).

Search. The field above the cards finds the callsign, registration or aircraft type anywhere in the text, upper or lower case and with or without the hyphen: `fin7`, `OH-LX` or `ohlx`, `AT75`. The number beside it shows how many of the flights match. The search on the full site's Approach History works the same way.

Both (from 3 November 2026). List: Both shows the landings and the departures in one list, newest first (Fig. 31). A departure's runway box is filled **light blue**; a landing's is framed as before. Each card keeps its own chips, second line and details — the ACI only for the landings, the trial ramps only for the departures — and the line at the top counts the departures, e.g. *235 flights (115 dep., blue runway)*.

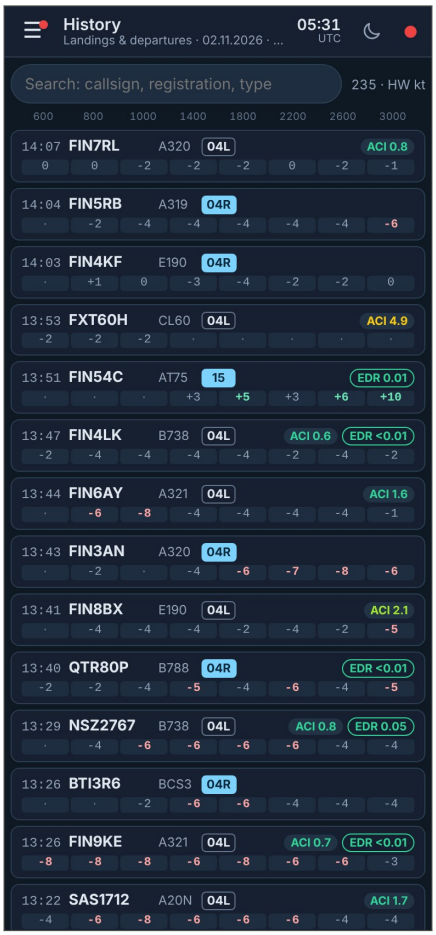


Figure 31. List: Both, headwind per band: the stored landings and departures of 2 November 2026 (a copy of the database, drawn in a test). The departures — here FIN5RB, FIN4KF and FIN54C — have their runway box filled light blue.

Options (Fig. 32): List Landings / Departures / Both (or METAR & reports, Section 9.1); the range 1 h – 1 day (default 3 h); or a **date**: the calendar field or ⟨ ⟩ (a day back / forward), and Live returns to the range; Show; and Lo · 8 or Hi · 13 height bands. A live range is updated every minute; a date is loaded once.

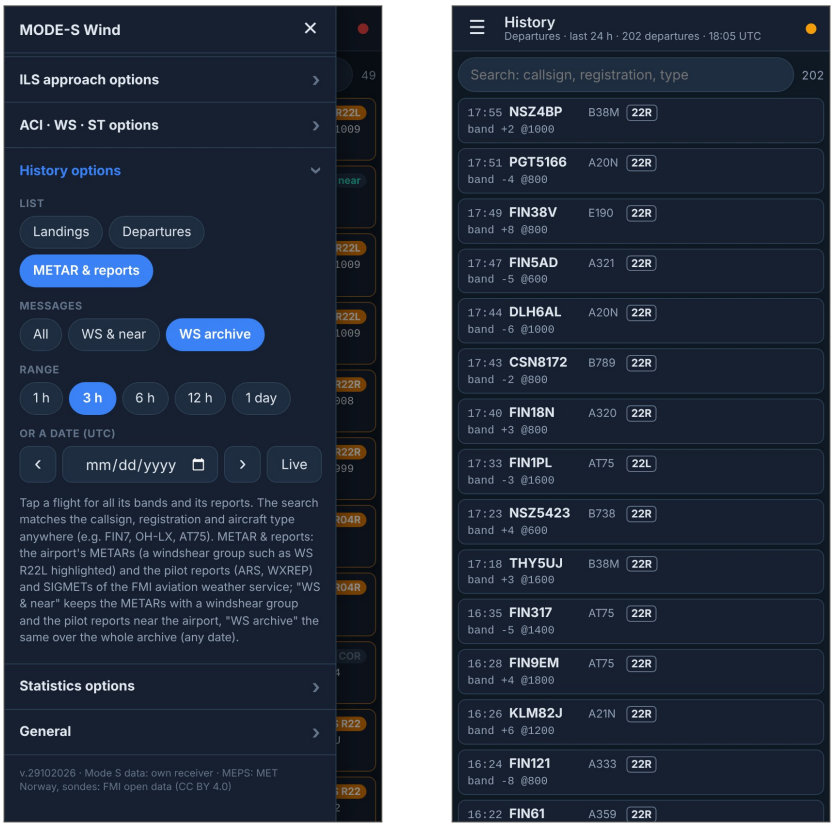


Figure 32. Left: the History options (here with METAR & reports chosen, which has Messages in place of Show and the height bands). Right: the departures of a day, Windshear shown — the band change of each climb-out against the earlier flights.

9.1 METAR & reports

From 29 October 2026 the History's List has a third choice, METAR & reports (Fig. 33): the METARs of Helsinki-Vantaa and the pilot reports and SIGMETs of the FMI aviation weather service, newest first, one card each — the time, the kind (`METAR`; `WXREP` teal, `ARS` red, `SIGMET` violet), for a pilot report the aircraft, what was reported and the height, the chips `WS R22L` (a METAR with a windshear group) and `near`, and the whole message with the windshear group in amber and the reported phenomena (`MOD WS`, `FBL TURB`, `INV` ...) in teal.

- **Windshear in a METAR**: in Finland the group `WS R22L` or `WS ALL RWY` is only in the METARs made by an observer — windshear that an aircraft or the tower reported on a landing or take-off below 1 600 ft. The automatic METARs of the night (`AUTO`) never have it.
- **WXREP** is a pilot report below the ARS criteria: light turbulence, light icing, light windshear, an inversion, the top of a CB. **ARS** is for moderate or severe turbulence or icing and the like.

Options: Messages All, WS & near (only the METARs with a windshear group and the pilot reports near the airport) or WS archive (the same over every stored date — also years back once the METAR history has been loaded); the range or a date as for the flights. The search finds text in the messages: `ws`, `r22l`, `turb`, a callsign.

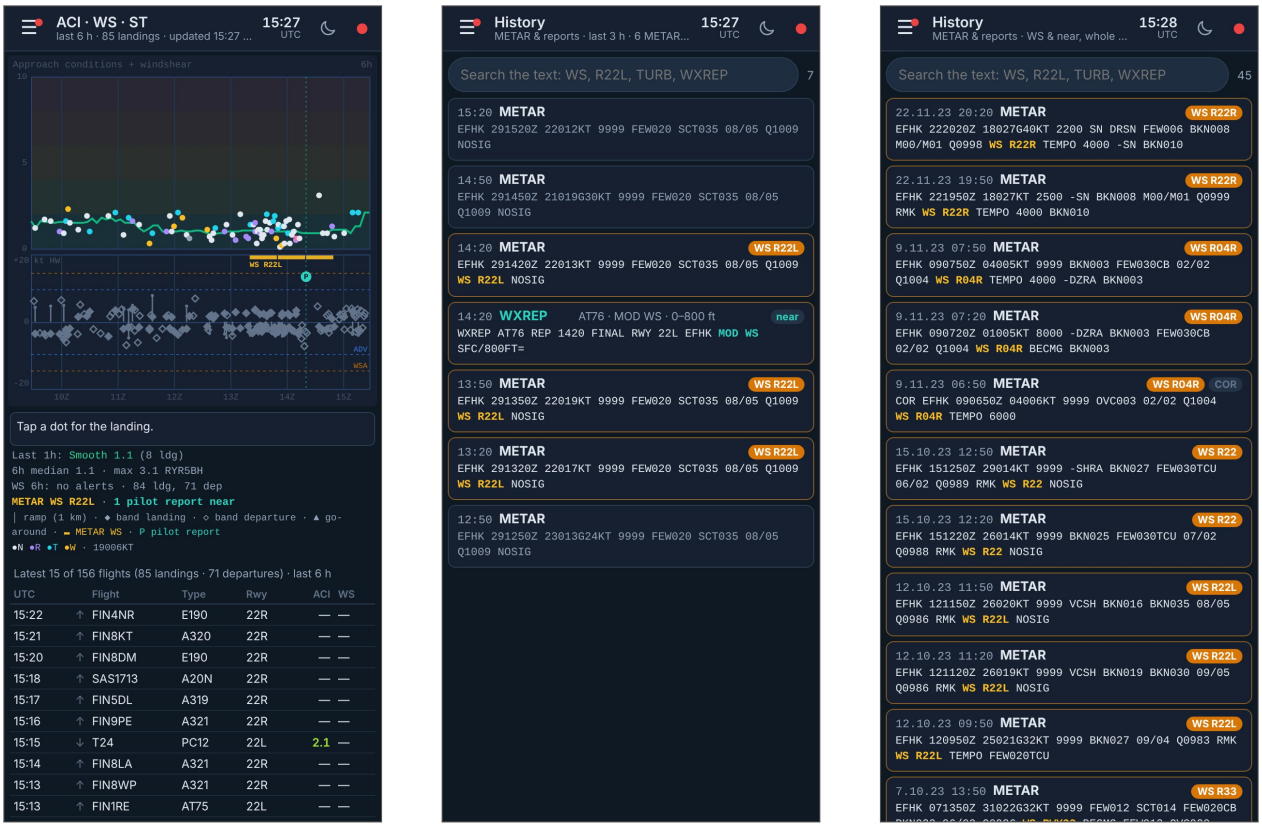


Figure 33. Left and middle: an example drawn with test messages — a METAR `WS R22L` for three half-hours and a WXREP `MOD WS SFC/800FT` on final 22L, on the ACI + windshear chart (amber bar, **P**) and in the History's METAR & reports. Right: WS archive — the real METARs of Helsinki-Vantaa with a windshear group in 2023, e.g. 22 November in snow and a southerly gale, `WS R22R`.

10 Statistics

The Statistics page of the full site made for a phone (from 28 October 2026): what the receiver hears, one panel under the other (Fig. 34). At the top six figures — messages per second, aircraft, aircraft with ADS-B and with EHS (the Comm-B registers that give wind and temperature), the MLAT and ADS-B positions and the age of the receiver's feed — and the time of the last update with the window. Below them the sections you have chosen, by default:

- **Message rate** — the messages per second as a chart (1 h, 24 h, 7 d above it; the selector chooses the message class). Its second tab, Data transfer, shows how much data the server receives from the receiver and sends to the users.
- **Downlink formats** — the share of each kind of Mode S reply, e.g. DF17 ADS-B, DF20/21 the Comm-B replies that carry the meteo registers, DF11 the all-call replies.
- **Meteo yield** — how many winds and temperatures per minute the replies give, and why the rest gave none.
- **Range rings** — how far the receiver hears aircraft in each direction, for three height layers, with a table per distance. *Tap* the rings for the farthest aircraft in that direction and its height.

Each panel's explanation is folded under About this panel; tap it to read it. The panels are the same as on the full site, so the *Mode S & ADS-B and the Statistics page* guide explains them in full.

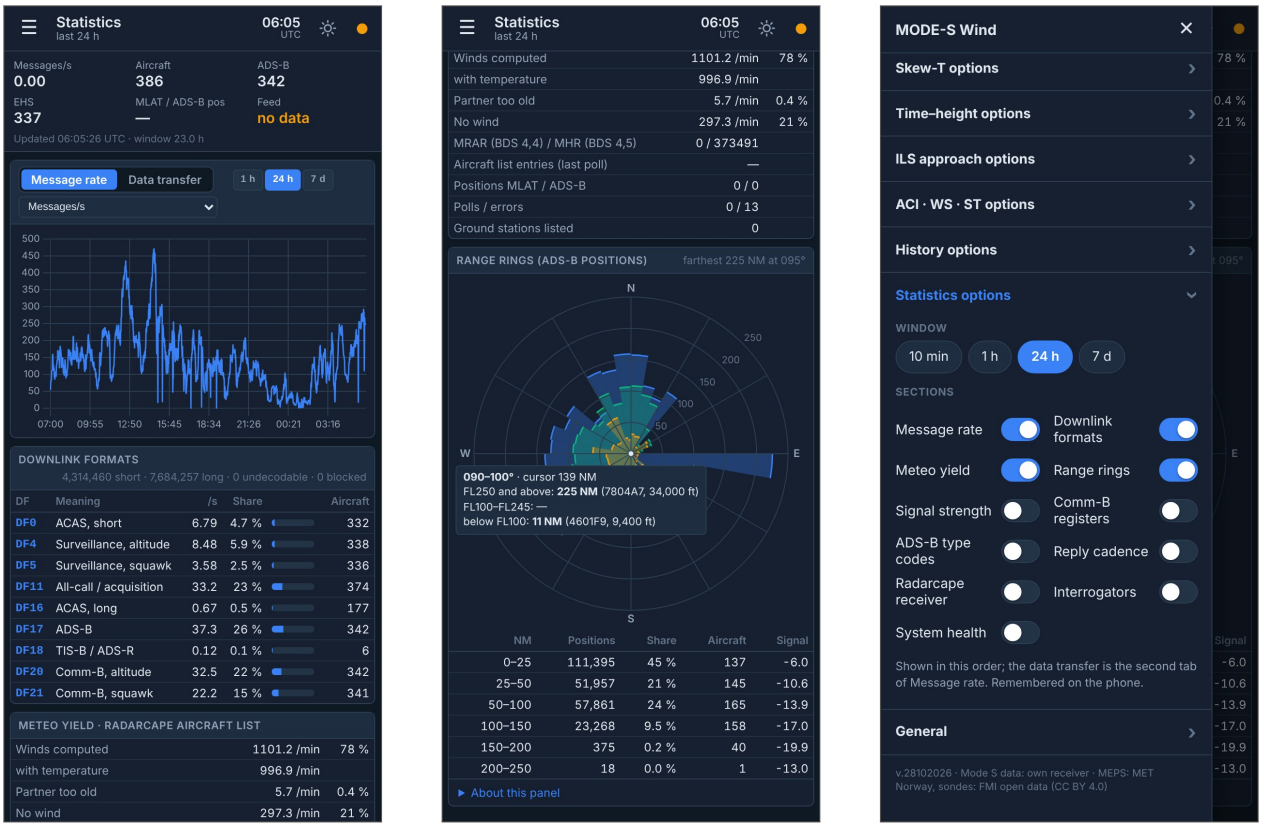


Figure 34. Statistics, last 24 h (a copy of the database of 28 October 2026 without a live feed, so the top figures show 0 messages per second and *no data*). Left: the summary, the message rate and the downlink formats. Middle: the range rings with the read-out of a tap towards the east. Right: the Statistics options — window and sections.

Options (Fig. 34, right): the **window** 10 min, 1 h, 24 h (default) or 7 d — the period of the tables, the downlink formats, the signal histogram and the range rings; and a switch for each **section**. Besides the four above you can switch on Signal strength (the histogram of the reply levels), Comm-B registers, ADS-B type codes, Reply cadence (how often each aircraft is interrogated), Radarcape receiver (its GPS, timestamps and temperature), Interrogators (the radars and WAM stations that ask, from the all-call replies), Aircraft as sensors (from 10 November 2026, below) and System health (the indicators of the health dot with their details). The sections are always shown in this order, and the choice is remembered on the phone.

Aircraft as sensors (from 10 November 2026, Fig. 35). Every aircraft that sends its heading and speeds is a wind sensor, and they are not equally good. This section compares the aircraft types: one row per type — the ten most common and, in amber below them, the propeller types — and *one dot per airframe, without its identity*; the white line is the type's median. The selector chooses what is compared: the *heading offset* the system has learnt to take off the airframe's heading, the *direction-dependent heading error* (large where the heading is sensed magnetically, as on the turboprops), the *flight-to-flight spread* of the offset, or the *wind difference from the others*. The table under the chart gives the medians per type; *tap* a dot for all the values of that airframe. The last three quantities are collected from 10 November 2026 on and fill up over the following days. The *Mode S & ADS-B and the Statistics page* guide explains the panel.

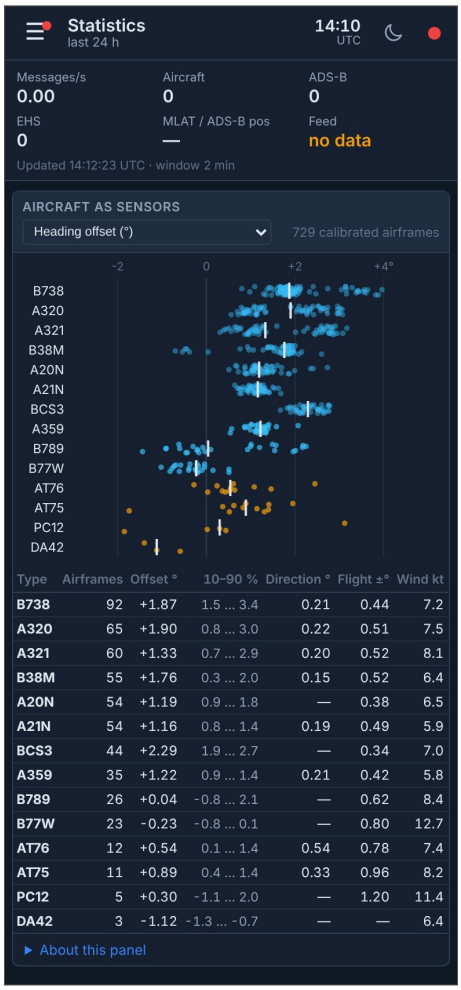


Figure 35. Aircraft as sensors, the heading offset: the common narrow-body jets between about +1 and +2°, the Boeing 787 and 777 and the turboprops (amber) nearer zero. From the station's observations of 23 October – 10 November 2026, replayed through the new code on a copy of the database without a live feed.

More receivers (from 2 November 2026). When MODE-S Wind listens to more than one receiver — the Radarcape **MODES-RX-01** in Tuusula and the AirSpy **MODES-RX-02** in Vantaa — a row of buttons under the top bar chooses whose statistics you see (Fig. 36). All receivers is the default: the replies of every receiver, each reply counted once — what the weather is computed from — with a *Receivers · merged* panel among the sections. A receiver's own button shows what that receiver alone hears, with its receiver panel (its feed, its clock lined up with the Radarcape's GPS time, the delay to the server, and how much of what it hears only it heard). Everyone sees All receivers; a receiver's own button is shown only to its owners — marked `yours` — and to the users with *Full view*, who see every receiver (a user can be both). The other receivers are then summed up in the merged panel (*1 other receiver*) and in the system health (*Other feeds*). The full site's Statistics page works the same way; the *Mode S & ADS-B and the Statistics page* guide explains the panels. Your choice of receiver is remembered on the phone.

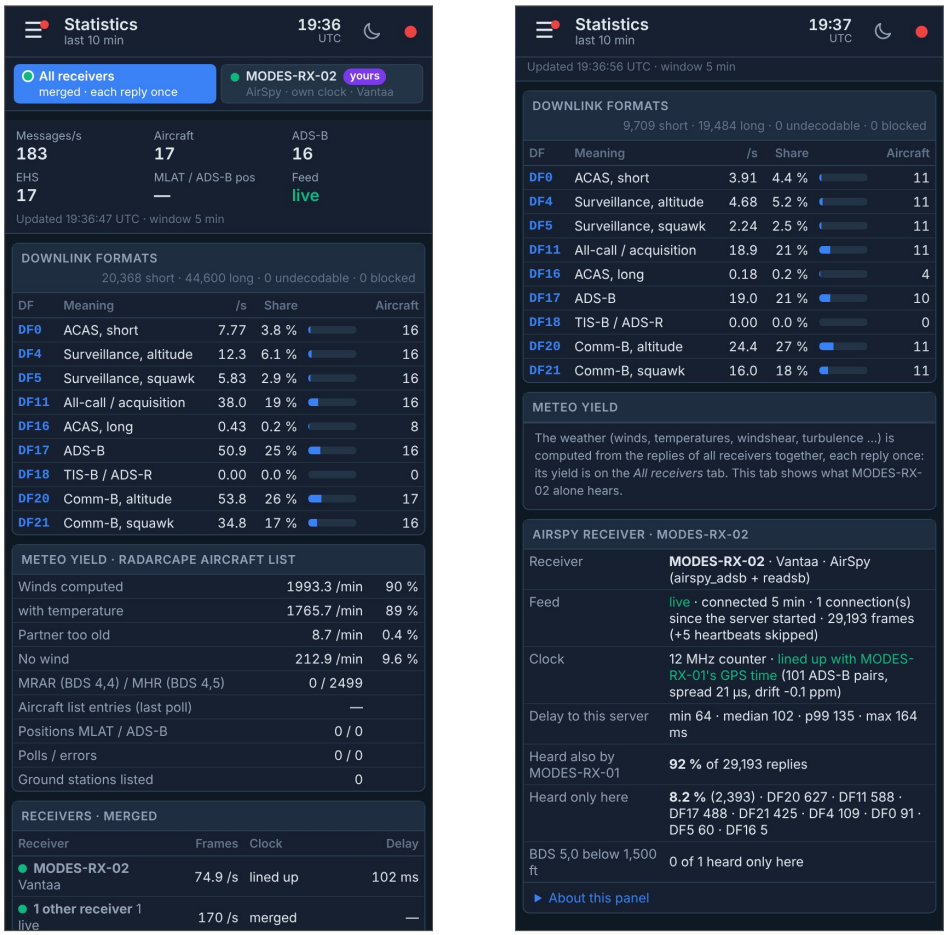


Figure 36. Statistics with two receivers, for the owner of MODES-RX-02 (a real 10-minute recording of both receivers of 2 November 2026 replayed in a test; sections Downlink formats, Meteo yield and the receiver panel, window 10 min). Left: All receivers — the merged replies, the meteo yield and the *Receivers · merged* panel with MODES-RX-02 and *1 other receiver*. Right: MODES-RX-02 (`yours`), scrolled to its receiver panel; two lines of the test set-up (its position and *readsb*'s own counts) are left out.

Updates and data. The view is updated every 30 seconds while it is shown (the full site's page every 10 seconds), and not at all while another view is shown. Each update is about 55 kB, so an hour of watching uses about 7 megabytes of mobile data — more than the other views; switch to another view when you have seen what you wanted. The chart program and the Statistics code are loaded the first time you open the view.

11 Weather

From 30 October 2026 the phone has the weather maps of the full site's *Weather* page: the **MEPS** model of MET Norway over Finland and the Baltic Sea, hour by hour, with a slider to move in time and ▶ to play them as an animation (Fig. 37). It shows the newest model run from **6 hours back** to **24 or 48 hours ahead**. All the hours are loaded when the view is opened — the strip above the slider fills as they arrive (grey: past hours, violet: forecast; pale: not loaded yet) and a `Loading 36 / 40 hours...` badge shows the progress — so the animation then runs smoothly without waiting for the network. The hour nearest to now is loaded first.

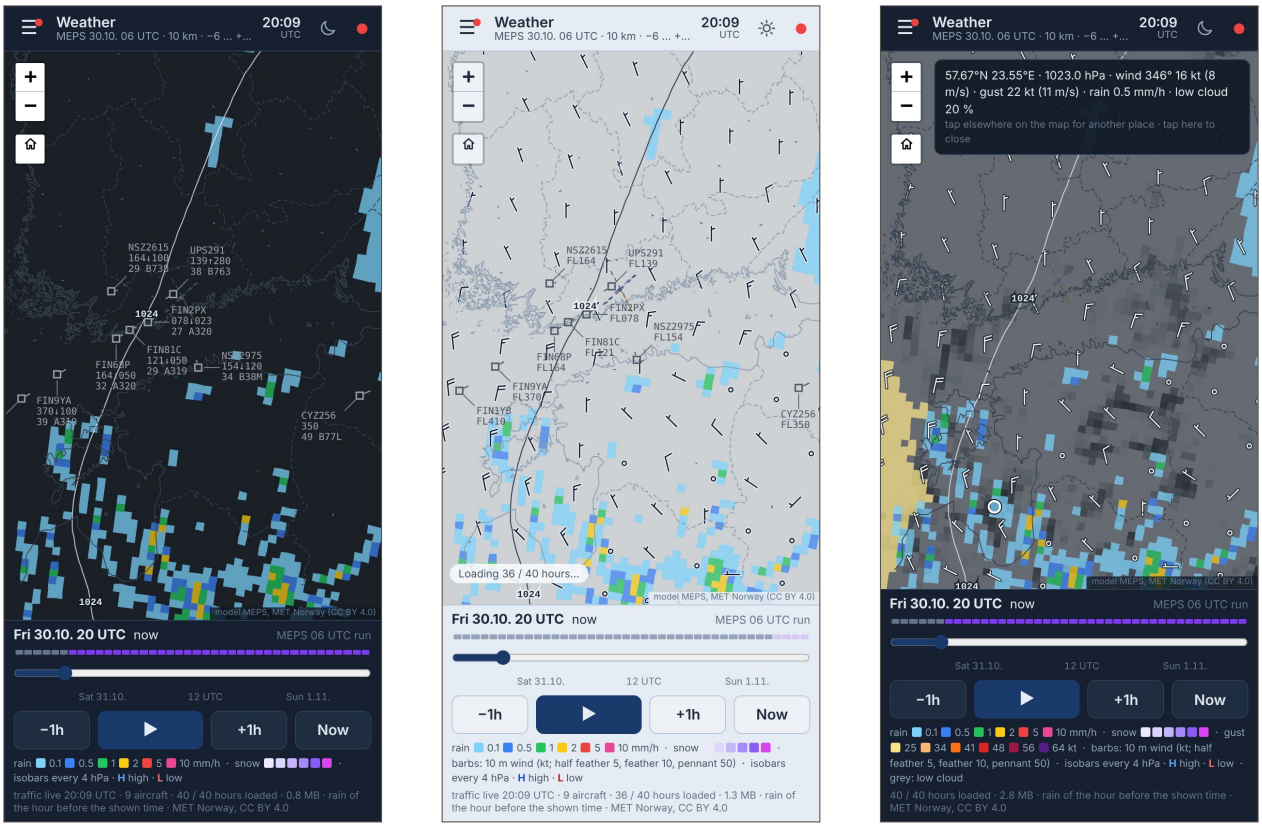


Figure 37. The Weather view at 20 UTC on 30 October 2026, MEPS 06 UTC run, 10 km grid. Left: Dark theme, rain and isobars with the air traffic on the ATC map with ATC labels. Middle: Light theme with wind barbs and standard labels (callsign and flight level), still loading (36 of 40 hours). Right: wind, gusts and low cloud, with the read-out of a tap on the map. The aircraft are a real snapshot of the production Live Map at 20:09 UTC on 30 October (the 9 with a position under a minute old); the weather is the stored model run, drawn in a test with a copy of the database. The street map's tiles cannot be loaded in that test, so the pictures use the ATC map style (on the phone the Street style shows the usual street map).

The panel under the map. The shown hour (*Fri 30.10. 20 UTC*) with how far it is from now (*now*, −3 *h*, or in violet +*12 h* for the forecast) and the model run; the strip and the slider with the days; −1h, ▶ / ■, +1h and Now; the legend of the shown layers; and a status line: the air traffic's time and number of aircraft, the hours loaded and the megabytes. The legend and the status line keep the tallest height they have had with the same options (from 5 November 2026), so the slider and the buttons stay in place when the radar's lines and text come and go between Now and the other hours. The rain and snow are those of the *hour before* the shown time, as on the full site. In landscape the panel is beside the map. **Following the clock** (from the evening of 3 November 2026): while the slider is on the present hour (*now* — as the view opens, after Now, or moved back to it), it moves on to the new present hour by itself when the clock passes the hour, and the hours of the slider move with it; on another hour (moved with the slider, −1h / +1h or ▶) it stays there until you tap Now.

Tap the map for the values of that place at the shown hour: position, pressure, wind, gust, rain or snow and cloud, as far as the layers are on (Fig. 37, right). A small circle marks the place, and the values follow the slider and the animation, so you can watch how the weather at one place changes. From 5 November 2026 the box has Meteogram here ›: it opens the Meteogram view (Section 13) at that place, with all its hours at once. Tap elsewhere for another place, or the box to close it. The box moves to the top when you tap the lower half of the map. The house button goes back to southern Finland; the map's position is remembered.

FMI radar (from 30 October 2026). Rain source adds the rain the Finnish weather radars observe — the national composite of the Finnish Meteorological Institute (FMI open data, CC BY 4.0): the rain rate of the last 5–15 minutes, about 2 km, over Finland and the nearby seas — as on the Live Map's weather panel. From 11 November 2026 it is FMI's renewed composite (FMI replaces its radar layers in November 2026): a quality-filtered product that shows fewer echoes than the old one did (the pictures here are of October, from the old one), and it reaches a little further, to Stockholm and northernmost Lapland. It is always the *newest* radar image, so it is shown only at the **present hour** (Now; the panel then says *now · radar 20:40*); at the other hours of the slider the model's rain is shown, and the status line says *FMI radar only at the present hour*. FMI radar shows the radar alone at that hour; Both the radar inside its coverage (dashed edge) and the model's rain outside it and, inside it, as white outlines (thin from 0.1, thick from 1 mm/h), so you can compare the forecast with what the radar sees (Fig. 38). A tap then gives the radar's value too: *radar 15.6 mm/h*,

radar dry or *outside the radar*. The rain of the radar is a rate at that moment, the model's the rain of the hour before the shown time. Opacity (from 1 November 2026, before Rain opacity and Radar opacity) makes all the coloured layers more or less transparent at once — the rain and snow, the radar, the gusts, the clouds and fog and the temperature colours — 75 % by default; the isobars, barbs, lines and numbers stay as they are. With Both the model's rain outside the radar area is drawn a third fainter than the radar, so that the radar stands out. The radar image is read every 4 minutes while the view is shown (about 10–150 kB, depending on the rain).

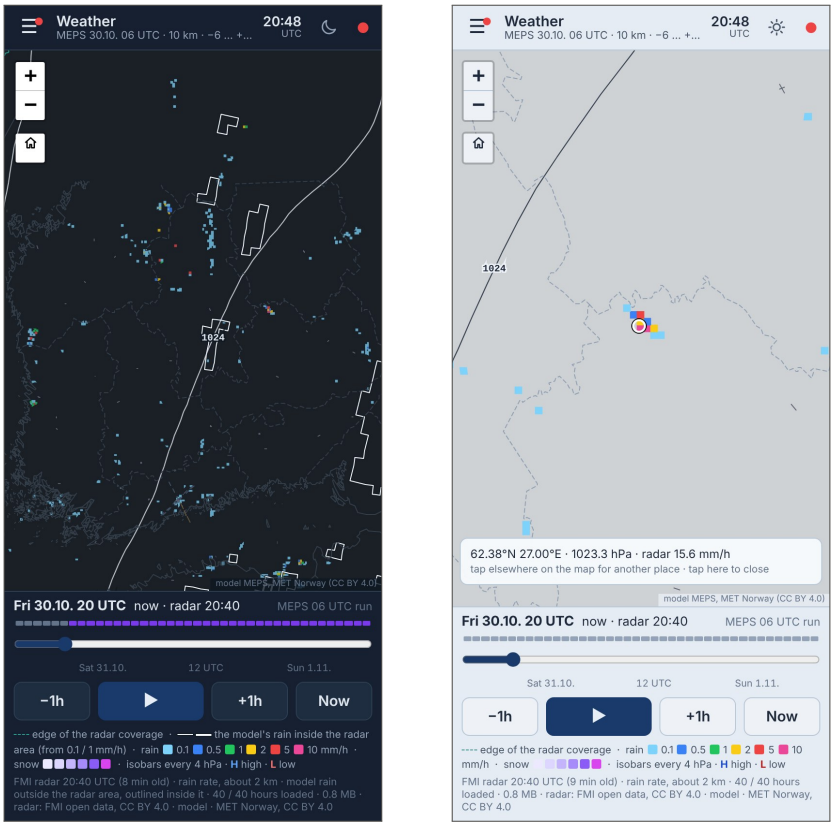


Figure 38. The FMI radar at the present hour: the composite of 20:40 UTC on 30 October 2026, read from FMI's open data in a test, with the stored MEPS 06 UTC run of the same day at 20 UTC. Left: Both, dark theme — light showers in central and eastern Finland (the radar's colours), the model's rain areas outlined in white, the dashed edge of the radar coverage. Right: FMI radar alone, light theme, Radar opacity 100 % (now Opacity), zoomed in near Pieksämäki, with the read-out of a tap on the strongest shower (15.6 mm/h). The ATC map style is used because the street map's tiles cannot be loaded in the test.

2.5 km detail near the airport (from 31 October 2026). MEPS itself is computed on a 2.5 km grid; the maps above use every fourth point of it (10 km), to keep the data small. Around Helsinki-Vantaa the view now uses the *full* grid: a box of about 300 × 300 km — from central Estonia to the latitude of Mikkeli and from about Turku to Hamina — with every 2.5 km point — when the map is zoomed in to **zoom 7 or closer** and the box is on the screen (Fig. 39). Inside the box showers, rain bands, fog patches and the clouds along the coast get their real shape instead of 10 km squares; a faint **dotted line** marks its edge, and outside it the 10 km map continues without a gap. The rain and snow, the clouds and fog and the gusts use the box; the isobars and the wind barbs stay those of the 10 km map. A tap inside the box gives its 2.5 km values, marked *2.5 km grid*, and the status line says *2.5 km near the airport*. Zoomed out further the box is not loaded or drawn. Detail near EFHK: Off switches it off. A finer grid shows the model's features more sharply, but it does not make the forecast for one place more exact: the model really resolves things of about 10–15 km.

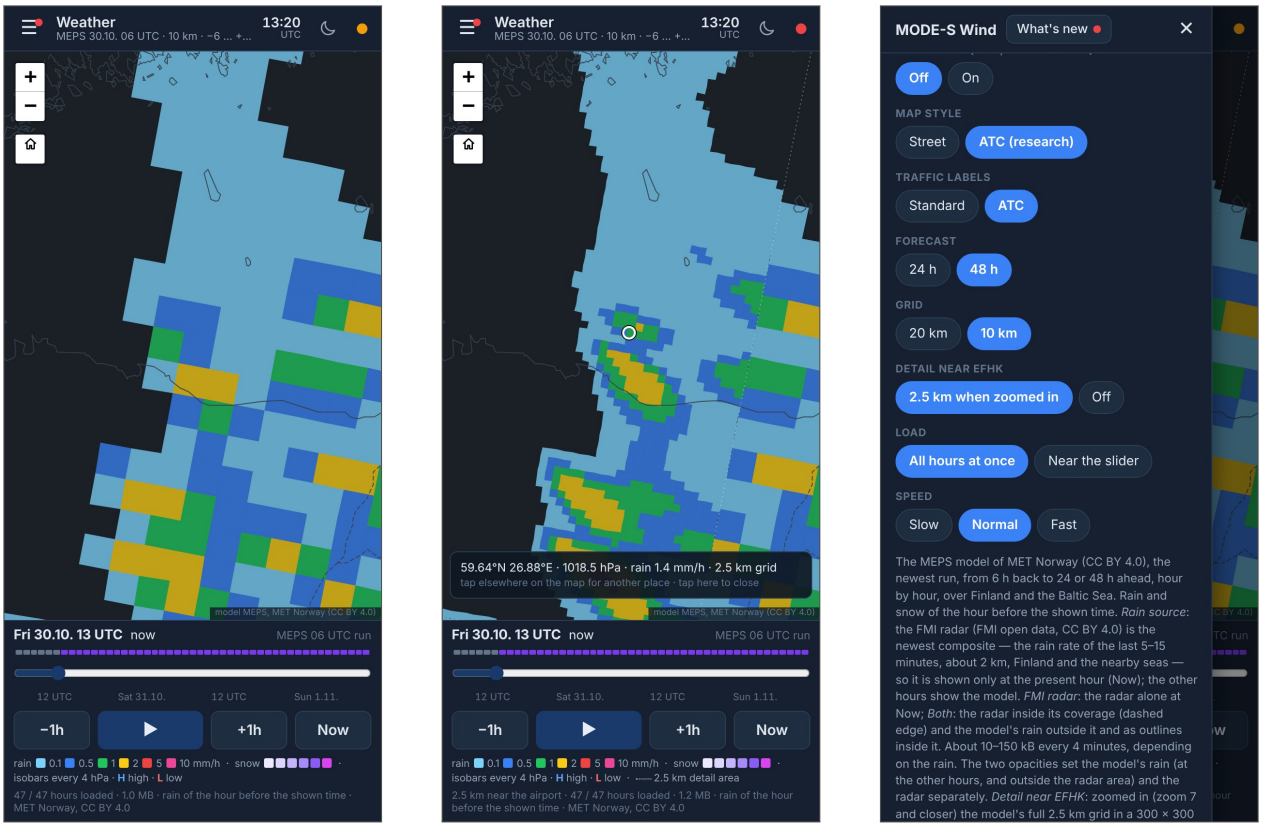


Figure 39. The 2.5 km detail: the stored MEPS 06 UTC run of 30 October 2026 at 13 UTC, a band of showers over the Gulf of Finland and the north coast of Estonia (zoom 8, ATC map style, drawn in a test). Left: Detail near EFHK: Off — the 10 km map. Middle: the default — the same hour on the full 2.5 km grid, with the read-out of a tap (*rain 1.4 mm/h · 2.5 km grid*); the dotted line at the right is the edge of the box, beyond it the 10 km squares. Right: the option in the menu.

Air traffic (research). With Air traffic: On the aircraft the receiver hears *now* are on the map — all of them, not only the approaches — updated every 6 seconds, with their last positions as dots and a one-minute line ahead. They are always the live traffic, whatever hour the slider shows. Traffic labels: ATC gives the three-line labels of the Live Map's ATC style (callsign; flight level with an arrow for climbing or descending and the selected level; ground speed in tens of knots and type), Standard just the callsign and the flight level. The labels are a little smaller than on the full site (10 px) and move out of each other's way. *Tap an aircraft* for its callsign, type, level, climb or descent, selected level, ground speed and track. Map style: ATC is the dark (or light) research display of the Live Map with the EFHK sectors and the national border instead of the street map; the aircraft symbols are then coloured as on the Live Map. It is a research display, not for air traffic control. With Map style: ATC (from 1 November 2026) a tap on an aircraft opens its label as the six-line **extended label**, as on the ILS view's map (Section 7) and on the full site's ATC maps: callsign, type and registration; level, selected level and vertical rate; ground speed, true and indicated airspeed and Mach; magnetic heading, track, drift and roll; the wind, the head- or tailwind and crosswind and the temperature; the distance and bearing from the airport (Fig. 40). Its values are read every 2 seconds while it is open; a tap on the label keeps it, a tap anywhere else on the map closes it. On the street map a tap shows the card as before.

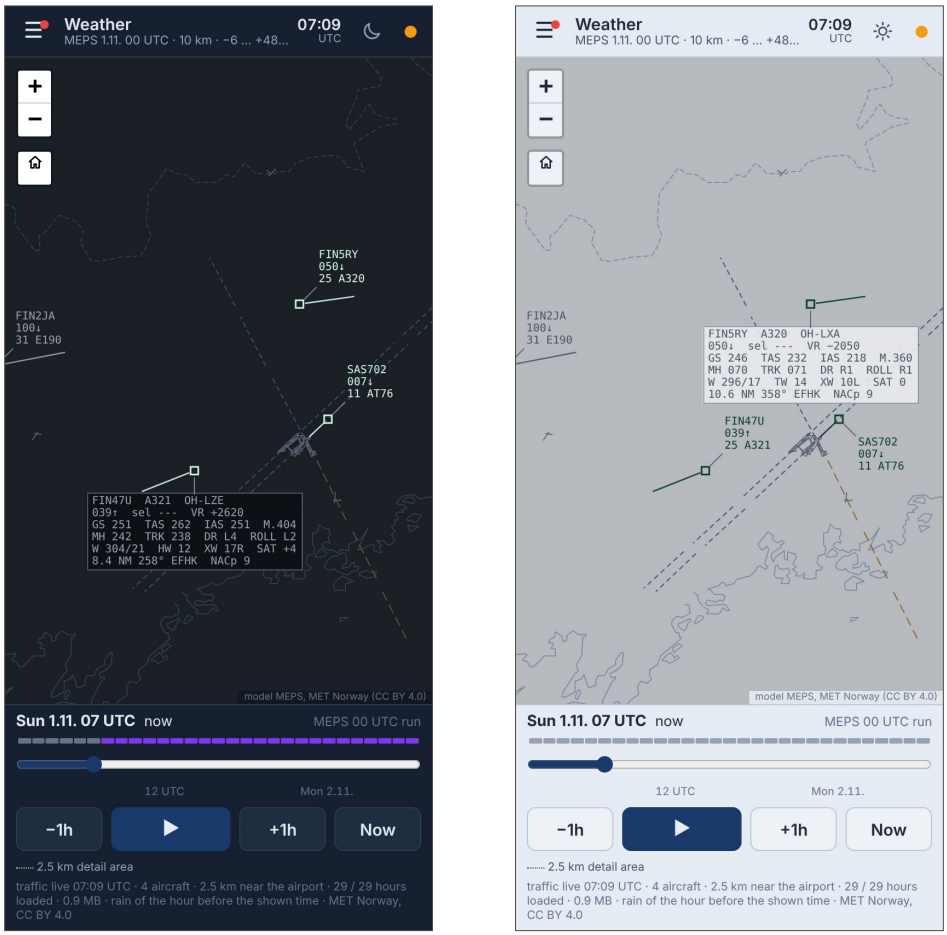


Figure 40. The extended label in the Weather view with Air traffic: On and Map style: ATC: the stored observations of 30 October 2026 08:29 UTC replayed in a test (positions, levels, speeds and the BDS 5,0 / 6,0 values as received). Left: FIN47U after take-off; right (Light theme): FIN5RY descending.

Receivers (from 2 November 2026). With Receivers (triangles): On — the default — the receivers of MODE-S Wind are small triangles on the map at their positions (to about 1 km), as on the full site's maps; with more receivers each in its colour and with its short name (`RX-01` blue, `RX-02` violet). Tap one for its card: the receiver's name, site and type. Off hides them.

Smooth shading (from 31 October 2026). The model's values are given on a grid, and drawn cell by cell they make a map of small squares. With Shading: Smooth (the default) the map is drawn smoothly instead: the rain and snow, the gusts and the FMI radar as areas whose edges follow each colour step between the grid points, and the clouds and fog as a smooth grey (or yellow) shading (Fig. 41). The values and the colour steps are the same, and so is the read-out of a tap; only the drawing between the grid points is smoother. A smooth edge is not finer data — a peak in a single cell can shrink to a small spot. Shading: Cells brings back the squares, e.g. to see the grid itself. The full site's Weather page has the same choice, and the Live Map's weather panel a Smooth tick box.

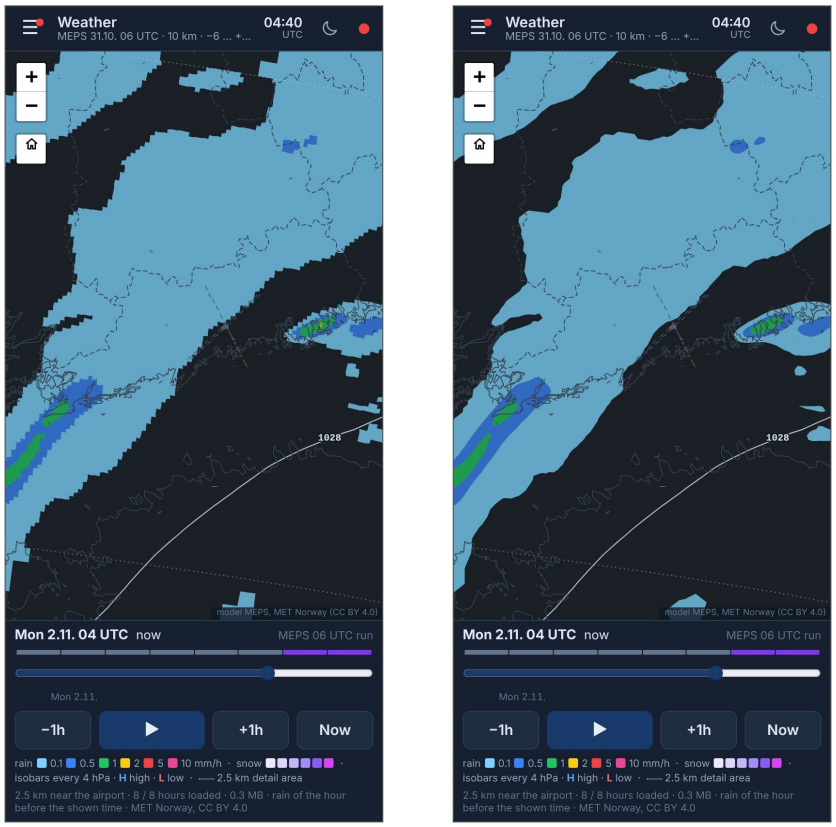


Figure 41. Smooth shading: the stored MEPS 06 UTC run of 31 October 2026, the rain of the hour to 04 UTC on 2 November (+46 h; drawn in a test, ATC map style, zoom 7). Left: Shading: Cells — every grid cell a square. Right: the default Smooth — the same values as smooth bands.

Temperature (from 1 November 2026). Temperature (2 m) adds the model's temperature near the ground, in whole degrees (Fig. 43). Colours paints it under the other layers in bands of 2 °C on a fixed scale — purple at −30, blue below freezing, cyan at 0 °C, then green, yellow, orange and red to +30 — with a thin line along **0 °C**. Isotherms draws lines instead: every 2 °C when zoomed in (every 4 °C further out), the 0 °C line thick, each labelled with its value. The lines go well with the rain, whose light blues would otherwise look like the cold colours. A tap gives the value, e.g. *2 m temperature +6 °C*; near the airport, zoomed in, the 2.5 km grid has its own values. The temperature is stored from the model runs of 1 November 2026 on; an earlier hour says *no temperature stored for this hour*. It adds about 6 kB per hour at 10 km.

Temperature numbers (from 3 November 2026 on its own). Temperature numbers: On writes the temperature in whole degrees over the whole map, inland and at sea, as bold numbers in one colour (light, dark in the Light theme) about 2 cm apart, so they stay readable over the rain and the lines (Fig. 42). It works with Temperature (2 m) Off — the numbers alone — as well as with Colours or Isotherms; before, it worked only with Colours. Zoomed in near the airport the numbers come from the 2.5 km grid. **How dense** (from the evening of 3 November 2026): Temperature numbers grid chooses how close together they are — Normal (about 2 cm apart, as before), Dense (the default, about 1.5 cm) or Very dense (about 1 cm). Inside the 2.5 km box the numbers are the box's own values on its own grid, so zoomed in they can come as close as its cells (before, they kept the spacing of the 10 km grid's points there).

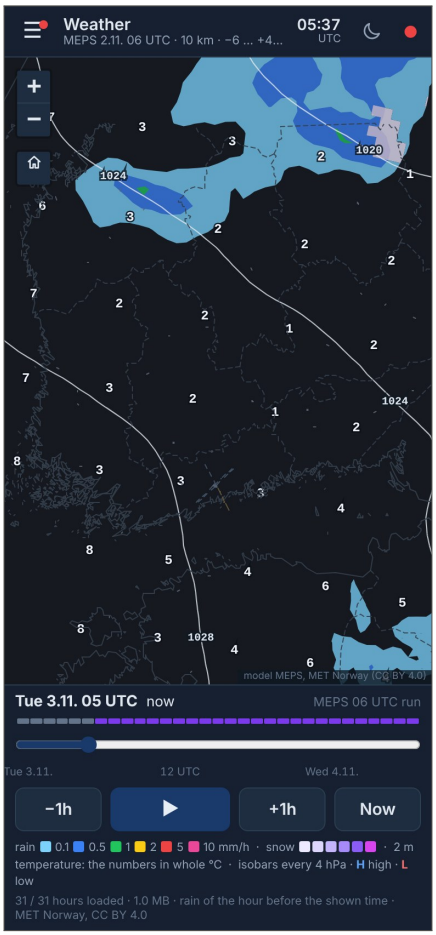


Figure 42. Temperature numbers: On with Temperature (2 m): Off, rain and isobars: the stored MEPS 06 UTC run of 2 November 2026, its forecast for 3 November 05 UTC (ATC map style, drawn in a test with a copy of the database) — +1 to +3 °C inland, +4 to +8 °C over the sea.

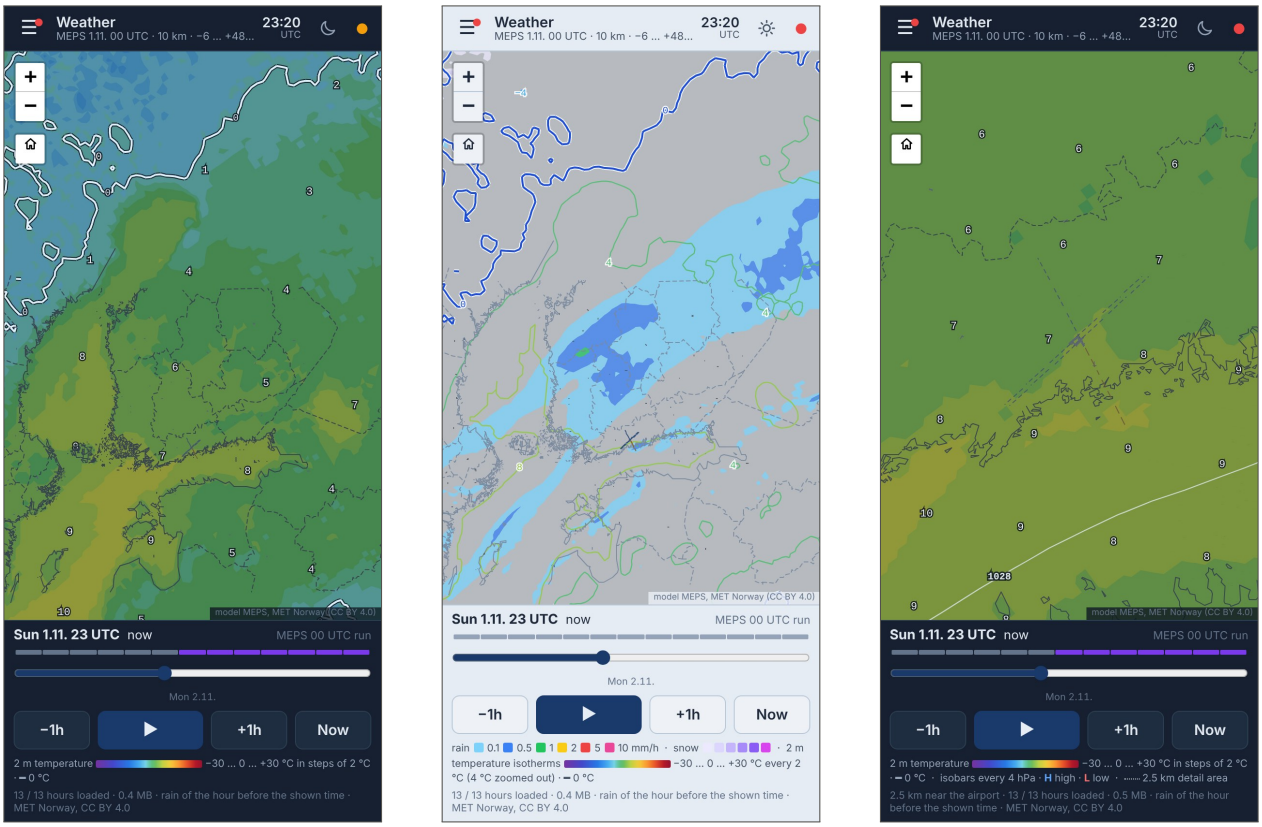
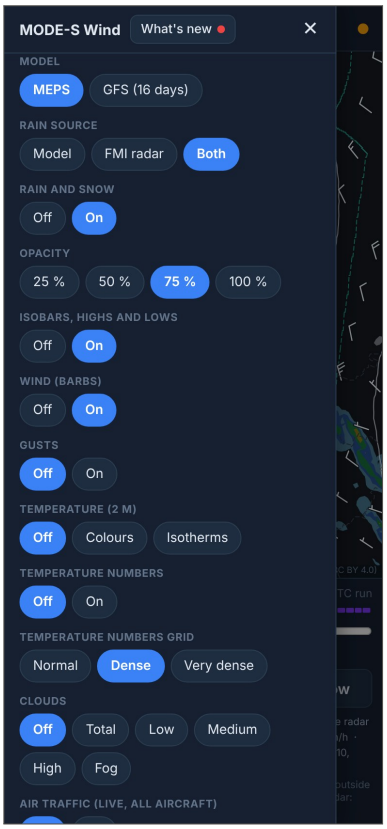


Figure 43. The 2 m temperature, the stored MEPS 00 UTC run of 1 November 2026 at 23 UTC (drawn in a test, ATC map style). Left: Colours with Temperature numbers — below freezing in the north-west, +4 to +9 °C in the south. Middle: Isotherms with the rain (Light theme). Right: zoomed in near the airport (2.5 km grid) with the isobars.



Option	Choices (*default*)
Model (with GFS)	*MEPS*, GFS (16 days)
Rain source	*Model* (the model shown), FMI radar, Both
Rain and snow	Off, *On*
Opacity	25, 50, *75*, 100 % — all the coloured layers
Isobars, highs and lows	Off, *On* — every 4 hPa, **H** / **L**
Wind (barbs)	*Off*, On — 10 m wind in knots
Gusts	*Off*, On — coloured from 25 kt
Temperature (2 m)	*Off*, Colours, Isotherms
Temperature numbers	*Off*, On — also alone
Temperature numbers grid	Normal, *Dense*, Very dense
Clouds	*Off*, Total, Low, Medium, High, Fog (grey)
Air traffic	*Off*, On — live, all aircraft
Map style	*Street*, ATC (research)
Traffic labels	Standard, *ATC*
Receivers (triangles)	Off, *On*
Forecast	24 h, *48 h* ahead (MEPS)
Grid	20 km, *10 km* (MEPS)
GFS forecast	48 h, 5 d, 10 d, *16 d* ahead
GFS grid	*0.5°*, 0.25°
Detail near EFHK	*2.5 km when zoomed in* (zoom 7+), Off
Shading	*Smooth*, Cells
Load	*All hours at once*, Near the slider (±6 h)
Speed	Slow, *Normal*, Fast (0.8 / 0.4 / 0.2 s per hour)

Figure 44. The Weather options in the menu (left, with Rain source: Both; on a server with GFS Model comes first, then Rain source and Rain and snow; the menu scrolls on to the other options) and their choices (right). The choices are remembered on the phone.

Data use. The phone loads only the area of Finland and the Baltic Sea (about 40 % of the model's grid). At 10 km each hour is about 20 kB with rain and isobars, about 50 kB with wind and gusts as well and about 70 kB with clouds too — e.g. 40 hours with rain and isobars 0.8 MB, with wind, gusts and low cloud 2.8 MB (measured). The 20 km grid takes about a third of that and draws faster on an older phone; Near the slider loads only the 13 hours around the slider and the rest when you move it. The model run is checked every 10 minutes; a new run is loaded when it arrives. The 2.5 km detail adds about 8 kB per hour with rain and isobars (about 30 kB with wind, gusts and clouds), only while zoomed in near the airport. The FMI radar adds about 10–150 kB every 4 minutes (at most about 2 MB an hour), only with Rain source FMI radar or Both. The air traffic adds about 1–3 MB an hour (more with heavy traffic), and nothing when it is off. The map program and the Weather code are loaded the first time you open the view; nothing is loaded while another view is shown.

Speed (from 3 November 2026). The map follows the slider smoothly: while you drag, it is drawn once per screen refresh with the newest hour instead of once for every step. The last four hours drawn are kept as pictures, and the hours just before and after the one shown are drawn ahead while you look, so +1h and −1h usually show the next hour at once (in a test about 0.03 s instead of 0.3–0.5 s with the temperature and low cloud on). Moving or zooming the map draws them again. The pictures and values are the same as before; the kept pictures use some of the phone's memory (about 10 MB each), and nothing more is downloaded.

Kept hours (from 12 November 2026). The hours you have loaded stay in the app's memory, sorted by model and grid. Go from MEPS to GFS and back, or from the 20 km grid to 10 km and back, and the hours are there at once — nothing is downloaded again. Turning a layer on (wind, gusts, temperature, a cloud layer) fetches only that layer for the hours already loaded; turned off and on again, it fetches nothing. Each hour is still checked against the server, so a newer model run replaces the old hour as before. The hours shown are always kept; the others are packed into a compact form and kept while they fit in about 25 MB of memory, the ones used last first — only a very large set would be fetched partly again. Closing the app or Reload empties the memory. The megabytes of the status line are those downloaded into the hours shown: when they came from memory, the number is what they cost when they were first fetched.

GFS to 16 days (from 5 November 2026, where the server has it). The first Weather option, Model, chooses between MEPS — everything above — and GFS, the global model of the US National Weather Service (NOAA/NCEP), for a longer look ahead (Fig. 45). With GFS the slider runs from 2 hours before now to its newest run's last hour, up to **16 days** ahead (GFS forecast: 48 h, 5, 10 or 16 days); the runs of 00 and 12 UTC arrive about 3½–5 hours after their start. GFS gives every hour to +120 h (5 days) and then every 3 hours, so there the buttons read −3h / +3h and the rain and snow are the mean of the 3 hours; −1d and +1d step a day. The time line adds the days (*+145 h (6 d 1 h)*), the line under the title gives the run, the grid and the range, and the map's corner says *model GFS, NOAA/NCEP*. GFS grid: 0.5° (default) or the model's own 0.25° (about 28 km north–south here). The same layers are drawn — rain and snow, isobars, wind, gusts, temperature and clouds, the radar at Now — except fog (GFS has none) and the 2.5 km detail; a tap on the map gives the **ceiling** (the lowest broken or overcast layer) instead of the cloud base. Switching keeps the hour shown when the other model has it. The data are small: the 175 hours of a run with rain and isobars about 0.3 MB, with wind barbs about 0.8 MB at 0.5°. GFS is for viewing — it is coarser than MEPS, and beyond about a week a rough guide only; the *Meteogram* (Section 13) shows one place's days from both models together.

Figure 45. The Weather view with Model: GFS (the stored GFS run of 5 November 2026 12 UTC, drawn in a test on the evening of 5 November; ATC map style, as the street map's tiles cannot be loaded in the test). Left: +1d four times — Monday 9 November 17 UTC, +96 h, isobars and wind barbs over a high. Middle: Wednesday 11 November 18 UTC (+145 h, after +120 h the buttons step 3 hours) — a low of 992 hPa over central Finland with snow on its north side, and a tap on the snow (*0.7 mm/h (mean of 3 h)*) with Meteogram here ›. Right: the top of the Weather options — Model, GFS forecast, GFS grid, and Rain source with Model: the rain of the model shown.

12 Mast

From 1 November 2026 the phone has a view of the **Kivenlahti mast** — the Finnish Meteorological Institute's 327 m mast in Espoo, 24 km south-west of the airport, which measures the temperature at eight heights from 2 to 297 m and the wind at four from 26 to 327 m every 10 minutes (FMI open data, CC BY 4.0). It shows the lowest 300 m — where the aircraft are few and the night inversions and the low-level jets form — beside what the aircraft, the MEPS model and the sounding say there (Fig. 46). One view that scrolls, refreshed every 5 minutes; the line under the title gives the mast's newest time.

From the top:

- **EFHK now** — the airport's box of the full site's Live Map: the MEPS model's weather of the hour at EFHK (wind, gust, temperature, pressure, rain, cloud, base, fog), the radar, and the mast's newest values with their difference to the model (Δ*MEPS*) and to the airport's aircraft at the same height (Δ*Mode S*) — e.g. *297 m +3.8 °C, +0.9, +0.2*: the mast 0.9 °C warmer than MEPS and 0.2 °C warmer than the aircraft. Tap a row for what it means; « makes it one line. EFHK now box: Off in the menu hides it.
- **Four tiles** — the wind at the top (327 m, about 1 200 ft MSL) with the gust; Δ*T* 297 – 2 m with the stability (violet: an inversion, warmer above; orange: unstable); the wind shear from 26 to 327 m in kt per 1 000 ft; the 2 m temperature and humidity.
- **Profile** 0–1 500 ft — the mast's temperatures (orange) and winds (barbs at the right) of one 10-minute step, with the airport's Mode S hour (white), MEPS (cyan) and the Jokioinen sounding where the menu switches them on. Tap a point or a barb for its values.
- **Boundary layer** · **temperature** and · **wind** — the Windshear page's boundary-layer charts for the Range (12 h, 24 h or 3 days) up to the Chart top (1 500, 3 000 or 5 000 ft): below the amber line (the mast's top) the mast's 10-minute steps, above it the aircraft's hourly cells — the temperature against the METAR (red warmer: an inversion), or the wind speed with barbs; the strip between them is the mast's Δ*T* top − 2 m per step. Mast in the charts: Fill (default, as described), Columns (the mast's hour means as a narrow column at the right of each hour, so the aircraft's cells below 1 200 ft can be seen beside it) or Off. Tap a chart for the cell; a tap on the mast also shows its profile of that

time (Now × returns to the newest).

- **Time series** — the mast's wind speed with gusts, the direction and the temperatures of its eight heights (the axis inverted, colder up) with the ΔT bars; the airport's Mode S value of each hour at the same height above the ground as hollow circles (Mode S hourly: Top — the 327 m wind and 297 m temperature — All or Off) and the METAR. Mast: 10 min, 1 h (a running hourly mean, closer to the aircraft's hours) or Off. Tap for the step's values and its profile.
- **Comparison with the mast** — Temperature or Wind over 7 d, 30 d or 90 d: for each height the mean difference and the RMS of the aircraft (Mode S), MEPS and the soundings against the mast, with how many were compared.

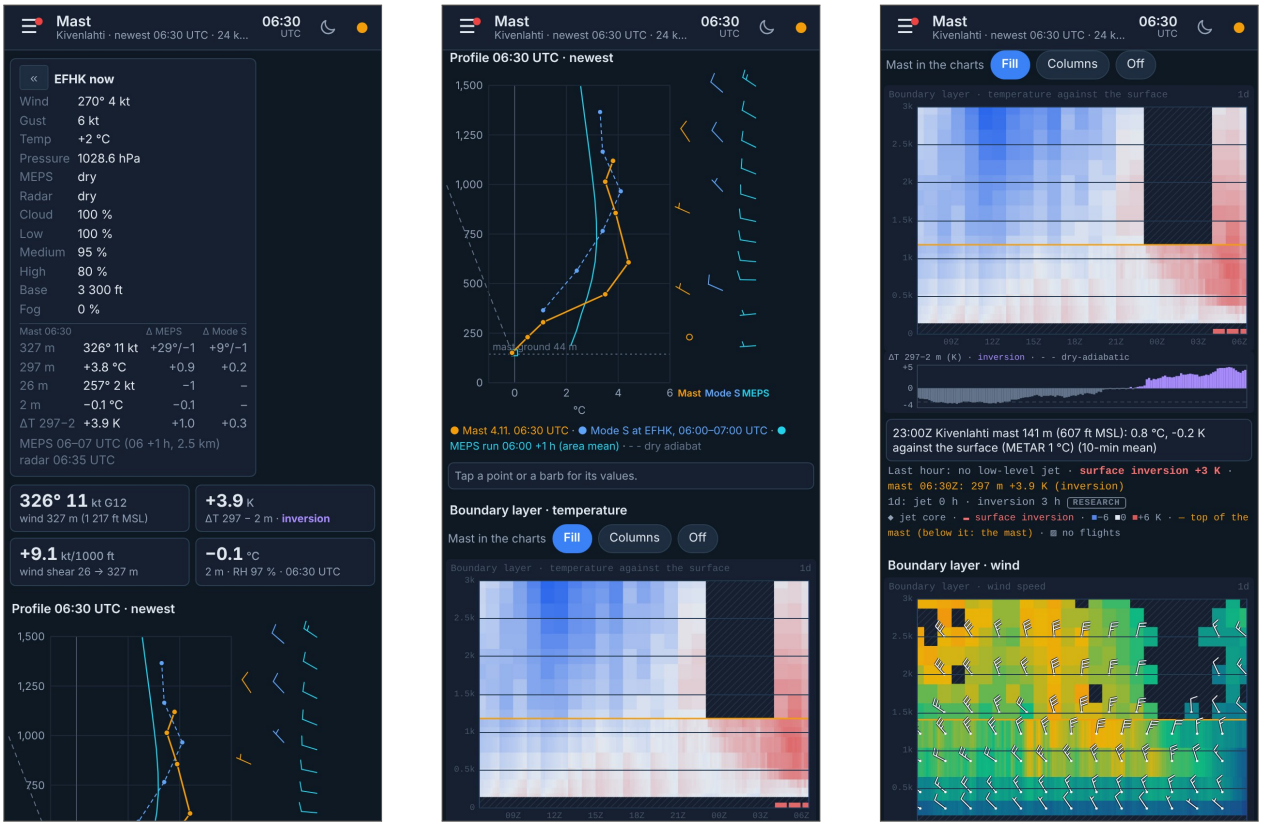


Figure 46. The Mast view at 06:30 UTC on 4 November 2026, the calm night before that day's front (the stored data, drawn in a test). Left: EFHK now — the airport's MEPS hour, the radar and the mast rows; the tiles show a ground inversion of +3.9 K from 2 to 297 m, 11 kt at the top and almost calm at 26 m. Middle: the profile — the mast (orange) about 0 °C at the ground and +4 °C at its top, the aircraft (white) and MEPS (cyan) — and the start of the boundary-layer temperature chart. Right: the temperature chart (24 h, mast Fill) after a tap, the ΔT strip turning violet in the evening, and the wind chart.

Figure 47. Further down, the same time. Left: the time series of the last 24 hours — wind, direction and the eight temperatures with the airport's Mode S hours (circles) and the ΔT bars. Middle: the comparison with the mast, temperature, 7 days — the aircraft and MEPS within a few tenths of a degree of the mast at 297 m, the aircraft colder than the mast lower down. Right: the Mast options.

Options (Mast options in the menu, Fig. 47, right): Range 12 h, *24 h* or 3 d; Chart top 1 500, *3 000* or 5 000 ft; EFHK now box Off or *On*; the switches Mode S, MEPS and Sonde in the profile (shared with the full site's Sounding page). The chips in the view (Mast in the charts, Mode S hourly, Mast, the comparison's) are remembered too.

The mast in the other views. Mast in the BL wind / temp charts in the EDR · WS · ACI · ST options and Mast in the BL wind / temp chart in the ILS approach options (Off by default) add the mast's hourly columns to those boundary-layer charts, as on the Windshear page.

Reading it. The mast stands 24 km from the runways, near the coast: on a clear calm night its ground inversion is usually like the airport's, but sea breezes, fog banks and showers can differ. Below about 1 000 ft the aircraft are few (the landings and departures only), so the mast fills the gap the Mode S data leave near the ground. This is research data, not an official observation: for flight operations use the METAR / ATIS.

13 Meteogram

From 5 November 2026 the phone has the full site's *Meteogram* (on the Weather page: Map | Meteogram) as its own view (Fig. 48): the model weather of **one place** over the days, every panel on the same time axis in UTC. It is the same drawing as on the full site, made smaller for a phone.

From the top: the chips, a line with the place and the model's grid cells, the values of the hour you tapped, then the panels:

- **Temperature** — the 2 m temperature in °C, red above and blue below 0 °C, with each day's highest and lowest.
- **Precipitation** — the rain (blue) and snow (violet) of each hour as bars in mm per hour, with the day totals in mm at the top (after +120 h GFS gives the mean of its 3-hour step).
- **Wind** — the 10 m wind (teal) and the gust (orange, dashed) in kt, with wind barbs along the top.
- **Pressure** — the sea-level pressure in hPa (off by default; switch it on in the menu).
- **Clouds** — rows of the high, medium and low cloud cover (and fog with MEPS), darker the more cover, with the cloud base (MEPS) or the ceiling (GFS) in feet along the bottom where there is cloud below about 20 000 ft.

The night is shaded (from the sun at the place), the forecast part is tinted and the red line is now; the days are named at the top. **Tap or drag** on the chart: a line marks the hour and the lines above the chart give all its values from each model, e.g. *Fri 6.11. 18 UTC · MEPS +36 h (2.5 km): +1.0 °C · wind 304° 7 kt (4 m/s) · gust 12 kt · dry …*

Figure 48. The Meteogram at EFHK with the stored MEPS run of 5 November 2026 06 UTC and the GFS run of 12 UTC (drawn in a test at 17:22 UTC). From the left: Both, 5 d, with a tap on Friday 18 UTC — MEPS bold to its last hour, GFS thin and dashed on to 10 November; Both, 16 d in the Light theme with Pressure on and a tap on 12 November; MEPS alone — the past 6 hours and its forecast, the 16 d and 5 d chips greyed; the Meteogram options in the menu.

Chips at the top:

- Place — EFHK (default); the **receiver sites** RX-01, RX-02 … — the same receivers as the triangles on the Weather map, at their position to 0.01° (about 1 km; the line under the chips gives the receiver's name); and Tapped — the place you last opened from the Weather view: tap the map there and Meteogram here › in its card (Section 11). The tapped place is kept until you open another.
- Model — MEPS (MET Norway's model: the past 6 hours and its forecast, about 2.5 days), GFS (NOAA's global model, from now up to 16 days ahead) or Both (default): GFS thin and dashed over its whole range and MEPS bold over its own hours, so you see where the two agree. The barbs, the clouds and the day totals come from MEPS where it has the hour. Only on a server that has the GFS forecast; otherwise MEPS only.
- Range — 2 d, 5 d (default), 7 d, 10 d or 16 d ahead with GFS and Both (7 and 10 days from 6 November 2026); MEPS has only its own forecast.

Options (Meteogram options in the menu, Fig. 48, right, and Fig. 50): a switch per panel — Temperature, Precipitation, Wind, Pressure (off by default) and Clouds — 850 hPa temperature and, from 7 November 2026, the further options below (all off by default) and, where the server has GFS, Typical error (on by default, below). All choices are remembered on the phone. The view is refreshed every 10 minutes while it is shown; a new model run comes in on its own. Each update is a few kB.

850 hPa temperature (from 7 November 2026; 850 hPa temperature in the Meteogram options, off by default). The temperature at 850 hPa, about 1 500 m (4 800 ft) up, is the usual level for telling the air mass: whether falling snow melts on the way down, cold or warm air coming in. Switched on, the temperature panel shows it on the same scale in fuchsia: the models' line (MEPS bold, GFS thin and dashed with Both) and, for the past hours near the receiver (within 150 km), **dots measured by the aircraft themselves** — MODE-S Wind's own hourly profile of the area, the temperatures the aircraft reported between 4 000 and 6 000 ft of pressure altitude taken to 850 hPa. So you see how the model's air mass compares with what the aircraft actually flew through. A tap adds *850 hPa −2.5 °C* to each model's values and *Measured 850 hPa −1.8 °C (15–16 UTC, 51 values)* for an hour with a dot. Model hours stored before 7 November 2026 have no 850 hPa value, so the line starts with the next run.

More to switch on (from 7 November 2026; each a switch in the Meteogram options, off by default — Fig. 50). The model's weather next to what was actually measured, and what next summer's thunderstorms need:

- Freezing level — a panel of the 0 °C level in ft above sea level: the models' line (cyan; MEPS bold, GFS dashed) and, for the past hours near the receiver, the level **measured** each hour (dots) — from the ground up: the EFHK METAR, the airport's climb-outs and landings, then the area's aircraft. A triangle at the ground means the whole column was below 0 °C. A thin layer that only just touches 0 °C aloft is not counted (it must reach +0.5 °C).
- Dew point & visibility — the dew point as a dotted green line in the temperature panel; a yellow strip at the panel's foot where the temperature is within 2 °C of it (darker within 1 °C): mist or fog is possible; and a *Vis* row under the clouds

where the visibility is below 5 km (darker below 1.5 km and 500 m).

- Thunderstorm (CAPE) — a thin strip coloured by the model's convective energy: yellow from 100, orange from 500, red from 1 000 and violet from 2 500 J/kg. Mostly empty in winter; for the summer's showers and thunderstorms.
- 850 hPa wind — a second row of barbs under the surface ones: the model's wind at 850 hPa (fuchsia) and, in the past hours, the wind the aircraft measured there (white; dark in the Light theme).
- Landings' winds — in the wind panel, the wind the landing aircraft met at 1 000 ft (squares) and 2 000 ft (triangles) above their runway, the median of each hour's landings — often much stronger than the 10 m wind. Near EFHK only.
- METAR — the EFHK observations as dots: the temperature (and with Dew point the dew point, open green), the mean wind (open) and the gust, the QNH. Near EFHK only.
- Sun times — the sunrise and sunset (UTC) at the place for each whole day. At 2 d they are on the day's label; at 5 d on a row of their own under the labels, with a sun sign at the left edge; from 7 d on two short rows, the sunrise above the sunset, and at 16 d — where a day is narrower than the text — at every second day (Figure 49; from 11 November 2026, until then only on the shortest range). The times change by only a few minutes from one day to the next. With the phone on its side the days are wider and the fuller forms are used.

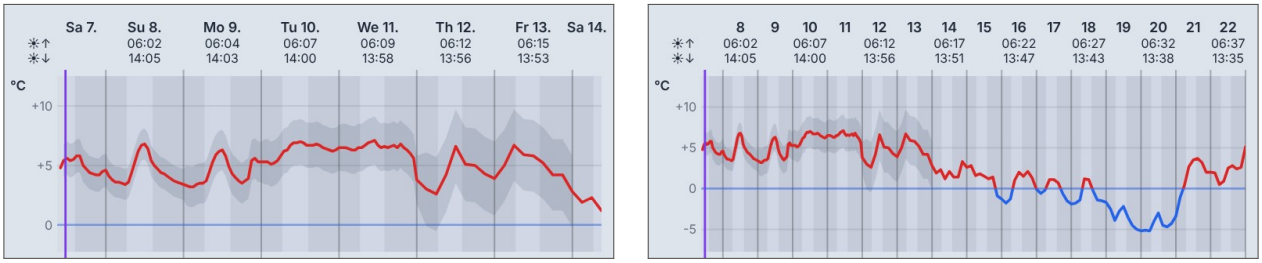


Figure 49. Sun times on the long ranges (the top of the chart; the live server's GFS run of 7 November 2026 00 UTC at EFHK, drawn in a test). Left: 7 d, the sunrise and sunset under each day. Right: 16 d, at every second day.

A tap gives all of it for that hour: *METAR 12:50: +5 / −1 °C · 310° 9 kt · Q1026 · vis ≥10 km · no ceiling*, *Measured 850 hPa −3.8 °C … · 850 hPa wind 003° 31 kt · freezing level 1 900 ft · landings 1 000 ft 327°/16 kt, 2 000 ft 348°/24 kt (20 landings)*, and each model's dew point, visibility, freezing level, CAPE and 850 hPa wind. Model hours stored before 7 November 2026 have none of the new model values; the measured ones go back as far as the chart.

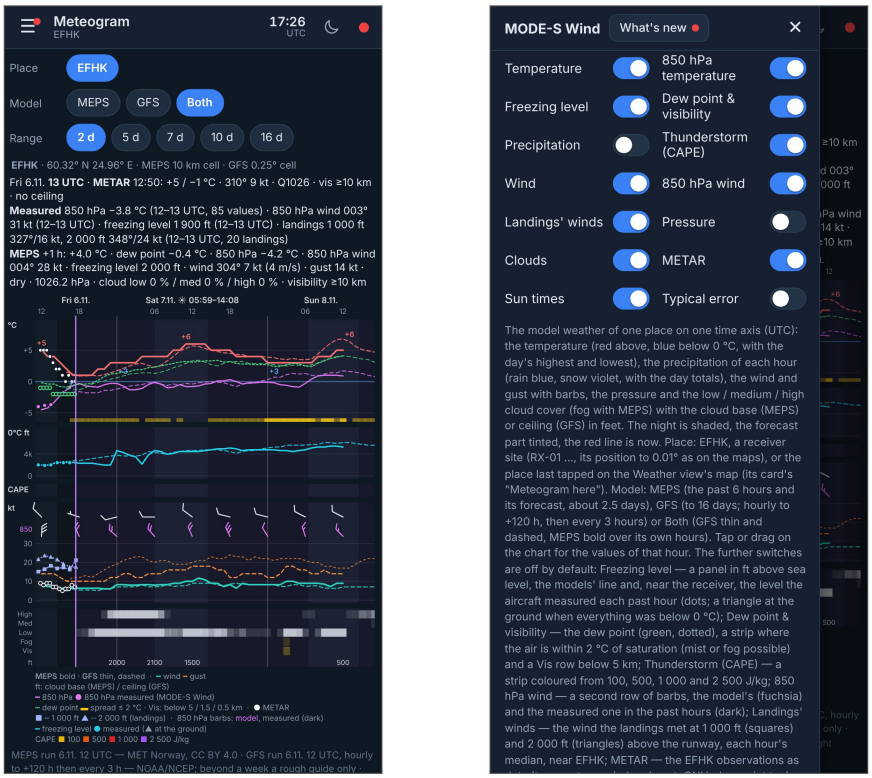


Figure 50. Left: the Meteogram at EFHK with all the new options on (Precipitation and Pressure off), Both, 2 d, with the real MEPS and GFS runs of 6 November 2026 12 UTC and the stored data of that day (drawn in a test at 17:26 UTC), and a tap on 13 UTC: the METAR, what the aircraft measured and MEPS's values. In the past hours the METAR temperatures (white) and dew points (open green), the measured 850 hPa temperatures (fuchsia) and freezing levels (cyan, about 2 000 ft — MEPS forecast 2 000–2 500 ft), the landings' winds at 1 000 and 2 000 ft (15–24 kt against a 10 m wind of 7–9 kt) and the measured 850 hPa barbs; ahead, the dew point close to the temperature (the yellow strip) and the 850 hPa wind from the north. *Sat 7.11.* with the sun symbol and *05:59–14:08*: the sunrise and sunset. Right: the Meteogram options in the menu.

Typical error (from 6 November 2026, where the server has GFS). How far can the forecast be trusted? The server keeps each run's forecast for the airport and checks it against what the EFHK METAR then reported. The meteogram draws a light **band** around the temperature and wind-speed lines of the forecast: at that lead time, 8 in 10 of the checked hours of the last 30 days were within it — narrow on the first day, wider further out. With Both the band follows MEPS over its own hours and GFS after them (Fig. 51); until enough MEPS runs have been checked, its width over the MEPS hours is the GFS error of the same lead day. A tap adds it to the values (*+2.8 °C (±1.9)*). The line under the chart sums it up: the temperature error on days 1, 3, 5 and 10 and *useful to about day N* — up to which day GFS has been clearly better than the usual weather of the season (the EFHK METARs of the same time of year since 2018). The band and the line appear for a lead day once enough runs have been checked; the full site's meteogram has the whole table. It is measured at the airport; for another place it is a guide.

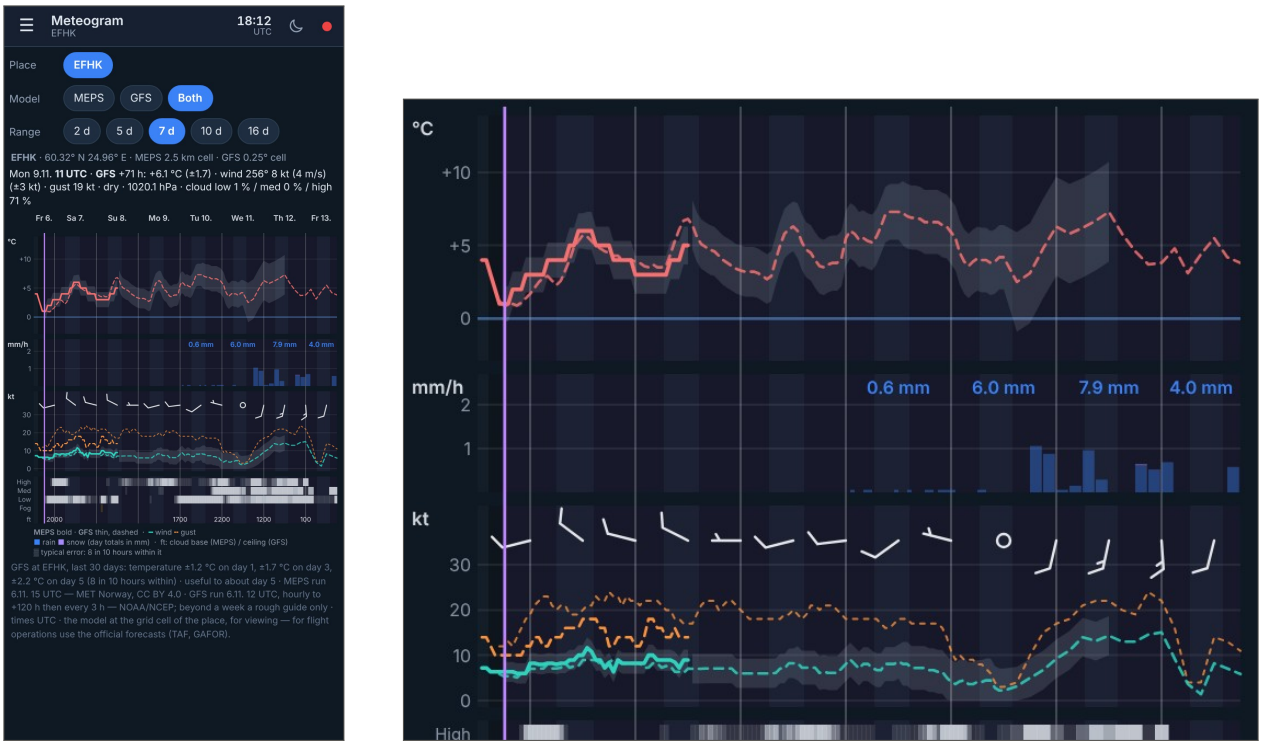


Figure 51. The typical-error band with real data: the live server's forecast and check of 6 November 2026 at 18:12 UTC (the MEPS run of 15 UTC, the GFS run of 12 UTC, and the check of up to 20 GFS runs against the EFHK METARs of the last 30 days), drawn in a test; Both, 7 d, dark theme. Left: the phone, with a tap on Monday 9 November 11 UTC — *GFS +71 h: +6.1 °C (±1.7) · wind 256° 8 kt (4 m/s) (±3 kt)*; the line under the chart: *±1.2 °C on day 1, ±1.7 °C on day 3, ±2.2 °C on day 5 … useful to about day 5*. Right: the temperature and wind panels enlarged — the grey band around MEPS (bold) over its hours and around GFS (dashed) after them, wider day by day. At that time the band reached to day 6 (the GFS run's +144 h, Thursday 12 November 12 UTC); from day 7 on fewer than 60 hours (and 6 runs) had been checked, so there is no band there yet. Only one MEPS run had been checked, so its hours use the GFS width.

Reading it. The values are those of the model's **grid cell** of the place: MEPS's 2.5 km cell near the airport (else its 10 km cell), GFS's 0.25° cell (about 28 by 14 km) — not a point forecast: a coast, a lake or a hill nearby can make the place differ. MEPS is the better guide for the next two days; GFS's coarser grid smooths the wind and the showers, and beyond about a week it is a rough guide only. The meteogram is for viewing — for flight operations use the official forecasts (TAF, GAFOR).

UTC or Finnish time (from 11 November 2026). The Meteogram options have a row Time: UTC (the default) or Finnish time — UTC + 2 h, in summer time + 3 h. With Finnish time the days of the chart change at Finnish midnight, the day's highest and lowest temperature and precipitation total are counted over the Finnish day (so they can differ a little from the UTC view), and the sun times and the values of a tapped hour are given in Finnish time: *Mon 9.11. 05 EET* for 03 UTC (*EEST* in summer time). The line under the chart then says *times in Finnish time (the runs' names UTC)*: the names of the model runs stay in UTC, as does the clock of the top bar. A chart that crosses the change of the clocks has one day of 23 or 25 hours. For users with the FIN right the row is Aika: UTC / Suomen aika, and the hour reads *klo 05* (Figure 52).

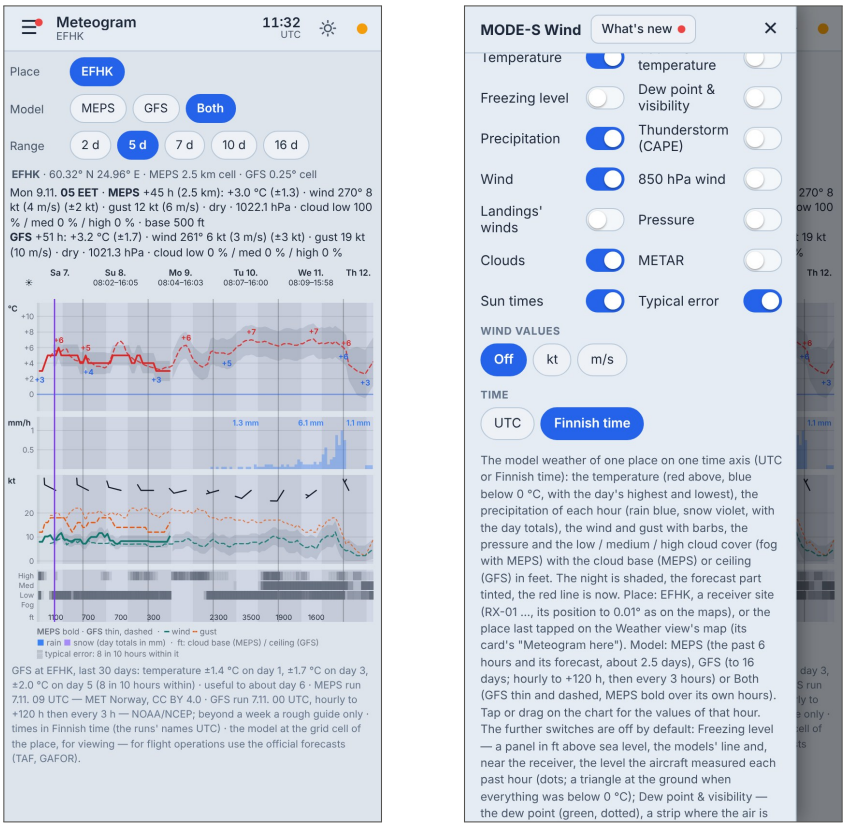


Figure 52. Time: Finnish time (the live server's runs of 7 November 2026, GFS 00 UTC and MEPS 09 UTC, at EFHK; drawn in a test). Left: Both, 5 d, Sun times on, with a tap on Monday 05 EET — the days change at Finnish midnight and the sun times are Finnish time. Right: the Time row in the Meteogram options.

The Meteogram guide. The whole meteogram — each panel, the two models, what the aircraft measure, the values of one hour and the typical error, with real data — is explained in its own user guide, *Meteograms* (from 7 November 2026), also in Finnish as *Meteogrammit (suomeksi)*. Open it from User guides in the menu's General section (Section 2); on the full site the ? next to the meteogram's controls opens it (the Finnish one for users with the FIN right).

14 GNSS (Full view)

From 3 November 2026 the full site's *GPS Quality* page has a short version on the phone (Fig. 53). It shows how the satellite navigation (GPS / GNSS) of the aircraft around Helsinki is doing, from what the aircraft themselves report in their ADS-B: the accuracy and integrity of their position (NACp, NIC), positions that freeze or stop, and checks against the multilateration (MLAT) position and the barometric altitude.

> **Full view only.** The GNSS view — like the GPS pages of the full site — is shown only to users with the *Full view* right, given by the maintenance user. Without it the GNSS tile is not in the menu and the GPS link is not on the full site; this guide stays open to everyone. Ask the author if you need the right.

The full site's GPS pages — *Quality*, *Episodes* and *Aircraft*, with the overview of all traffic by day, time of day, direction, altitude, type and airline — have their own user guide, *The GPS pages* (from 8 November 2026), in User guides.

From the top:

- **Now** — the number of aircraft with degraded GPS at this moment, in the chosen zone, with the time. The bar at the left is **green** when there are none, **amber** when some are degraded and **red** when at least one is *severe*: its position is unusable (NACp 0, NIC 0, or no own position at all).
- **Four figures of the last 24 hours** — the degradation *events* (one per aircraft and signal each 5-second check), the *aircraft affected*, the *severe* aircraft-hours and the *peak hour*.
- **Degraded aircraft per hour** — a bar per hour over the Window (24 h, 48 h or 7 days): red the severe aircraft, amber the milder ones. Tap a bar for the hour: how many of the aircraft heard were degraded, the severe ones and those with a spoofing indicator, e.g. *07:00–08:00 UTC · 27 of 35 aircraft degraded (77 %) · severe 26 · spoofing indicator 0*. With 1 m or 3 m the chart is **degraded aircraft-hours per day** (Fig. 54, left): each aircraft counted once in every hour

it was heard, so a day's bar sums its 24 hours; days before 9 October 2026, when the severity was not yet recorded, are tan (*Not split*). Tap a bar for the day, e.g. *Tue 27.10. · 699 of 1 085 aircraft-hours degraded (64 %) · severe 655 · 74 738 events*.

- **FL band heatmap** (optional, from 3 November 2026) — the GPS Quality page's heatmap: a row per flight level band (FL010–030 at the top, FL300+ at the bottom) and a column per day (Fig. 54). Days: 14 d or 1 m. Cells: Events — the degradation events, each row coloured against its own busiest day; or % aircraft — the share of the band's aircraft-hours that were degraded, one scale for all rows (0, under 10, 25, 50 and 50 % or more; recorded from 6 October 2026). The cells are colour only, without numbers; tap a cell for the day and band: its events, the degraded and severe aircraft-hours and the mix of the signals (*NACp 48 % · ADS-B 47 % · NIC 5 %*). Switch it on in the menu (FL band heatmap: On).
- **Live — degraded GPS** — the aircraft now: callsign, flight level, NACp, NIC and the signals — `NACp` or `NIC` low, `Freeze` (the position stands still while the aircraft flies), `Gap` (no position for a while), `ADS-B` (its own position lost). A severe aircraft is marked red.
- **Spoofing indicators** of the window — the aircraft flagged by each check: *ADS-B / MLAT mismatch* (its own position far from where the receivers' timing places it), *GNSS–baro at limit* and *GNSS–baro jump* (the difference between its satellite and barometric altitude at its largest value, or jumping), and *position jump* (further than its speed allows). `2+` counts the aircraft with two or more *independent* indicators — much stronger evidence than one, as a single MLAT mismatch is more often an MLAT error. Below, the 12 newest flagged aircraft with the time, callsign, level and indicators; a gold mark at the left means it was flagged while reporting normal GPS quality, the strong spoofing case.

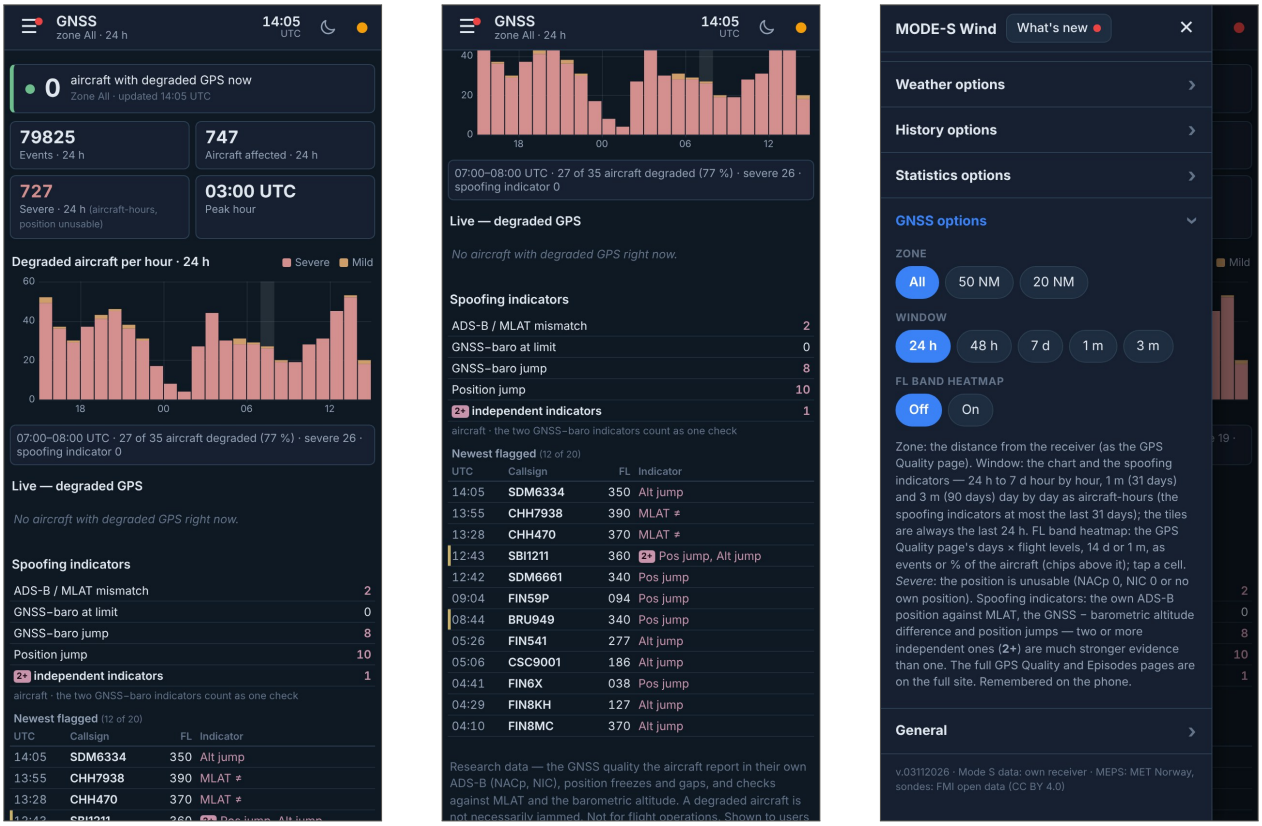


Figure 53. GNSS, zone All, 24 h (the stored data of 2 November 2026 shown in a test as at 14:05 UTC, without a live feed — so the live list is empty). Left: the top — no aircraft degraded at that moment, the 24-hour figures and the degraded aircraft per hour with the read-out of a tap on 07–08 UTC. Middle: scrolled down — the spoofing indicators and the newest flagged aircraft (one with `2+`). Right: the GNSS options (Window up to 3 months, the FL band heatmap switch).

Options (Fig. 53, right): Zone — All aircraft heard, or only those within 50 NM or 20 NM of the airport, as on the GPS Quality page; Window — 24 h (default), 48 h or 7 d hour by hour, 1 m (31 days) or 3 m (90 days) day by day, for the chart and the spoofing indicators (these at most the last 31 days — they are recorded from 9 October 2026; the four figures are always the last 24 hours); FL band heatmap — Off (default) or On. The choices, also the heatmap's chips, are remembered on the phone.

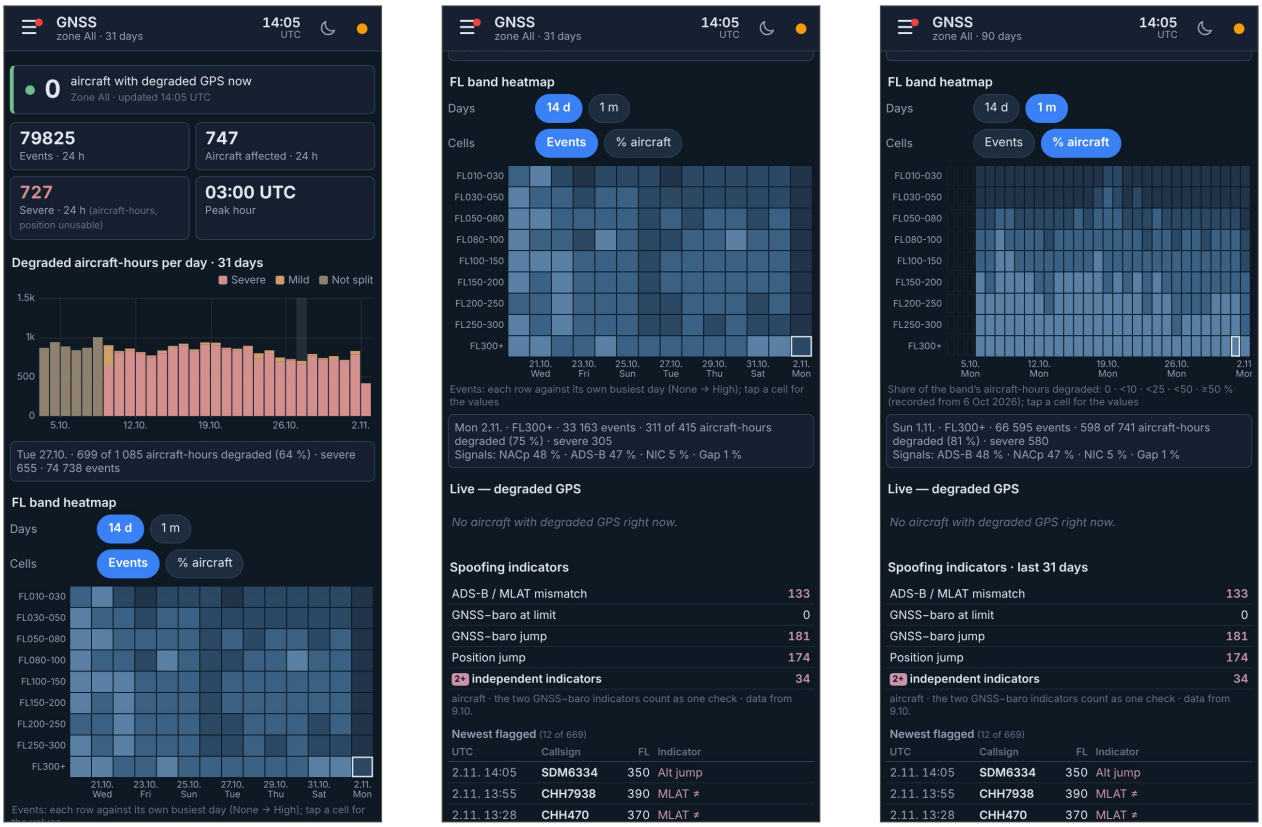


Figure 54. GNSS with 1 m and the FL band heatmap on (the stored data of 2 November 2026 shown in a test as at 14:05 UTC). Left: degraded aircraft-hours per day over 31 days — the first days tan (no severity split before 9 October), a tap on 27 October. Middle: the heatmap, 14 days, Events, with a tap on FL300+ on 2 November. Right: Window: 3 m, the heatmap with 1 m and % aircraft; the spoofing indicators say *last 31 days*.

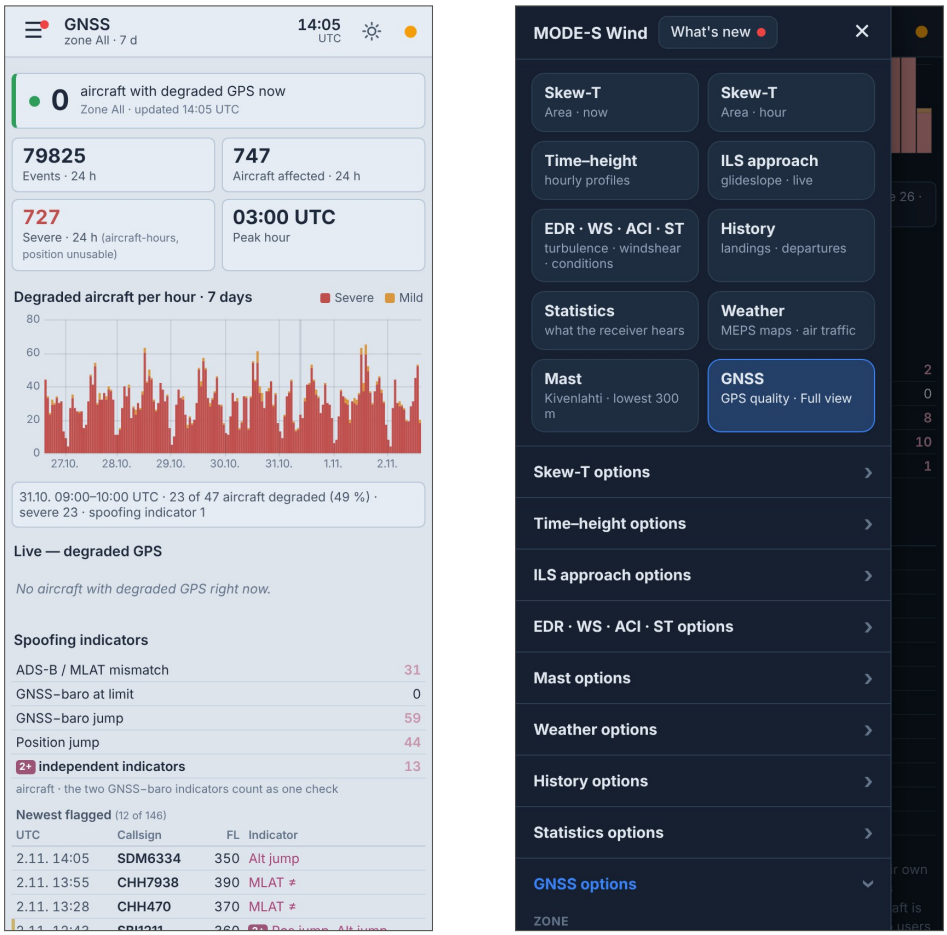


Figure 55. Left: Window: 7 d in the Light theme — the week to 2 November 2026 14 UTC hour by hour, with a tap on 31 October 09–10 UTC (stored data, as in Fig. 53). Right: the menu with the ninth tile, GNSS · GPS quality · Full view.

Reading it. A degraded aircraft is not necessarily jammed: an old or faulty receiver on board, a turn at the edge of coverage or a single bad report can show up too. GNSS jamming in this region typically shows as many aircraft degraded at the same time, mostly at cruise levels and often for hours; the hourly chart shows such periods at a glance. The weather of MODE-S Wind is protected from it: the heights of the pressure surfaces use only aircraft with good GNSS outside their degradation episodes, and MLAT keeps aircraft placed when jamming takes away their own positions. This is research data, not for flight operations. The full site's *GPS* pages (Quality and Episodes) have the heatmap of the flight levels, the 10-minute view and each degradation episode on a map.

Updates and data. The figures and the live list are updated every 30 seconds while the view is shown (about 15 kB each; about 45 kB with the FL band heatmap), the chart and the spoofing indicators every 2 minutes (about 8 kB for 24 h, 50 kB for 7 days or more).

15 Pilot (Pilot right)

From 5 November 2026 the phone has a view for pilots: what the aircraft *before you* measured at Helsinki-Vantaa — by runway and by height — for the briefing on the ground, the turnaround and the debrief after the flight (Fig. 56). Everything comes from the aircraft's own Mode S replies, with the MEPS model, the Kivenlahti mast and the METAR where they help.

Research only, and the Pilot right only. The Pilot view is not an official briefing and must not be used for flight operations: use the ATIS / METAR, TAF, NOTAMs and the official weather briefing. It is shown only to users with the *Pilot* right, given by the maintenance user on its own (a pilot does not need Full view). Without it the Pilot tile is not in the menu.

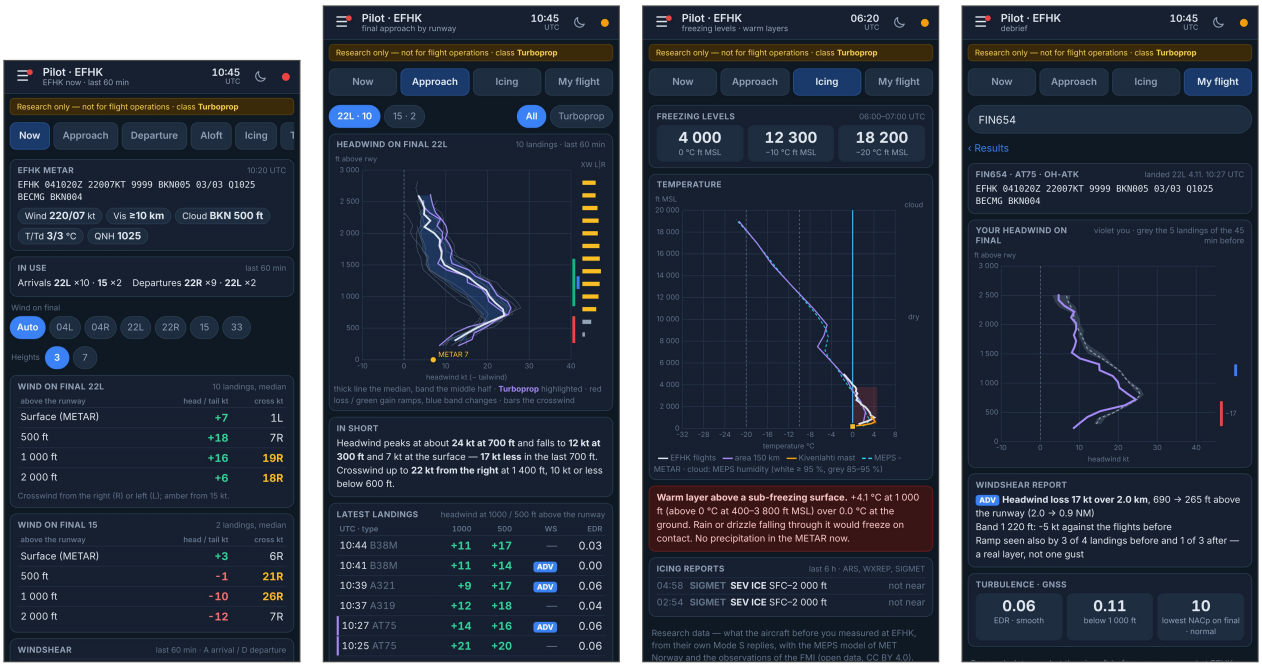


Figure 56. The Pilot view with the stored data of 4 November 2026 (drawn in a test). From the left: Now at 10:45 UTC — the METAR, the runways in use, the Wind on final bar on Auto, the wind on final of 22L and 15 and the windshear reports of a frontal passage; Approach — the headwind on final of the ten landings on 22L, the two ATRs highlighted; Icing at 06:20 UTC — a warm layer of up to +4 °C above a ground at 0 °C; My flight — an ATR's landing at 10:27 UTC with its 17-kt headwind loss.

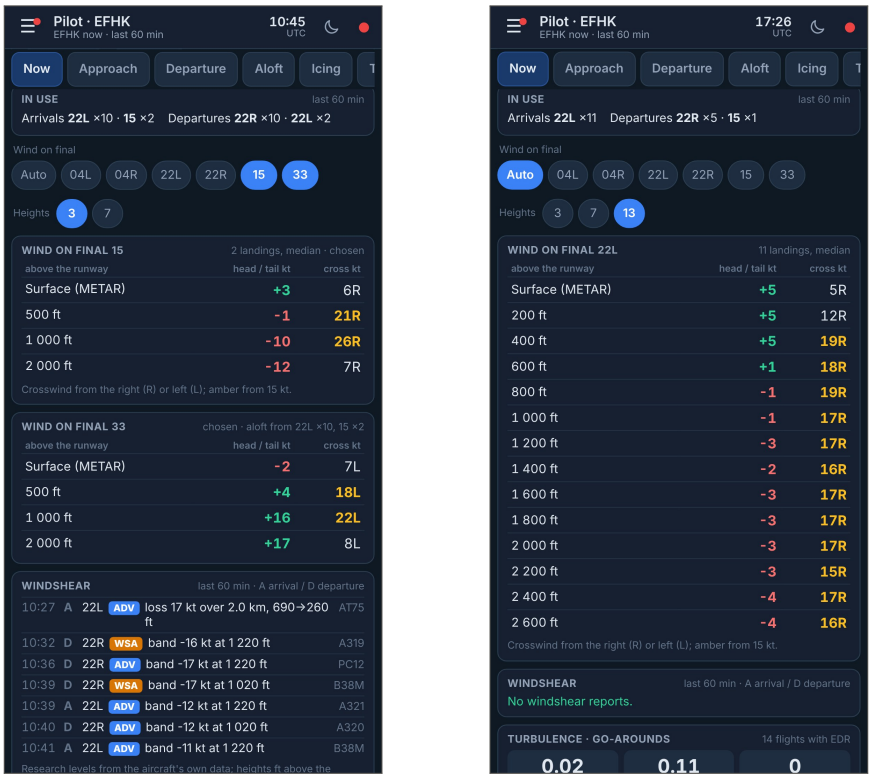


Figure 57. Left: Now with the runways chosen by hand (the same time as Fig. 56): 15 and 33. Runway 15 had two landings of its own — a tailwind of 10 kt and 26 kt from the right at 1 000 ft; 33 had none, so its winds aloft come from all twelve landings, turned to its heading: +16 kt head and 22 kt from the left at 1 000 ft. Right: Heights 13 (from 7 November 2026) with the stored data of 6 November 2026 at 17:26 UTC (drawn in a test): the wind on final of the eleven landings on 22L every 200 ft — a 12-kt crosswind from the right at 200 ft rising to 19 kt at 400 ft, and the light headwind turning into a tailwind of 1–4 kt from 800 ft up.

Figure 58. Now with a surface inversion (from 8 November 2026), the stored data of 6 November 2026 at 21:20 UTC (drawn in a test): +4.3 K on a calm, clear night, the warmest air (+3.7 °C over −1 °C at the surface) already at about 400 ft above the airport — the top as it is given from 10 November 2026 —, the wind turning from 260/06 at the surface to 321/20 kt at 800 ft; it had lasted since 15 UTC, strongest (+6.1 K) at 20 UTC. No inversion aloft in the bands of that hour, which reached 2 800 ft.

Figure 59. Now with an *inversion aloft* (from 9 November 2026), the stored data of 7 November 2026 at 05:40 UTC (drawn in a test): no surface inversion, but +4.0 K from about 2 600 to 4 200 ft above the airport — −1.9 °C at its base and +2.1 °C at its top, a warm layer above 0 °C over colder air, hence the icing note; the cloud of the METAR (BKN 2 100 ft) lies just under it; the wind hardly changes across it (308/26 below, 315/23 kt above). It had been there since 04 UTC. The grey line tells how high the hour's temperatures reached (here 4 800 ft; from 10 November 2026 up to 10 000 ft MSL).

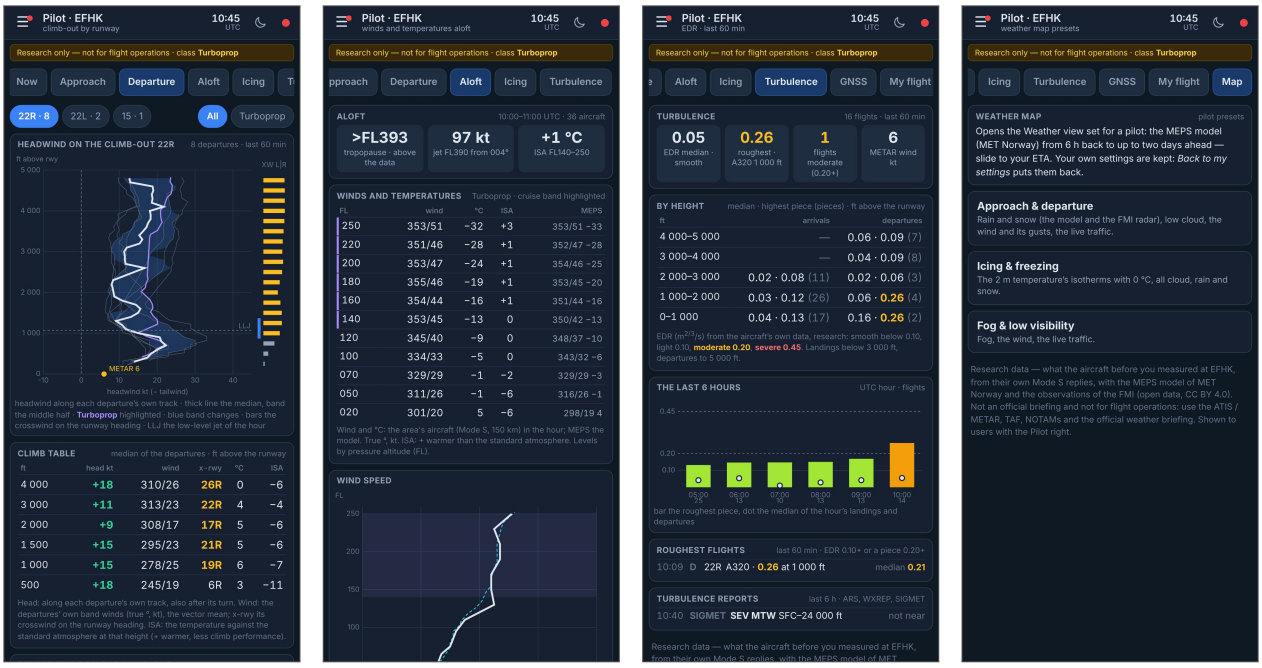


Figure 60. The five tabs added in the evening of 5 November 2026, with the stored data of 4 November 2026 at 10:45 UTC (drawn in a test). From the left: Departure — the climb-outs of 22R, the headwind falling off above the low-level jet at about 1 100 ft, a strong crosswind from the right; Aloft for a turboprop — the north wind of the front, the cruise band FL140–250 highlighted, temperatures close to the standard atmosphere; Turbulence — mostly smooth with light pieces below 3 000 ft, one A320 with a moderate piece at 1 000 ft; Map — the three Weather presets.

The frame. An amber banner *Research only — not for flight operations* with your aircraft class, and nine tabs in a row that scrolls sideways (the last five from the evening of 5 November 2026, Fig. 60):

- **Now** — the newest EFHK METAR, raw and in big chips (wind, visibility, cloud, temperature / dew point, QNH); from 8 November 2026 under it the **boundary layer** (Fig. 58): a **surface inversion** when the aircraft of the last hour found the air at 600–1 000 ft at least 0.5 K warmer than the METAR — its strength (*weak* under 2 K, *moderate*, *strong* from 5 K) and its top above the airport (from 10 November 2026 the top of the warmest layer, where the temperature stops rising; until then the highest level still clearly warmer than the ground, which read about 1 000 ft too high when the air above cooled slowly), the temperature at the surface and the warmest air above it, the METAR wind against the aircraft's wind just above the top, and since when it has lasted with its strength hour by hour — or *No surface inversion*; from 9 November 2026 also an **inversion aloft** (Fig. 59) — a layer higher up, above the surface-inversion zone, in which the temperatures of the landings and departures rise by at least 1 K over at least 400 ft (to 5 000 ft MSL and, from 10 November 2026, to 10 000 ft when the hour's climb-outs — now followed that high — reach it; above about 3 000 ft from the climb-outs of jets only, so often empty at night; a grey line gives the top of the hour's data) — with its base and top above the airport (*to at least* when the warming reaches the top of the data), its strength (*weak* under 2 K, *moderate*, *strong* from 4 K), the temperature at its base and top, the wind below and above it with their difference, since when it has lasted and a note: haze and low cloud are often trapped under it and the air above it is smoother, but the wind can change across it; a warm layer above 0 °C over colder air below 0 °C adds an icing note — or *No inversion aloft up to … ft*; and a **low-level jet** when there is one; the runways in use by the arrivals and departures of the window; the **wind on final** of the busiest arrival runways — the median headwind (green, red for a tailwind) and crosswind (`R` from the right, `L` from the left, amber from 15 kt) at the surface (the METAR) and at 500, 1 000 and 2 000 ft above the runway, from the band winds of the window's landings. From 6 November 2026 the bar Wind on final above these cards lets you **choose the runways** yourself — Auto (the runways in use, as above) or up to three of `04L … 33`, e.g. when your turboprop is given 15 while the jets land on 22L (Fig. 57). A chosen runway with landings in the window uses its own landings (*2 landings, median · chosen*); one without uses the band winds of all the window's landings, turned to its heading and counted from its threshold (*chosen · aloft from 22L ×10, 15 ×2*); the surface wind always comes from the METAR. The choice is kept on the phone (and on the Pilot page); Auto returns to the runways in use. Heights 3 / 7 / 13 beside it gives the wind at three heights (500, 1 000 and 2 000 ft), at seven: 300, 500, 750, 1 000, 1 500, 2 000 and 2 500 ft above the runway, or — from 7 November 2026 — at thirteen: every 200 ft from 200 to 2 600 ft, the spacing of the stored band winds (Fig. 57, right). The landings' band winds reach about 2 700 ft, so the top rows may rest on fewer of them; the lowest one, 200 ft, comes from the band nearest to it, as no band below the threshold is ever flown. Then the **windshear** reports of arrivals (`A`) and departures (`D`) with their levels and heights above the runway (a dashed `WSA?` is unverified: the aircraft's GNSS was degraded); and

the turbulence (EDR median and highest) with the go-arounds.

- **Approach** — the headwind on final of every landing on one runway in the window, in **ft above the runway** (EFHK 179 ft): each flight a thin line, the median thick with the middle half as a band, and your aircraft class in violet. All or your class only (then with the class's own median). The yellow dot is the METAR's surface headwind; at the right edge the reports' headwind *loss* (red) and *gain* (green) ramps and band changes (blue), and the crosswind per 200 ft as bars. *In short* puts it in words — where the headwind peaks and how much of it is left at the surface, and the crosswind above and below 600 ft — and *Latest landings* lists the newest with their headwind at 1 000 and 500 ft, level and EDR. The runway buttons switch the runway.
- **Departure** — the climb-outs of one runway in the window, as Approach does for the landings: the headwind of every departure up to 5 000 ft above the runway — along its own track, also after its turn — with the median and the middle half, your class in violet (All or your class), the METAR's surface headwind, band changes (blue), the crosswind on the runway heading as bars at the right, and the hour's **low-level jet** as a dashed line `LLJ`. The **climb table** gives at 500, 1 000, 1 500, 2 000, 3 000 and 4 000 ft above the runway the median headwind, the wind (true degrees and kt, from the departures' own measurements), its crosswind on the runway (`x-rwy`), the temperature and its **ISA deviation** (+ warmer than the standard atmosphere: less climb performance). Below, the temperature on the climb-out with the ISA line, the boundary layer of the hour in words (a low-level jet: the headwind or tailwind changes through its height; a surface inversion: warmer air above, usually smooth below) and the latest departures with their headwind at 1 000 and 3 000 ft, level and EDR.
- **Aloft** — the winds and temperatures by flight level, like a WINTEM chart: from the aircraft within 150 km in the hour, with the MEPS model beside (and the Jokioinen sounding when one of the last 15 hours is stored). The levels follow your class — FL020–250 for a turboprop, FL050–390 for the regional jets and narrowbodies, FL050–450 for the widebodies — and the cruise band is highlighted (turboprop FL140–250, regional jet FL250–390, narrowbody FL300–390, widebody FL310–410). The tiles give the **tropopause** (`>FL393`: higher than the data reach), the **jet** (the strongest wind above 15 000 ft, from 60 kt) and the mean ISA deviation in the cruise band; the charts the wind speed and the ISA deviation against the flight level.
- **Icing** — the heights of 0, −10 and −20 °C (the jets also −40 °C) in ft MSL; the temperature from the ground up — the airport's landings and departures (white), the area within 150 km (violet), the Kivenlahti mast (orange), the MEPS model (cyan, dashed) and the METAR — with the 0 °C line, 0–20 000 ft for a turboprop and 0–30 000 ft for the jets; MEPS cloud from its humidity at the right (white 95 % or more, grey 85–95 %). A red box warns of a **warm layer above a sub-freezing surface**: air above 0 °C over a ground at or below about 0 °C — rain or drizzle falling through it freezes on contact (freezing rain); it says whether the METAR reports precipitation. Below, the icing reports of the last 6 hours (ARS, WXREP and SIGMET with icing or freezing rain), near EFHK or not.
- **Turbulence** — the EDR of the window's landings and departures: the median, the roughest piece with its aircraft type and height, the flights with a moderate piece (0.20 or more) and the METAR gusts; **by height** per 1 000 ft above the runway (the median, the roughest piece and how many pieces; landings below 3 000 ft, departures to 5 000 ft; amber moderate, red severe); **the last 6 hours** as bars (the roughest piece of each hour, the dot its median, the number of flights below); the roughest flights and the turbulence reports of the last 6 hours (ARS, WXREP and SIGMET with TURB or MTW).
- **GNSS** — the GNSS interference as counts only: the aircraft with degraded GPS now and the severe ones (no position accuracy, or a frozen position), with their FL bands; the degraded aircraft-hours of the last 24 hours as bars; and the landings at EFHK degraded on final and the departures degraded on the climb-out in the last 6 and 24 hours. From 11 November 2026 the Area buttons at the top choose which aircraft are counted — Coverage (everything the receiver hears), 50 NM (the default) or 20 NM from the airport — for the counts of now and the 24-hour bars alike; the two cards name the area, and the choice is remembered. An aircraft is in an area by its last position, so one that has lost its position altogether is counted only in Coverage. The landings and departures are the same in every area. The GPS pages and the GNSS view with the aircraft themselves are for the *Full view* right.
- **My flight** — type a callsign or registration (e.g. `FIN654` or `OH-ATK`): the landings and departures of the last 7 days. Tap one for its debrief: its own headwind on final (or on the climb-out) against the median and middle half of the flights of the 45 minutes before it in the same direction; its windshear report — the ramp with its heights and distances, the band change, and whether the landings before and after saw the same; and its turbulence (EDR, all and below 1 000 ft) with the lowest GNSS accuracy (NACp) on final. ⟨ Results goes back to the list.
- **Map** — three presets of the Weather view: Approach & departure (rain and snow from the model and the FMI radar, low cloud, the wind and its gusts, the live traffic), Icing & freezing (the 2 m isotherms with 0 °C, all cloud, rain and snow) and Fog & low visibility (fog, the wind, the traffic). A tap sets the Weather options as the menu would and opens the view; slide to your ETA. Your own settings are kept: Back to my settings (shown after the first preset) puts them back.

Pilot options in the menu: Aircraft class — Turboprop, Regional jet, Narrowbody or Widebody: highlighted on Approach and Departure, the levels and cruise band of Aloft and the heights of Icing; and Window — 30 min, 60 min or 2 h for Now, Approach, Departure and Turbulence. They are remembered on the phone, as are the tab, the runways, the filter and the search. The view refreshes itself every minute (not My flight or Map).

On the full site (from the evening of 5 November 2026) the same tabs are on the **Pilot** page — *Pilot* in the top bar, for users with the Pilot right only (Fig. 61): the tabs in a bar as on the Sounding page, the aircraft class and the window beside them, and the cards in one column from top to bottom on the left as on the phone (from 6 November 2026 — before, side by side). Map there opens the Weather page with the preset. The class, the window and the tab are shared with the phone view in the same browser.

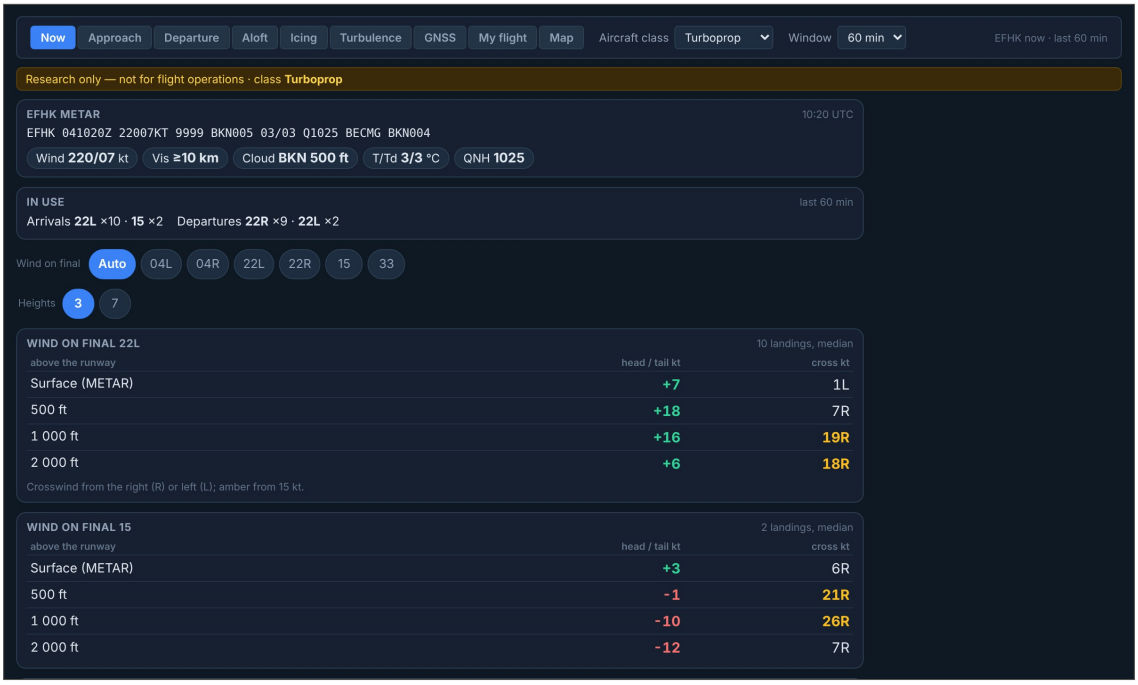


Figure 61. The full site's Pilot page, Now for a turboprop (the stored data of 4 November 2026 at 10:45 UTC, drawn in a test): the tabs with Aircraft class and Window, the METAR, the runways in use, the runway choice and the wind on final — from 6 November 2026 the cards run from top to bottom in one column on the left as on the phone, and the page scrolls on to the windshear and the turbulence.

16 When something looks wrong

- **The page does not load, or asks for the password again and again.** The phone must reach MODE-S Wind as a computer does; check the network and the address (with `/app`). The login is remembered for 30 days; log in again once after your password has been changed, or when the phone's browser data has been cleared.
- **`Could not load ...`** under a view: the server did not answer in time. The view tries again at its next update, or switch the view and back.
- **The health dot is amber or red.** Open the menu: General says which part. MODE-S Wind keeps working with what it has — e.g. without new MEPS data the Skew-T has no model line, the rest is unchanged.
- **`No aircraft on the approach`**: there is no traffic in the approach corridors of the chosen runway; choose All.
- **`MEPS: no model data for this time`** or `no sounding stored`: the model run or the radiosonde for that time has not arrived (or is not available); the aircraft profile is not affected.
- **After an update of MODE-S Wind** the phone may still show the old page from its cache: reload the page in the browser, or close the home-screen app and open it again.
- **Weather: `No weather maps stored yet.`** or the strip stays pale: the MEPS run has not been stored yet or the network is slow; the view tries again with the next run check (10 min), or open it again. With an old phone choose 20 km or Near the slider. `FMI radar not reachable`: FMI's service did not answer; the view tries again within a minute or a few, and shows the model meanwhile.
- **No GNSS tile in the menu**, or `The GNSS data needs the Full view right`: the GNSS view is for users with the *Full view* right only (Section 14).
- **No Pilot tile in the menu**, or `The Pilot view needs the Pilot right`: the Pilot view is for users with the *Pilot* right only (Section 15).
- **Text too small?** Use landscape, or the phone's own zoom (pinch); the diagrams are drawn at the phone's full resolution.

17 Quick reference

Item	Meaning
`/app`	the mobile view; same login as the full site, remembered for 30 days
≡	menu: views as tiles, options of each product, General
Red dot on ≡	a new version: open What's new at the top of the menu (or in General)
Time at the top	the current time in UTC (Finland: + 2 h in winter, + 3 h in summer)
Moon / sun / half circle	theme Dark / Light / Auto (follows the phone); a tap steps to the next
Dot at the top right	system health: green working, amber late / fallback, red stopped
Tap / drag	values at that point (the mouse hover of the full site)
Swipe up / down	scroll a view taller than the screen
Grey line and barbs	aircraft profile (Skew-T), with the wind values
Hodograph (Skew-T)	the wind of each layer as a point, joined from the ground (red) up (violet); rings = speed in kt, clockwise with height = veering, dashed = the two hours before
Thin orange lines (Time–height)	isentropes, potential temperature every 5 K
Hatched right part (Time–height)	the MEPS forecast after the present: a forecast, not a measurement
Calendar	one row per UTC day, one column per UTC hour; tap an hour for its values, tap it again to open the day in Time–height
Pink / teal	Jokioinen radiosonde / MEPS weather model
Green / amber / red dot (ILS)	aircraft on / above / below the glideslope (±300 ft)
Squares on the map (ILS)	aircraft; colour = wind source, framed = whose barbs are drawn; tap to follow
`ADV` / `WSA` / `MBA`	windshear advisory / windshear alert / microburst alert
`HW` / `TW` / `AP`	headwind / tailwind component; autopilot engaged
ACI	Approach Conditions Index 0–10: below 2 smooth, light, choppy, rough, 8+ very rough
Updates	Skew-T now 2 min, time–height 10 min, ILS 5 s, EDR · WS · ACI · ST and History 1 min, Statistics and GNSS 30 s, Weather air traffic 6 s; only the shown view, only with the screen on
Blue / red cells (ST)	less / more headwind than the flights of the previous 45 min
Card chips (History)	ACI value, windshear level, *trial* (departure ramps, no alert), GA go-around; for the reports `WS R22L` and `near`
Amber bar / **P** (WS chart)	a METAR with a windshear group / a pilot report near the airport
Strip / slider (Weather)	hours loaded: grey past, violet forecast, pale not yet; ▶ plays them
Dotted box (Weather)	the 2.5 km detail near the airport, zoom 7 and closer
Radar (Weather)	the newest FMI radar image, only at Now; dashed: the edge of its coverage; white outlines (Both): the model's rain
Tap the map (Weather)	the weather of that place, following the slider, and Meteogram here ›; tap an aircraft or a receiver triangle for its details
Model: GFS (Weather)	NOAA's global model to 16 days, every 3 hours after +120 h; −1d / +1d step a day; for viewing
Mast	the Kivenlahti mast's lowest 300 m: EFHK now with Δ MEPS / Δ Mode S, the tiles, the profile, the boundary layer with the mast below the amber line, the time series, the comparison
Meteogram	one place (EFHK, a receiver site or the tapped place) over the days: temperature, precipitation, wind, pressure, clouds; MEPS bold, GFS thin and dashed; tap or drag for the values
About this panel (Statistics)	the panel's explanation, folded; tap to open
All receivers / `yours` (Statistics)	every receiver's replies, each once / your own receiver's tab (owners and Full view only)
GNSS (Full view only)	degraded GPS now (green none, amber degraded, red severe), per hour or (1 m / 3 m) per day, the FL band heatmap, the live list and the spoofing indicators; `2+`: two or more independent indicators
Pilot (Pilot right only)	EFHK now, the headwind on final and on the climb-out by runway (ft above the runway, your class in violet), winds and temperatures aloft, icing (freezing levels, warm layer above a sub-freezing surface), turbulence, GNSS counts, My flight and the weather map presets; also the full site's Pilot page; research only
Settings	remembered on the phone; Skew-T options and theme shared with the full site in the same browser

Other guides (User guides in the menu, and the User guides tab of the full site's *What's new* window): *Reading the Atmosphere Profile (Skew-T)*; *The Sounding and Wind Map pages*; *Windshear detection and the Windshear page*; *Mode S & ADS-B and the Statistics page*; *Meteograms* and *Meteogrammit (suomeksi)*; *The GPS pages — Quality, Episodes and Aircraft*.